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Robust Learning over Streaming Feature Spaces

Yubo Cai, Vassilis Digalakis Jr., and Michael Lingzhi Li

Working draft; in preparation for submission to Operations Research.

This draft studies supervised prediction when features become available sequentially and in an uncertain order. It develops permutation-aware losses, robust optimization formulations, and gradient- and sampling-based methods, with synthetic experiments. Its connection to continual learning is the study of predictive performance under evolving information availability. The setting uses complete training data and sequential feature revelation at prediction time.

Suggested reading: Sections 2-4 (paper pp. 7-17) for the setting and robust objectives; Section 5 (pp. 17-24) for algorithmic development.

2 / THEORY AND COMPUTATIONAL METHODS PDF pp. 46-88

Optimizing Lyapunov Certificates via Stability-Preserving Quadratization for Polynomial Systems

Yubo Cai and Gioele Zardini

Preprint, 2026. arXiv:2609.18871

This paper develops optimization-based Lyapunov certificates for polynomial dynamical systems. It combines an exact fixed-gain semidefinite formulation, nonsmooth outer optimization, conditional stationarity analysis, and numerical benchmarking. Included to demonstrate mathematical derivation, optimization algorithm design, and computational evaluation, it provides methodological background for studying stability and optimization in learning dynamics.

Suggested reading: Sections 3-4 (paper pp. 7-18) for formulation and algorithms; Section 5 (pp. 19-24) for experiments; Appendix C (pp. 34-38) for technical derivations and convergence analysis.

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Robust Learning over Streaming Feature Spaces

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Abstract.

Key words: streaming feature space, sequential feature arrival, feature arrival uncertainty, feature acquisition, missing data

1. Introduction

As data availability increases, there is a meteoric rise in using machine learning (ML) algorithms to guide decision-making across areas including healthcare, retail, transportation, supply chain, and many more. Machine learning algorithms aim to predict the value or prevalence of a certain quantity of importance $y \in \mathbb{R}$ with the available information, enabling decision-makers to make informed choices that optimize outcomes. Traditionally, these algorithms assume a fixed and complete input of p features $\boldsymbol{x} \in \mathbb{R}^p$ to predict a downstream target y.

However, in many real-world scenarios, the information available at decision time often varies on a case-by-case basis. For example, when a patient arrives in the emergency room, the available information may vary depending on the nurse's triage, the patient's own measurements, and the availability of test results. To address missing features, the most common approach, referred to as impute-then-predict (ITP), is to impute any missing values among the p features and utilize the trained ML algorithm $f(\cdot)$ as usual.

Recent work increasingly points towards the limitations of such methods. In particular, ITP faces two difficulties:

• Imputation is most commonly done without knowledge about the downstream task, so even an optimal imputation might not be optimal for the downstream prediction.

• Imputation-then-predict means that the ML algorithm f is fixed no matter the feature order, implying that different feature orders should be treated identically. This assumption might not hold in many real-world scenarios where the features available at decision time should affect the downstream prediction methodology.

In response to these limitations, we propose learning over *streaming* feature spaces, where $\boldsymbol{x} \in \mathbb{R}^p$ refers to the space of *potential* features that can be utilized for the downstream prediction. Instead of assuming the features are all present at decision time, we assume that the features are revealed over time in an uncertain order that can be identified by a permutation $\sigma \in S_p$, where S_p is the symmetric group of size p.

Depending on the uncertainty of σ and the goal of the decision-maker, we consider two regimes:

1. In the first regime, the decision-maker has no control over the feature arrivals and wishes to learn a ML model that optimizes the average or the worst-case performance across all possible feature arrivals.

2. In the second regime, the decision-maker has some level of control over the feature arrivals and wishes to learn a ML model that optimizes the average or the worst-case performance across the uncontrollable portion of the feature arrivals.

These two regimes address different practical scenarios. For example, in an emergency room, the attending doctor might have limited control over the information available and must make immediate recommendations based on varying sets of data. Conversely, in outpatient clinics, doctors often have strong control over the next tests to sequentially discover information in the optimal order for diagnosis or treatment.

We present a unifying framework that combines both regimes under a three-stage optimization problem. This framework encompasses many existing problems as special cases, including the feature acquisition problem, the robust learning problem, and the unconstrained data learning problem. We develop a general alternating optimization algorithm to solve this problem. Our empirical evaluation using synthetic datasets, real-world datasets from the UCI ML repository, and the MIMIC-IV medical database—a large real-world dataset of emergency patients—demonstrates significant improvements in performance and robustness compared to traditional methods. This highlights the potential for our framework to be applied in various dynamic and high-stakes settings.

1.1. Contributions and Outline

Formally, our contributions are as follows:

1. **Conceptual Innovation:** We formally introduce the concept of learning in streaming feature spaces, where the set of features used in the learning process can change over time. This represents a significant departure from traditional static feature space assumptions commonly used in ML. We provide a theoretical analysis demonstrating that the conventional imputation-then-learning approach cannot guarantee optimal solutions in such streaming settings. This underscores the need for novel learning paradigms that can adapt to feature space dynamics.

2. **Framework for Average-Case Optimization:** We propose a new framework for learning under streaming feature spaces when the policy-maker aims to optimize for *average-case* performance across all possible feature permutations, which may occur at different rates. We show that this problem can be efficiently approximated using a stochastic gradient descent scheme. This approach offers significant flexibility, allowing it to be easily integrated with a wide range of existing ML algorithms.

3. **Framework for Worst-Case Optimization:** We extend our framework to scenarios where the policy-maker has some level of control over the set of possible feature permutations and seeks to optimize for the *worst-case* performance across all potential cases. This extension is crucial for applications where robustness and reliability under the most adverse conditions are paramount. We develop an alternating optimization algorithm that efficiently solves this problem.

4. **Empirical Validation:** We validate our proposed framework through extensive empirical evaluation using the MIMIC-IV medical database, a large-scale real-world dataset. We demonstrate the practical applicability and effectiveness of our approach in a complex, dynamic environment. Our results show significant improvements in performance and robustness compared to traditional methods, highlighting the potential for our framework to be applied in various dynamic and high-stakes settings.

[**Vassilis:**

What is special in our work compared to, eg, standard imputation?

The fundamental defining aspect of our problem is that we have missing features during prediction time. The model is trained on full features. So, we have the following cases:

- On one extreme, when the model is allowed to change as features arrive, we effectively get the oracle approach. Such models can be independent (as in the oracle approach we discussed) or linked (e.g., as policies etc). Here, during training, we can train using all possible subsets of features and hence be prepared for any permutation

- On the other extreme, when the model is not allowed to change as features arrive, we effectively get one model. One approach here is to do imputation: we train imputation models (eg MICE) during training (where data is fully observed), and we deploy them during testing. The prediction model here during deployment is fixed.

- In between, we can have models for different partitions of the feature set. Or we can have indicators depending on what features have arrived — which would serve as some form of missingness bias correction

Qs:

- What is the gap between the oracle and the imputation approach?

- Can we improve on that gap by exploiting the fact that we know that feature will arrive sequentially? (1) If we know the order? (2) If we pick the order? (3) If we know/pick the order but there is uncertainty? (4) If we do not know/pick the order?

OUTLINE:

Impute-then-predict: works in the linear case, suboptimal beyond that

General + principled framework to do impute-and-predict simultaneously

- Permutation-aware loss function: easy to implement

- Algorithm for optimizing over models + permutations

– Expand dataset to all 2^p subsets

– Dictionary: which permutation corresponds to which row

– For each step in the training process, find permutation that corresponds to our loss function at that step

– Update the gradient

- Stochastic framework for scalability

]

1.2. Related Literature

[**Vassilis:** Cite and discuss Wu et al. (2012)]

The problem setting that we consider in this paper is related with a few problems that have received considerable attention in the ML literature: these include the feature acquisition problem, the incremental feature ranking problem, and learning with missing data.

Feature Acquisition. Sequential feature acquisition and selection are critical in scenarios where features have different costs and are acquired incrementally under constraints such as budget or performance criteria. Early works like Saar-Tsechansky and Provost (2007) explored strategies for acquiring feature values based on their expected improvement in model performance per unit cost. This concept has evolved, with Xu et al. (2013) and Kusner et al. (2014) exploring how to balance the computational cost of feature extraction against classifier accuracy. They proposed using a tree of classifiers, where test inputs traverse through paths optimized for specific sub-partitions of the input space.

Nan and Saligrama (2017) extended this by adaptively approximating high-cost models in regions where low-cost models suffice, while Kachuee et al. (2018a) used a denoising autoencoder to handle unknown features and assess their informativeness. The role of reinforcement learning in this context has been highlighted by He et al. (2012) and Kachuee et al. (2018b), particularly in disease diagnosis applications (Kao et al. 2018, He et al. 2022). Recently, Liu et al. (2023) framed feature selection and acquisition as nonlinear discrete optimization problems aimed at minimizing prediction errors within a given budget.

Incremental Feature Ranking. Incremental feature ranking methods are essential in environments like streaming text applications, where the feature space is not initially known. For instance, Katakis et al. (2006) developed an incremental feature ranking method followed by a learning algorithm such as KNN that is capable of making predictions based on any subset of features. Similarly, Masud et al. (2010) introduced a feature space conversion technique that transforms heterogeneous feature spaces into a unified homogeneous space, enhancing the interoperability of data from varied sources. Additionally, Gomes et al. (2013) explored incremental feature selection strategies that dynamically store only relevant features and employ a dynamically weighted ensemble of models to optimize performance. For a more comprehensive overview of these evolving techniques, see the review by Kaya et al. (2020), which elaborates on the advancements and applications of dynamic feature space methodologies.

Learning with Missing Data. Research in ML and statistics has traditionally focused on addressing missing data during the model induction phase (Little and Rubin 2019, Hastie et al. 2009). Most imputation methods rely on the assumption of Missing Completely at Random (MCAR), where missing values occur independently of the data. While these methods work well under MCAR, they can be unreliable when data is Missing Not at Random (MNAR)—which is commonly the case in the setting we consider in our paper (Little 1988, Jaeger 2006).

As noted by Saar-Tsechansky and Provost (2007), there is a significant gap in methodologies specifically tailored for dealing with missing values at the time of prediction; their work provides a comparative analysis of the efficacy of various approaches, particularly for models like classification trees. The predominant strategies for managing missing features at prediction time involve value-based imputation methods, whereby a value for the missing data point is directly estimated based on the available data, and distribution-based imputation methods, whereby the conditional distribution of the missing feature is estimated and used to inform predictions (Troyanskaya et al. 2001). The Missing Indicator Method (MIM) is commonly used to handle missing data and has shown effectiveness both empirically and theoretically; recently, Selective MIM (SMIM) was introduced Van Ness et al. (2023), outperforming MIM in high-dimensional settings and offering a more effective approach for complex datasets.

An alternative strategy involves the use of reduced-feature models, which only incorporate features guaranteed to be present at test time, thereby eliminating the need for imputation. This approach is relatively underexplored, with some foundational ideas discussed by Schuurmans and Greiner (1995) and Friedman et al. (1996). These models adapt dynamically to the pattern of available features, using different models for different patterns of missing data. In particular, the lazy decision tree methodology (Friedman et al. 1996) relies on the construction of a bespoke decision tree for each instance at test time, ensuring that the tree only uses available features. Conversely, eager decision trees (Gavankar and Sawarkar 2017) attempt to anticipate all possible scenarios of missing data during the training phase, an approach that can be computationally prohibitive and less practical.

Recently, Bertsimas et al. (2024) view prediction with missing data as a two-stage adaptive optimization problem: their development of adaptive linear regression models, which adjust regression coefficients based

on available features, allows for simultaneous learning of imputation rules and downstream models, showing a significant improvement in prediction accuracy, particularly in settings where data is strongly not missing at random.

2. Problem Statement: What is a Streaming Feature Space?

The term "*streaming feature space*" describes the following supervised learning setting. Each data point has potential features $\boldsymbol{x} \in \mathbb{R}^p$ that are utilized to predict a target $y \in \mathbb{R}$. For the training process, we have access to a full dataset $D = (\boldsymbol{x}_i, y_i)_{i=1}^n$ that is drawn i.i.d. from an unknown distribution $(\boldsymbol{x}, y) \sim \mathcal{D}$. During deployment (and testing), the features are revealed sequentially over time in a (possibly uncertain) order. The feature arrival order is described by a permutation $\sigma \in \mathcal{S} \subset S_p$, where S_p is the symmetric group of size p and $\mathcal{S}$ is the set of feasible permutations. We denote by $\sigma(j)$ the jth feature to be revealed in the process.

When a feature is not yet revealed, we denote the missing value as $\emptyset$, and we define $\mathbb{R}_\emptyset^p = (\mathbb{R} \cup \emptyset)^p$ as the feature space augmented with the missing values. We formalize this sequential feature reveal by considering a masking function $s(\boldsymbol{x}, \sigma, j) : \mathbb{R}^p \times S_p \times [p] \to \mathbb{R}_\emptyset^p$ defined as follows:

$$s(\boldsymbol{x}, \sigma, l)_j := \begin{cases} x_j, & \exists\, k \leq l \;\; s.t. \;\; \sigma(k) = j, \\ \emptyset, & \text{otherwise,} \end{cases}, \tag{1}$$

i.e., feature j is revealed if and only if it is within the set of the first l elements of permutation σ. We use the shorthand $\boldsymbol{x}^{\sigma,l} := s(\boldsymbol{x}, \sigma, l)$ whenever it is clear.

We define a *ML model* $f \in \mathcal{F}$, where $\mathcal{F}$ denotes the class of all ML models under consideration, as a function mapping $f : \mathbb{R}_\emptyset^p \to \mathbb{R}$. In the ideal scenario, f would consist of 2^p separate "sub-models," one for each possible subset of features, and select the appropriate model based on the feature arrival order. We refer to this approach as the "oracle model." The oracle model can be represented as a decision tree whose first p levels identify the observed feature subset (each level i tests whether feature i is available), after which the tree routes to a sub-model that makes predictions using only the available features. In practice, training the oracle model is computationally infeasible and statistically challenging due to the exponential increase in the number of sub-models with the number of features and the extensive data requirements for training such a multitude of models, respectively. Consequently, we are interested in training practically-sized models.

The goal is to find a ML model that minimizes loss throughout the feature reveal process. Concretely, consider a data point $(\boldsymbol{x}, y) \sim \mathcal{D}$ and a model f producing predictions $\hat{y}^{\sigma,j} = f(\boldsymbol{x}^{\sigma,j})$ at each step j. Given a permutation $\sigma \in \mathcal{S}$ and a loss function $\ell : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\geq 0}$ measuring the quality of prediction $\hat{y}^{\sigma,j}$ for target y, we define the *permutation-aware loss* function as $\ell_p(y, \boldsymbol{x}, f, \sigma) = \sum_{j=1}^{p} \ell(y, f(\boldsymbol{x}^{\sigma,j}))$. Then, we seek a model that minimizes the expected permutation-aware loss:

$$\min_{f \in \mathcal{F}} \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}} \left[\ell_p(y, \boldsymbol{x}, f, \sigma)\right]. \tag{2}$$

In practice, given a dataset D, we approximate the expected loss with the empirical loss:

$$\min_{f \in \mathcal{F}} \sum_{(\boldsymbol{x}_i, y_i) \in D} \ell_p(y_i, \boldsymbol{x}_i, f, \sigma). \tag{3}$$

To simplify notation, we denote the aggregated permutation-aware loss over the training set as $\mathcal{L}(f, \sigma; D) = \sum_{(\boldsymbol{x}_i, y_i)\in D} \ell_p(y_i, \boldsymbol{x}_i, f, \sigma)$.

REMARK 1. The *oracle model*, which trains a separate predictor for each possible subset of features, minimizes the permutation-aware loss function.

REMARK 2. In Equation (2), we assume the feature arrival order σ is known and take the expectation over the data distribution. We will drop this assumption and treat the uncertain feature arrival order case in Section 4.

3. Limitations of Impute-Then-Predict in Streaming Feature Spaces

To address the problem presented in Section 2, machine learners commonly resort to imputation-based approaches. As we argue in this section, the standard ITP paradigm leads to suboptimal performance. To simplify presentation, we relegate all proofs to Appendix A.

We assume the data is drawn from a distribution $(\boldsymbol{x}, y) \sim \mathcal{D}$ and the feature arrival order is described by a fixed permutation $\sigma \in \mathcal{S}$. We consider the squared loss $\ell(y, f(\boldsymbol{x}^{\sigma,j})) = (y - f(\boldsymbol{x}^{\sigma,j}))^2$. First, we characterize the optimal predictor in each step of the feature reveal process. To do so, we adapt the following well-known result:

LEMMA 1. *For any $j \in [p]$, the optimal predictor after j feature arrivals according to permutation $\sigma \in \mathcal{S}$ is given by $f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \mathbb{E}[y|\boldsymbol{x}^{\sigma,j}]$.*

Since the optimal predictor changes throughout the feature reveal process, using the optimal predictor in each step requires 2^p models—one for each feature subset. As we already outlined, training this many models can be infeasible in practice, hence why machine learners typically resort to imputation-based approaches.

The standard ITP paradigm relies on training the optimal model for the full feature set, denoted by $f^\star : \mathbb{R}^p \to \mathbb{R}$, finding an imputation for the missing features in each step of the feature reveal process, and finally plugging the imputed data into $f^\star$. Next, we characterize the optimal imputation in each step of the feature reveal process. Denoting by $\boldsymbol{x}^{\sigma,-j}$ the features in $\boldsymbol{x}$ that have not yet been revealed after j arrivals according to permutation σ, we again adapt a well-known result:

LEMMA 2. *For any $j \in [p]$, the optimal imputation after j feature arrivals according to permutation $\sigma \in \mathcal{S}$ is given by $\hat{\boldsymbol{x}}^{\sigma,-j} = \mathbb{E}[\boldsymbol{x}^{\sigma,-j}|\boldsymbol{x}^{\sigma,j}]$.*

Thus, the ideal version of ITP is $f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}) = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}]$. As we show next, even the ideal ITP predictor, which relies on the optimal imputation of Lemma 2, may lead to suboptimal performance compared to the optimal predictor of Lemma 1:

PROPOSITION 1. *For any $j \in [p]$, it holds that $\mathbb{E}\left[(y - f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}))^2\right] \leq \mathbb{E}\left[(y - f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}))^2\right]$. Equality holds if and only if $f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}) = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]$ almost surely.*

Proposition 1 establishes a non-strict upper bound on the difference in expected squared error between the optimal predictor and the ITP predictor. In what follows, we examine specific conditions under which the ITP approach is optimal and, conversely, conditions under which a strict performance gap exists, thereby motivating the development of alternative prediction methods. The following Corollary 1 provides a concrete example of the former case: for linear-Gaussian data, the optimal predictor $f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j})$ and the ITP predictor $f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j})$ are equivalent.

COROLLARY 1. *Assume a linear data generating process $y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon$ with Gaussian features $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu} \in \mathbb{R}^p$ is the mean vector, $\boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$ is the covariance matrix, $\boldsymbol{\beta} \in \mathbb{R}^p$, and $\epsilon \sim \mathcal{N}(0, \sigma^2_\epsilon)$ is independent of $\boldsymbol{x}$. Then, for any $j \in [p]$, it holds that $\mathbb{E}\left[(y - f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}))^2\right] = \mathbb{E}\left[(y - f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}))^2\right]$.*

Proposition 2 provides the specific conditions under which ITP is suboptimal. Under such regimes, we expect our proposed approach to be most beneficial.

PROPOSITION 2. *For any $j \in [p]$, it holds that $\mathbb{E}\left[(y - f^{\star}_{\sigma,j}(\boldsymbol{x}^{\sigma,j}))^2\right] < \mathbb{E}\left[(y - f^{\star}(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}))^2\right]$ if and only if **(i)** $\mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}, \boldsymbol{x}^{\sigma,-j}]$ is nonlinear in $\boldsymbol{x}^{\sigma,-j}$ (**nonlinearity**) and **(ii)** $Var(\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}) > 0$ (**non-degeneracy**).*

[**Vassilis:** This is incorrect (specifically, the part of the proof in red). We could turn proposition 2 into a corollary: if E[y|x sigma j] is strongly convex or strongly concave in x sigma j, then under the nonlinearity and non-degeneracy conditions, we indeed have strict inequality. This way we have one corollary for equality and one for strict inequality — kind of as examples.

@MICHAEL: can you at some point refine this result?]

Numerical Experiments

Experiment 1: *What is the gap between the oracle and ITP?*

To numerically verify the limitations of the ITP approach, we compare the performance of the optimal predictor $f^{*}_{\sigma,j}(x^{\sigma,j})$ with the ITP predictor $f^{*}(x^{\sigma,j}, \hat{x}^{\sigma,-j})$ under different data-generating processes and feature arrival orders. Specifically, we consider a linear process, where the target y is a sparse linear combination of the features plus noise, and a nonlinear process, where the target y includes quadratic terms, all pairwise interactions, and squared terms, plus noise. We evaluate a worst-case arrival order, where features with the weakest correlation with the target y are revealed first, and a random order. We use nonlinear models (boosted trees) to predict the target variable. To approximate the optimal imputation, we use MICE (Multiple Imputation by Chained Equations) (Van Buuren and Groothuis-Oudshoorn 2011), which iteratively models each feature as a function of the others to handle missing values. We provide additional details on the experiments in Appendix E.

The results, presented in Figure 1, confirm the theoretical predictions. When the data-generating process is linear, the ITP approach performs similarly to the optimal predictor, regardless of the feature arrival order. When the data-generating process is nonlinear, the ITP approach is suboptimal, with the performance gap increasing for the worst-case arrival order.

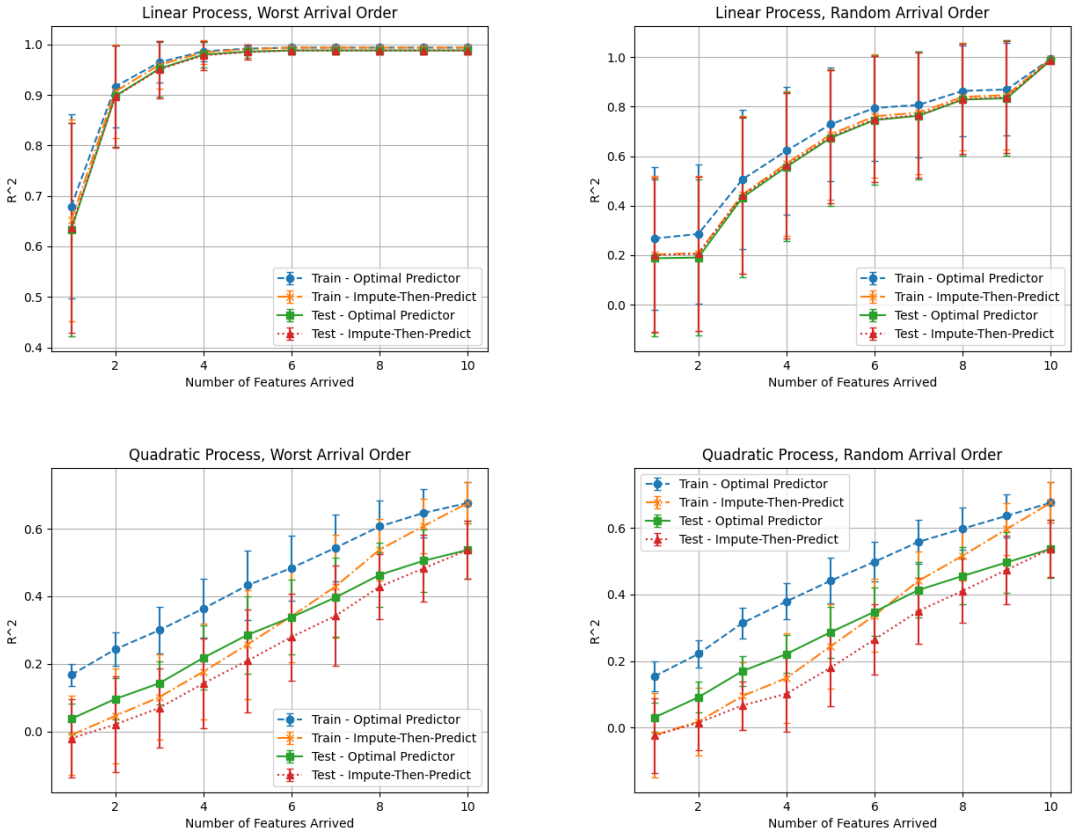


Figure 1 Comparison of the optimal and ITP predictors across different processes and feature arrival orders.

4. A Framework for Robust Learning over Streaming Feature Spaces

So far, we have disregarded any uncertainty around or control over the feature arrival order either by assuming that the permutation σ is fixed and known, or by minimizing the expected loss over all permutations. In this section, we incorporate feature arrival uncertainty considerations into our modeling and develop a practical framework for robust learning over streaming feature spaces.

In some scenarios, a decision-maker might have full control over the permutation σ, whereas in other scenarios the decision-maker might have no control. More commonly, the decision-maker have some level of control over the incoming permutation σ but are exposed to a range of possible permutation sequences. To formalize this characterization, we consider a K-sized set of subsets of $\mathcal{S}$, denoted $\mathcal{A} = \{A_1, \cdots, A_K\}$. The decision-maker can choose among the K subsets, but once the decision-maker chooses subset A_k, any permutation $\sigma \in A_k$ can occur. The choice over the K-sized set of subsets and the size of each subset thus represents how much control the decision-maker has and how much robustness the decision-maker needs to

consider. We are interested in both helping the decision-maker to choose among the K subsets, and learning a ML model that performs well under any permutation within the chosen subset:

$$\min_{f\in\mathcal{F}} \min_{A_k\in\mathcal{A}} \max_{\sigma\in A_k} \mathcal{L}(f,\sigma). \tag{4}$$

The framework described by Equation(4) encompasses many existing problems as special cases:

- If we assume that the decision-maker has full control over the permutation sequence, i.e. the set of subsets are all the unit permutations $\mathcal{A}=\{\sigma\}_{\sigma\in\mathcal{S}}$, then the maximization problem degenerates and we have:

$$\min_{f\in\mathcal{F}} \min_{\sigma\in\mathcal{S}} \mathcal{L}(f,\sigma). \tag{5}$$

By doing so, we recover the *feature acquisition problem*.

- If we assume that the decision-maker has no control over the permutation sequence, i.e., $\mathcal{A}=\{\mathcal{S}\}$, then the minimization problem becomes degenerate, and the optimization problem can be written as:

$$\min_{f\in\mathcal{F}} \max_{\sigma\in\mathcal{S}} \mathcal{L}(f,\sigma). \tag{6}$$

- In many scenarios, the decision-maker can select a particular permutation sequence but needs to defend against up to B misorderings in the sequence. For any permutation $\sigma_0\in\mathcal{S}$, we consider a "ball" $\mathcal{B}_B(\sigma_0)=\{\sigma : d(\sigma,\sigma_0)\le B\}$, where $d:\mathcal{S}\times\mathcal{S}\mapsto\mathbb{R}_{\ge 0}$ measures the "distance" between two permutations, capturing the dissimilarity between the corresponding feature reveal sequences. Then, we have $\mathcal{A}=\{\mathcal{B}_B(\sigma)\}_{\sigma\in\mathcal{S}}$ and the optimization problem can be written as:

$$\min_{f\in\mathcal{F}} \min_{\sigma_0\in\mathcal{S}} \max_{\sigma\in\mathcal{B}_B(\sigma_0)} \mathcal{L}(f,\sigma). \tag{7}$$

Equation (7) reflects the robust optimization view of the problem, where $\mathcal{B}_B(\sigma_0)$ is the uncertainty set.

To measure the distance between permutations, we use the *Levenshtein edit distance4* Levenshtein (1966), defined as the minimum number of edit operations required to transform one permutation into another. For permutations where all elements are present, the only meaningful edit operation is substitution, which corresponds to reordering elements. Formally, given two permutations $\sigma_1,\sigma_2\in S_p$, the Levenshtein distance is $d_{\text{Levenshtein}}(\sigma_1,\sigma_2)=\sum_{i=1}^{p}\mathbf{1}[\sigma_1(i)\neq\sigma_2(i)]$, where $\mathbf{1}[\cdot]$ is the indicator function, and $\sigma_1(i)$ and $\sigma_2(i)$ represent

the elements at the i-th position in σ_1 and σ_2, respectively. This distance measures the number of mismatched positions between σ_1 and σ_2, penalizing mismatches linearly.

[**Vassilis:** Theoretical backing?

1. Show that, by using the permutation-aware loss function together with the optimal imputation, we improve on average (this would be the first result actually supporting our method

2. Prove the magnitude of the gap, on the loss function with the particular sigma chosen in the min-min formulation, between min-min formulation and standard formulation (just minimize loss without picking permutation) when assuming a particular permutation model.

3. Prove the magnitude of the gap between the min-max formulation, and the max loss of the standard formulation under worst case permutation.]

Numerical Experiments

The numerical experiments in Section 3 verified the existence of a gap between the oracle predictor and the ITP predictor. Acknowledging such a gap, we now turn to the question of whether we can improve upon the standard ITP predictor by exploiting our knowledge of the fact that features will arrive sequentially.

Experiment 2: *Can we improve by using the permutation-aware loss for known feature arrival order?*

The first experiment investigates whether we can improve by using the permutation-aware loss function—which, when the model is allowed to adapt depending on the features that have arrived, recovers the oracle. To evaluate the effectiveness of the permutation-aware loss function, we compare the performance of the standard ITP predictor with the permutation-aware ITP predictor. The standard ITP predictor is trained on the full dataset and imputes missing features during inference. In contrast, the permutation-aware predictor is trained using the permutation-aware loss function—which results in an expanded dataset that explicitly accounts for sequential feature arrivals by replicating training samples with increasing subsets of features. We test these approaches under a quadratic data-generating process and random and worst-case feature arrival orders. For prediction, we use a nonlinear model (boosted trees) and perform imputation using MICE.

The results, presented in Figure 2, show that the permutation-aware approach improves generalization compared to the ITP approach. Across both feature arrival orders, the permutation-aware loss leads to a

better test R^2 score, particularly for early feature arrivals. The performance gap is more pronounced under the worst-case feature arrival order, where the permutation-aware loss significantly improves robustness by allowing the model to learn representations that generalize better as features arrive sequentially.

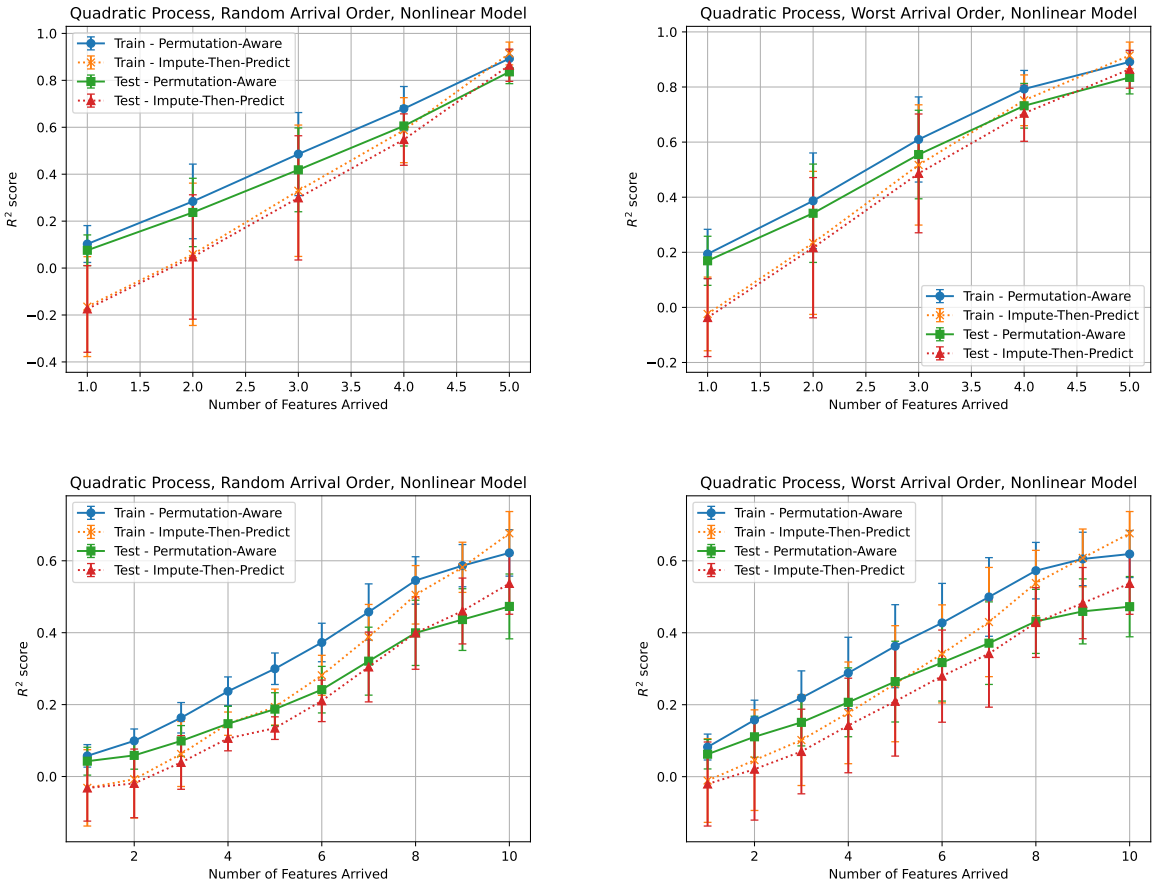


Figure 2 Comparison of standard and permutation-aware ITP under different feature arrival orders.

Experiment 3: *Can we improve by choosing the feature arrival order?*

In this experiment, we investigate whether selecting an optimal feature arrival order can further improve performance compared to using a fixed arrival order. Instead of relying on a predetermined feature arrival order, we sample a feasible set of possible permutations and select the best one based on validation performance. The goal is to determine whether learning under an optimal arrival order, chosen from a feasible set, leads to better predictive performance. We compare the standard ITP predictor with the permutation-aware ITP predictor, with the additional flexibility of selecting the feature arrival order. Specifically, for each feature

arrival permutation in the feasible set, we train a separate permutation-aware model using the expanded dataset that explicitly accounts for sequential feature arrivals. We then select the model corresponding to the best-performing feature arrival order on a validation set. We evaluate these approaches under a quadratic data-generating process and use a nonlinear model (boosted trees) for prediction. Feature imputation is handled via MICE. After selecting the best feature arrival order, we compare the performance of both approaches under this order on held-out test data.

The results, presented in Figure 3, show that selecting the best feature arrival order leads to significant improvements in generalization performance. The permutation-aware predictor trained using the best-selected order outperforms the standard ITP predictor, particularly in early feature arrivals. This suggests that strategically choosing a feature arrival order can mitigate the adverse effects of missing features and improve predictive accuracy.

Experiment 4: *Can we improve by robustly choosing the feature arrival order when there is uncertainty?* This experiment addresses scenarios where the feature arrival order is subject to perturbations (e.g., due to dynamic data collection). We investigate whether selecting a robust feature arrival order—one that minimizes the worst-case prediction error across potential perturbations—can improve generalization in the face of uncertainty compared to non-robust approaches. We compared the standard ITP predictor with the permutation-aware predictor trained under the selected robust feature arrival order. First, we randomly sampled a feasible set of feature orders $\mathcal{S}$ in (7), and for each element $\sigma_0 \in \mathcal{S}$, we constructed an uncertainty ball $B_L(\sigma_0)$ defined previously. We evaluated the worst-case performance across all orders within each ball $B_L(\sigma_0)$ and selected the feature arrival order $\bar{\sigma}_0$ that exhibited the best worst-case performance, ensuring robustness. Subsequently, we constructed an expanded dataset based on $\bar{\sigma}_0$ and trained the permutation-aware model on it. Following a similar evaluation setup as in Experiment 3 (quadratic data-generating process, nonlinear model, etc.), we computed the R^2 scores for both permutation-aware model and the standard ITP predictor across all perturbed orders within the uncertainty set $B_L(\bar{\sigma}_0)$.

The results in Figure 4 indicate that selecting a robust feature arrival order leads to better generalization and more stable performance in the face of uncertainty, both in terms of average performance across the

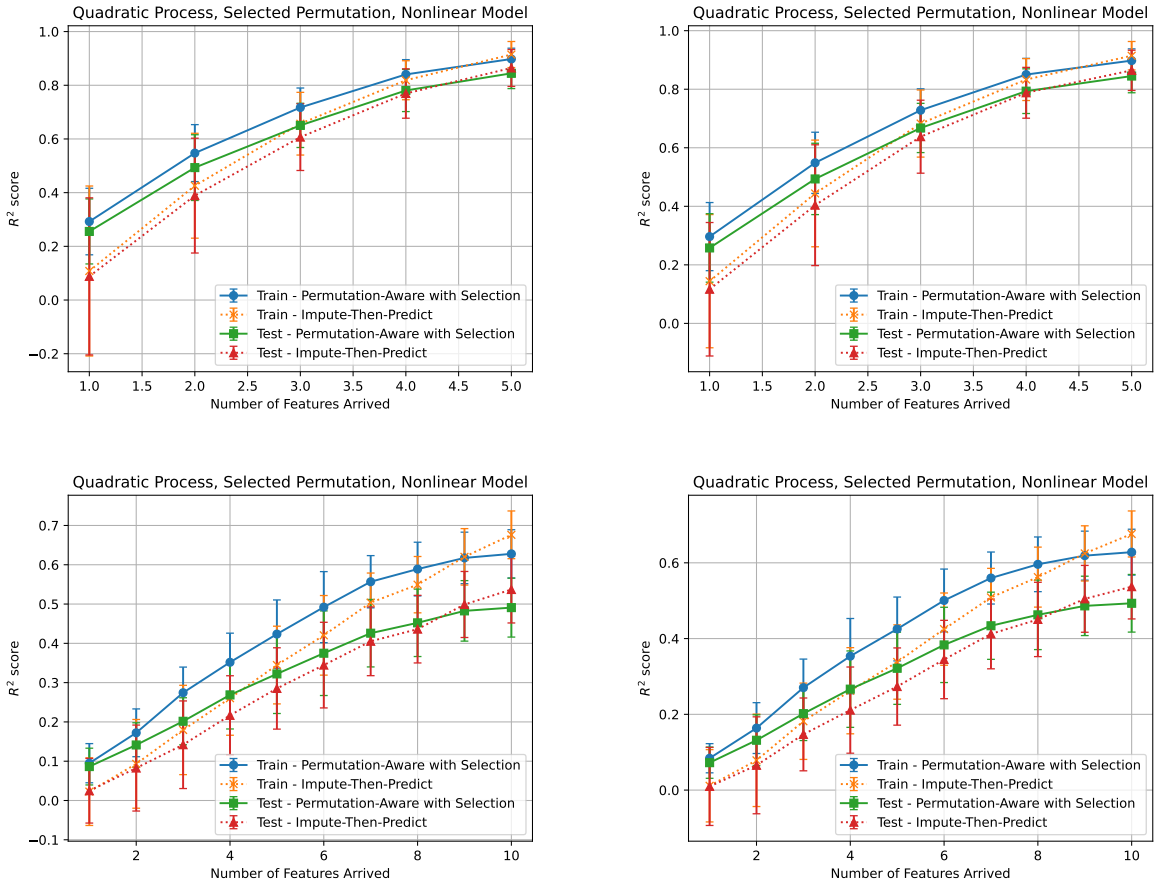


Figure 3 Comparison of standard ITP and best permutation-aware ITP, where the feature arrival order is selected from a feasible set of 10 (left) or 30 (right) candidate permutations.

uncertainty set and in the worst-case scenario. This demonstrates that the robust selection strategy effectively mitigates the adverse effects of feature arrival perturbations, ensuring more reliable predictions even under challenging conditions.

Experiment 5: *Can we improve by robustly choosing the feature arrival order when we have no control but considering only a random subset of all feasible permutations?*

[**Vassilis:** Additional experiments:

(1) Can we improve by robustly choosing the feature arrival order when we have no control?

(2) between 1 and 2^p models: missing indicator-based extended model (bias correction for each feature)

(3) weighted permutation-aware loss: (what if we care more about specific stages of the arrival process?)

]

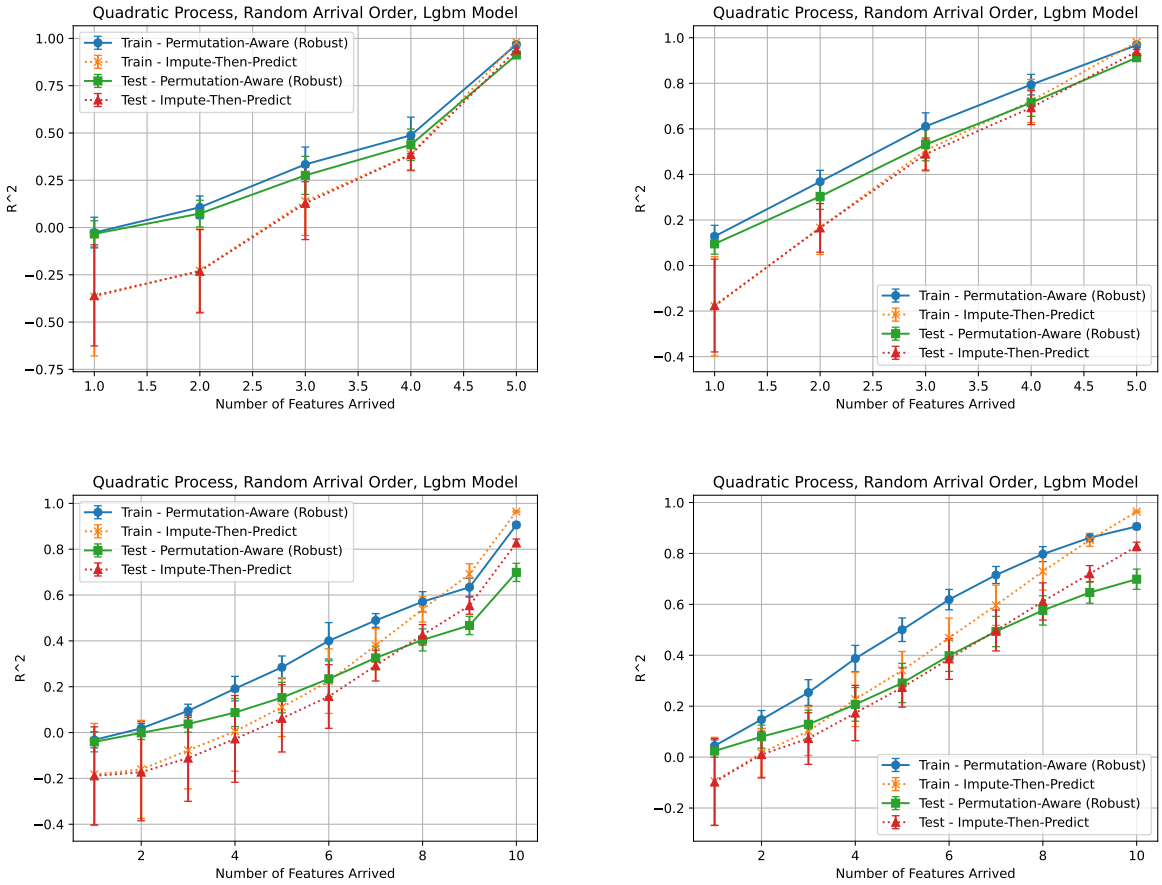


Figure 4 Comparison of standard ITP and robust permutation-aware ITP, where the left/right shows the average/worst-case performance within the uncertainty set $B_L(\bar{\sigma}_0)$.

5. An Algorithm for Robust Learning over Streaming Feature Spaces

5.1. Analysis of the Optimization Landscape

Yubo: I find the x-axis labels "1, 2, 3, 4, 5, 6" in Figure 5 to be somewhat abstract and potentially misleading. It would be clearer if the x-axis represented a parameter space, perhaps denoted as $\boldsymbol{\Theta} = \boldsymbol{\theta_1} \times \boldsymbol{\theta_2} \times \cdots \times \boldsymbol{\theta_n}$, which would align better with the concept of optimizing over model parameters. Additionally, the caption should explicitly state that the plotted curves represent Mean Squared Error (MSE) loss, or the specific loss function used for the illustration, to provide full clarity to the reader. For future iterations or supplementary materials, it would be highly beneficial to illustrate this concept with a simpler, concrete example. Consider a linear regression model, such as $y = \beta_1 x_1 + \beta_2 x_2$. We could then demonstrate the loss landscapes for different feature arrival orders, e.g., $\{x_1, x_2\}$ and $\{x_2, x_1\}$. Visualizing these individual 3D loss landscapes and then

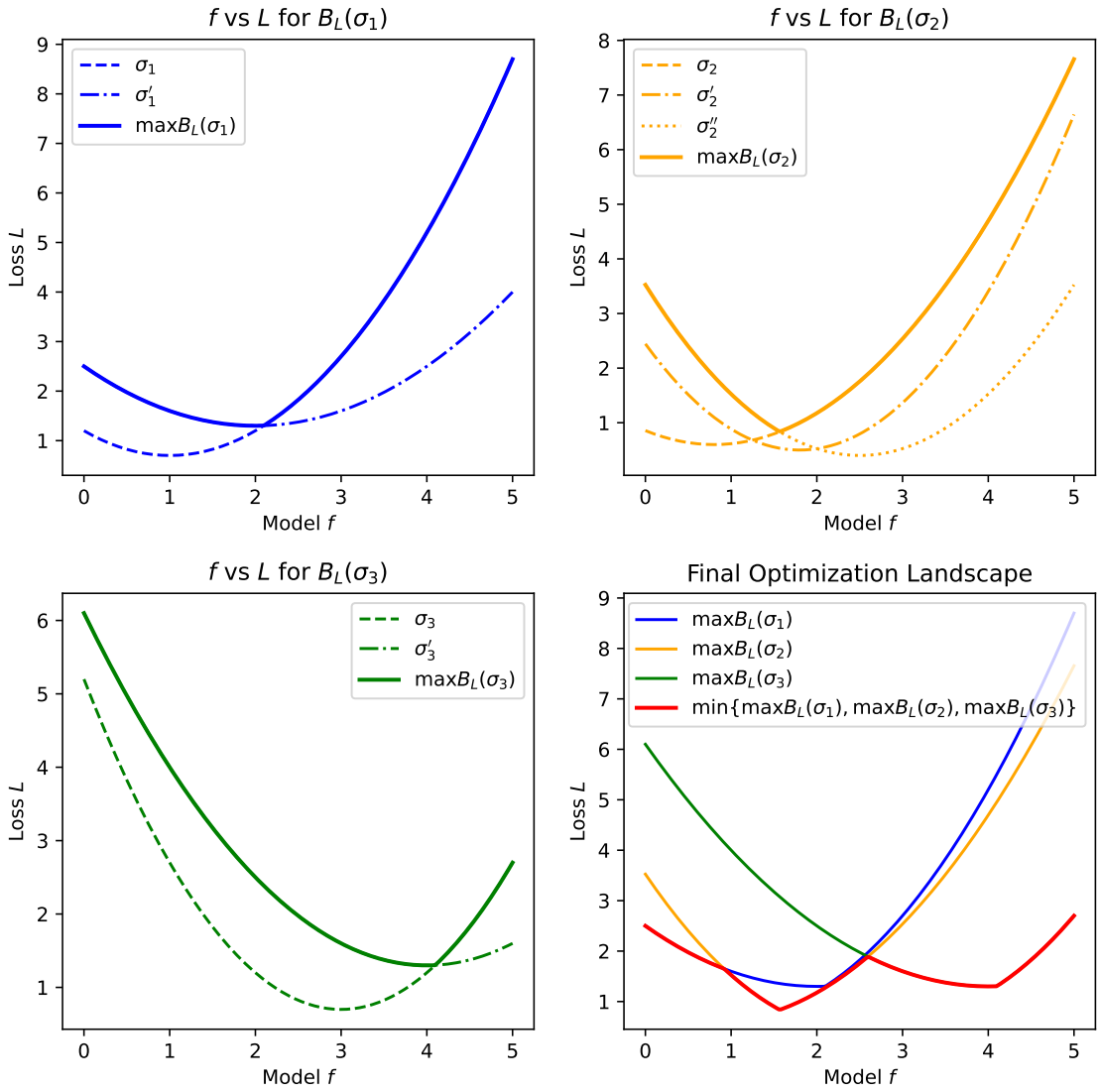


Figure 5 Illustration of the optimization landscape in robust learning over streaming feature spaces. In Graph 1, we fix a base permutation σ_1 and plot the loss curves corresponding to two perturbations $\sigma_1, \sigma_1' \in \mathcal{B}_B(\sigma_1)$; the bold curve shows the pointwise maximum, i.e., the worst-case loss for σ_1. Graphs 2 and 3 show a similar construction for a second and a third base permutation σ_2 and σ_3, with three and two perturbations, respectively. Graph 4 compares the worst-case losses for σ_1, σ_2, and σ_3; the bold red curve is the pointwise minimum between these, which defines the function ultimately minimized during training.

showing how they combine to form the minimax landscape (Graph 4 in the figure) would significantly enhance understanding. This would naturally be a 3D visualization, which might require more complex

plotting tools. This simple example would ground the abstract concepts in a more intuitive setting, making the "layered minimax structure" more tangible.

The optimization landscape induced by robust learning over streaming feature spaces can be understood as a layered minimax structure. For each base permutation σ_0, we evaluate the worst-case loss across all nearby permutations in its perturbation set $\mathcal{B}_B(\sigma_0)$, resulting in a pointwise maximum curve over the model class. These robustified loss curves capture the effect of local permutation uncertainty on performance. The learning objective is then to minimize the worst-case loss across all such base permutations, i.e., to find a model f that performs best under the most favorable base permutation when accounting for its local worst-case behavior. This leads to a loss surface defined by the pointwise minimum over a collection of maximum-loss curves, yielding a piecewise-smooth, nonconvex structure amenable to gradient-based optimization. Figure 5 visualizes the optimization landscape under a toy setting.

5.2. Gradient-based Optimization Approach

To solve the general version of the problem, $\min_{f\in\mathcal{F}} \min_{\sigma_0\in\mathcal{S}} \max_{\sigma\in\mathcal{B}_B(\sigma_0)} \mathcal{L}(f,\sigma)$, we propose a gradient descent-based method that exploits the piecewise structure of the loss landscape. We define an iterative procedure $f^{(k)} \mapsto f^{(k+1)}$ by taking gradient descent steps against the worst-case permutation within a local perturbation set. The direction of steepest descent is taken with respect to the most adversarial permutation in the most favorable local neighborhood.

We give the full procedure in Algorithm 1. At each iteration k, given the current model $f^{(k)}$, we first enumerate all feasible base permutations $\sigma_0 \in \mathcal{S}$. For each σ_0, we evaluate the inner worst-case loss $\max_{\sigma\in\mathcal{B}_B(\sigma_0)} \mathcal{L}(f^{(k)},\sigma)$. We then select the base permutation $\sigma_0^{(k)}$ that minimizes this quantity and identify the adversarial permutation $\sigma^{(k)} \in \mathcal{B}_B(\sigma_0^{(k)})$ that realizes the maximum. Finally, we perform a gradient descent step on the loss induced by this worst-case permutation: $f^{(k+1)} = f^{(k)} - \eta \cdot \nabla_f \mathcal{L}(f^{(k)},\sigma^{(k)})$.

5.3. Convergence of the Gradient-based Algorithm

Here, we establish that the gradient–based min–min–max descent (Algorithm 1) converges to a stationary point of the robust permutation-aware objective $\mathcal{R}(f) = \min_{\sigma_0\in\mathcal{S}} \max_{\sigma\in\mathcal{B}_B(\sigma_0)} \mathcal{L}(f,\sigma)$. We require four standard assumptions, stated formally below.

Algorithm 1: Gradient-Based Minimax Descent for Robust Learning over Streaming Feature Spaces

Input: Dataset $\{(x_i, y_i)\}_{i=1}^n$, loss function ℓ, model class $\mathcal{F}$, learning rate $\eta > 0$, permutation set $\mathcal{S}$, perturbation radius B, initial model $f^{(0)}$

1: Initialize $k \leftarrow 0$

2: **repeat**

3: **Worst-case base permutation selection:** $\sigma_0^{(k)} \leftarrow \arg\min_{\sigma_0 \in \mathcal{S}} \max_{\sigma \in \mathcal{B}_B(\sigma_0)} \mathcal{L}(f^{(k)}, \sigma)$

4: **Adversarial permutation selection:** $\sigma^{(k)} \leftarrow \arg\max_{\sigma \in \mathcal{B}_B(\sigma_0^{(k)})} \mathcal{L}(f^{(k)}, \sigma)$

5: **Gradient update:** $f^{(k+1)} \leftarrow f^{(k)} - \eta \cdot \nabla_f \mathcal{L}(f^{(k)}, \sigma^{(k)})$

6: $k \leftarrow k + 1$

7: **until** convergence (e.g., $\|\nabla_f \mathcal{L}(f^{(k)}, \sigma^{(k)})\| < \varepsilon$ or improvement below threshold)

Output: Final model $f^{(k)}$

ASSUMPTION 1 **(Convex, smooth loss)**. *For every (x_i, y_i) and permutation σ, the map $f \mapsto \mathcal{L}(f, \sigma)$ is convex and continuously differentiable with L-Lipschitz gradient.*

ASSUMPTION 2 **(Bounded gradients)**. *There exists $G > 0$ such that $\|\nabla_f \mathcal{L}(f, \sigma)\| \leq G$ for all f and all σ.*

ASSUMPTION 3 **(Finite uncertainty sets)**. *The set of base permutations $\mathcal{S}$ is finite and, for every $\sigma_0 \in \mathcal{S}$, its perturbation ball $\mathcal{B}_B(\sigma_0)$ is also finite.*

ASSUMPTION 4 **(Step size)**. *The learning rate satisfies $0 < \eta < 2/L$.*

Then, we establish the following result for the general case of Algorithm 1 (proof in Appendix B):

THEOREM 1 **(Stationary-point convergence)**. *Under Assumptions 1–4, the iterates $\{f^{(k)}\}$ generated by Algorithm 1 satisfy $\lim_{k\to\infty} \operatorname{dist}\big(0, \partial^C \mathcal{R}(f^{(k)})\big) = 0$, where $\partial^C \mathcal{R}$ denotes the Clarke subdifferential of $\mathcal{R}$. Consequently every accumulation point $\bar{f}$ of the sequence is Clarke-stationary: $0 \in \partial^C \mathcal{R}(\bar{f})$.*

Although $\mathcal{R}$ is generally non-smooth and non-convex (pointwise $\min$ of convex functions), Algorithm 1 guarantees that its gradient steps drive the Clarke-subgradient norm to zero. Hence the procedure terminates

only at parameter settings where no descent direction exists and converges to a stationary point of the non-smooth robust objective $\mathcal{R}$.

[**Vassilis:** Discuss types of stationary points here]

When the decision–maker has no control over feature–arrival order (Problem (6)), the robust problem reduces to the convex minimization $\min_{f\in\mathcal{F}} \mathcal{J}(f) := \max_{\sigma\in\mathcal{S}} \mathcal{L}(f,\sigma)$. Because $\mathcal{J}$ is the pointwise maximum of finitely many convex L-smooth functions, $\mathcal{J}$ itself is convex, L-smooth and G-Lipschitz.

COROLLARY 2 **(Global convergence in the min–max case).** *Let Assumptions 1–4 hold, and consider Algorithm 1 without the worst-case base permutation selection step. Then, for all $K \geq 1$,*

$$\mathcal{J}\big(\hat{f}_K\big) - \mathcal{J}(f^\star) \leq \frac{\|f^{(0)} - f^\star\|^2}{2\eta K} + \frac{\eta G^2}{2}, \qquad \hat{f}_K := \frac{1}{K}\sum_{k=0}^{K-1} f^{(k)},$$

where $f^\star$ denotes a (global) minimizer of $\mathcal{J}$. Choosing $\eta = D/(G\sqrt{K})$ with $D := \|f^{(0)} - f^\star\|$ yields the classical $O(1/\sqrt{K})$ convergence rate. If, in addition, $\mathcal{J}$ is μ-strongly convex, Algorithm 1 with constant step size $\eta \in (0, 2/L)$ enjoys linear *convergence of the iterates: $\|f^{(k)} - f^\star\| \leq (1-\eta\mu)^{k/2} D$.*

5.4. Fast, Provable Approximation of the Gradient

In Section 5.3, we established convergence guarantees for our gradient-based optimization procedure. However, a key computational bottleneck lies in evaluating the gradient $\nabla_f \mathcal{L}(f^{(k)}, \sigma^{(k)})$. As the number of features p grows, the space of feasible permutations $\mathcal{S} \subseteq S_p$ can grow factorially large, making the exact minimization computationally prohibitive.

To address this challenge, we employ a stochastic smoothing technique that approximates the non-differentiable min-objective with a smooth surrogate via the log-sum-exp transform. For simplicity, we would first illustrate this in the min-min setting (i.e. when $\mathcal{B}_B(\sigma_0) = \emptyset$), although the technique naturally extends to the general framework as shown in Appendix XXX. We consider the following relaxation:

$$\mathcal{J}(f) = \min_{\sigma} L(f,\sigma) \approx \mathcal{J}_\tau(f) := -\tau \log\left(\sum_{\sigma\in\mathcal{S}} \exp(-L(f,\sigma)/\tau)\right)$$

where $\tau > 0$ is a temperature parameter. As $\tau \to 0$, the approximation becomes exact:

$$\lim_{\tau\to 0} \mathcal{J}_\tau(f) = \mathcal{J}(f).$$

The function $\mathcal{J}_\tau(f)$ is differentiable, and its gradient can be written as:

$$\nabla_f \mathcal{J}_\tau(f) = \sum_{\sigma \in \mathcal{S}} P_\tau(\sigma \mid f) \cdot \nabla_f \mathcal{L}(f, \sigma), \quad \text{where} \quad P_\tau(\sigma \mid f) := \frac{\exp\left(-\mathcal{L}(f,\sigma)/\tau\right)}{\sum_{\sigma' \in \mathcal{S}} \exp\left(-\mathcal{L}(f,\sigma')/\tau\right)}.$$

This expression reveals that the gradient is a *weighted average of permutation-specific gradients*, where the weights form a Boltzmann (softmax) distribution over permutations. Therefore, if we can efficiently sample from $P_\tau(\sigma \mid f)$, we obtain an unbiased stochastic estimator of $\nabla_f \mathcal{J}_\tau(f)$. In Algorithm 2, we outline such an sampling process utilizing the Gumbel-Max trick (Luce et al. 1959), and Theorem 2 confirms both its validity and efficiency:

THEOREM 2 **(Sequential Gumbel-Max Sampling for Permutations)**. *Let $\mathcal{S} = S_p$ be the full symmetric group over p features, and note that the permutation-aware loss $\mathcal{L}(f, \sigma)$ can be decomposed as*

$$\mathcal{L}(f, \sigma) = \sum_{j=1}^{p} \sum_{(\boldsymbol{x}_i, y_i) \in D} \ell(y_i, f(\boldsymbol{x}_i^{\sigma, j})),$$

Then, Algorithm 2 samples permutations exactly from the Boltzmann distribution $P_\tau(\sigma \mid f)$ using $O(np^2)$ operations.

Proof. **Computational Complexity.** At each position $j = 1, \ldots, p$ the algorithm evaluates an incremental loss for the $p - j + 1$ unrevealed features. Computing one incremental loss requires summing ℓ over all n data points, hence the total work is $\sum_{j=1}^{p} n\,(p - j + 1) = O(np^2)$.

Sampling Distribution. We proceed by induction on the prefix length ℓ.

Base case $\ell = 0$. Before the first iteration the prefix is $P = \emptyset$. With empty prefix $P = \emptyset$ the algorithm computes for every $i \in [p]$ $s_i = c_i/\tau + G_i$ where $c_i = \sum_{(\boldsymbol{x}, y) \in D} \ell\big(y, f(\boldsymbol{x}^{\{i\},1})\big)$ and $G_i \sim \mathrm{Gumbel}(0, 1)$. By the Gumbel–Max identity of Luce et al. (1959),

$$\Pr\big(\sigma(1) = i\big) = \frac{\exp(-c_i/\tau)}{\sum_{j=1}^{p} \exp(-c_j/\tau)} = P_\tau\big(\sigma(1) = i \mid f\big),$$

so the first element is selected with the correct Boltzmann probability.

Inductive step. Assume that for some j $(1 \le j < p)$ every prefix of length $j - 1$ produced by the algorithm is distributed according to $P_\tau(\sigma_{1:j-1})$. Let $P = \sigma_{1:j-1}$ be a particular such prefix. For each unrevealed $i \notin P$ the score $s_i = c_i/\tau + G_i$ uses

$$c_i = \sum_{(\boldsymbol{x}, y) \in D} \ell\big(y, f(\boldsymbol{x}^{P \cup \{i\}, j})\big).$$

Applying the Gumbel–Max identity conditional on P,

$$\Pr\big(\sigma(j)=i \mid P\big)=\frac{\exp(-c_i/\tau)}{\sum_{i'\notin P}\exp(-c_{i'}/\tau)}=P_\tau\big(\sigma(j)=i \mid \sigma_{1:j-1}=P\big).$$

Therefore

$$\Pr\big(\sigma_{1:j}=P+[i]\big)=P_\tau\big(\sigma_{1:j-1}=P\big)\,P_\tau\big(\sigma(j)=i \mid P\big)=P_\tau\big(\sigma_{1:j}=P+[i]\big),$$

and the claim extends to length j. By induction the probability of generating any full permutation equals $P_\tau(\sigma \mid f)$, hence the sampler is exact. □

Algorithm 2: Sequential Gumbel-Max Sampling of Permutations

Input: Model f, data D, temperature τ

Initialize prefix $P \leftarrow \emptyset$, permutation $\sigma \leftarrow []$
for $j = 1$ to p **do**
  for each $i \in [p] \setminus P$ **do**
    Sample $G_i \sim \mathrm{Gumbel}(0,1)$
    Compute $c_i \leftarrow \sum_{(\boldsymbol{x}_i,y_i)\in D} \ell(y_i, f(\boldsymbol{x}_i^{P\cup\{i\},j}))$
    Set $s_i \leftarrow c_i/\tau + G_i$
  end for
  Let $i^* \leftarrow \arg\min_i s_i$
  Append i^* to σ; update $P \leftarrow P \cup \{i^*\}$
end for
return σ

The following proposition shows that the estimated gradient error is small:

PROPOSITION 3 **(Gradient Estimation Error)**. *Assume that Assumption 2 holds. Then for each f there exists $v \in \partial_f \mathcal{J}(f)$ such that:*

$$\|\nabla_f \mathcal{J}_\tau(f) - v\| \le G\tau \log|S|$$

Proof. Pick $\sigma^\star \in \arg\min_\sigma \mathcal{L}(f,\sigma)$ and set $v := \nabla_f \mathcal{L}(f,\sigma^\star)$. Define the auxiliary probabilities

$$p_\sigma := \frac{e^{-\mathcal{L}(f,\sigma)/\tau}}{\sum_{\sigma'} e^{-\mathcal{L}(f,\sigma')/\tau}}, \qquad \sum_\sigma p_\sigma = 1.$$

Note that then we have $\nabla_f \mathcal{J}_\tau(f = \sum_\sigma p_\sigma \nabla_f \mathcal{L}(f,\sigma)$. Therefore, using the triangle inequality and $\sum_\sigma p_\sigma = 1$, we have

$$\left\|\nabla_f \mathcal{J}_\tau(f) - v\right\|_2 \leq \sum_\sigma p_\sigma \left\|\nabla_f \mathcal{L}(f,\sigma) - \nabla_f \mathcal{L}(f,\sigma^\star)\right\|_2 \overset{\text{Ass. 2}}{\leq} G \sum_\sigma p_\sigma \left|\mathcal{L}(f,\sigma) - \mathcal{L}(f,\sigma^\star)\right|. \tag{8}$$

Now we can rewrite the defining equation of p_σ as: $\mathcal{L}(f,\sigma) = -\tau(\log p_\sigma + \log Z)$ where $Z = \sum_{\sigma'} e^{-\mathcal{L}(f,\sigma')/\tau}$. Therefore, we have: $\mathcal{L}(f,\sigma) - \mathcal{L}(f,\sigma^\star) = -\tau(\log p_\sigma - \log p_{\sigma^\star})$. Insert this into Eq. 8, we have:

$$\left\|\nabla_f \mathcal{J}_\tau(f) - v\right\|_2 \leq G\tau \sum_\sigma p_\sigma \left|\log p_\sigma - \log p_{\sigma^\star}\right| \leq G\tau \sum_\sigma p_\sigma \left(-\log p_\sigma\right) \leq G\tau \log |S|.$$

where the final bound comes from the upper bound on Shannon entropy. □

Proposition 3 shows that the estimation error for the approximated gradient depends on the *logarithm* of $|S|$. Given that in the general case we have $|S| \sim 2^p$, the approximation error is therefore on the order of $O(\beta\tau p)$. Thus, for any required estimation accuracy ϵ, we only need to take $\tau = O\left(\frac{\epsilon}{\beta p}\right)$.

6. Real-World Case Studies

[**Vassilis:** Yubo: refine experiments using UCI datasets. We already have 5, so maybe no need to use MIMIC. We will need to present them in detail and as case-studies.]

7. Conclusion

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Appendix A: Technical Proofs for Section 3

Proof of Lemma 1 We want to minimize the expected squared loss at each step $j = 1, \ldots, p$:

$$f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) := \arg\min \mathbb{E}\left[(y - f(\boldsymbol{x}^{\sigma,j}))^2\right].$$

We expand the squared loss:

$$\begin{aligned}
\mathbb{E}\left[(y - f(\boldsymbol{x}^{\sigma,j}))^2\right] &= \mathbb{E}\left[\left(y - \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] + \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] - f(\boldsymbol{x}^{\sigma,j})\right)^2\right] \\
&= \mathbb{E}\left[\left(y - \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]\right)^2\right] + 2\mathbb{E}\left[\left(y - \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]\right)\left(\mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] - f(\boldsymbol{x}^{\sigma,j})\right)\right] \\
&\quad + \mathbb{E}\left[\left(\mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] - f(\boldsymbol{x}^{\sigma,j})\right)^2\right].
\end{aligned}$$

Since $\mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]$ captures all the information in $\boldsymbol{x}^{\sigma,j}$ that is relevant to predicting y, the residual $(y - \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}])$ is uncorrelated with any function of $\boldsymbol{x}^{\sigma,j}$, so the cross-term vanishes: $2\mathbb{E}\left[(y - \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}])\left(\mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] - f(\boldsymbol{x}^{\sigma,j})\right)\right] = 0$. Thus, we are left with:

$$\mathbb{E}\left[(y - f(\boldsymbol{x}^{\sigma,j}))^2\right] = \mathbb{E}\left[\left(y - \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]\right)^2\right] + \mathbb{E}\left[\left(\mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] - f(\boldsymbol{x}^{\sigma,j})\right)^2\right].$$

To minimize this expression, note that the first term $\mathbb{E}\left[\left(y - \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]\right)^2\right]$ is independent of $f(\boldsymbol{x}^{\sigma,j})$, and the second term $\mathbb{E}\left[\left(\mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] - f(\boldsymbol{x}^{\sigma,j})\right)^2\right]$ is minimized when $f(\boldsymbol{x}^{\sigma,j}) = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]$. Therefore, the optimal predictor at any step j in the feature reveal process is:

$$f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}].$$

□

Proof of Lemma 2 We are given a set of features $\boldsymbol{x}^{\sigma,j}$ that have been revealed according to the permutation σ, and a set of missing features $\boldsymbol{x}^{\sigma,-j}$. The goal is to find the optimal imputation $\hat{\boldsymbol{x}}^{\sigma,-j}$ for the missing features by minimizing the expected imputation error:

$$\hat{\boldsymbol{x}}^{\sigma,-j} = \arg\min_{\hat{\boldsymbol{x}}} \mathbb{E}\left[\left(\boldsymbol{x}^{\sigma,-j} - \hat{\boldsymbol{x}}\right)^2 \mid \boldsymbol{x}^{\sigma,j}\right].$$

Expanding the squared error gives:

$$\begin{aligned}
\mathbb{E}\left[\left(\boldsymbol{x}^{\sigma,-j} - \hat{\boldsymbol{x}}\right)^2 \mid \boldsymbol{x}^{\sigma,j}\right] &= \mathbb{E}\left[\left(\boldsymbol{x}^{\sigma,-j} - \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}] + \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}] - \hat{\boldsymbol{x}}\right)^2 \mid \boldsymbol{x}^{\sigma,j}\right] \\
&= \mathbb{E}\left[\left(\boldsymbol{x}^{\sigma,-j} - \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}]\right)^2 + 2\left(\boldsymbol{x}^{\sigma,-j} - \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}]\right)\left(\mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}] - \hat{\boldsymbol{x}}\right)\right. \\
&\quad \left. + \left(\mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}] - \hat{\boldsymbol{x}}\right)^2 \mid \boldsymbol{x}^{\sigma,j}\right] \\
&= \mathbb{E}\left[\left(\boldsymbol{x}^{\sigma,-j} - \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}]\right)^2 + \left(\mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}] - \hat{\boldsymbol{x}}\right)^2 \mid \boldsymbol{x}^{\sigma,j}\right],
\end{aligned}$$

where, for the second equality, we again rely on the fact that the residual $\boldsymbol{x}^{\sigma,-j} - \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}]$ is uncorrelated with $\mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}] - \hat{\boldsymbol{x}}$, so the cross-term vanishes. The first term in the final expression does not rely on the imputation $\hat{\boldsymbol{x}}$. To minimize the second term, we choose $\hat{\boldsymbol{x}} = \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}]$. Therefore, the optimal imputation is $\hat{\boldsymbol{x}}^{\sigma,-j} = \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}]$. $\square$

Proof of Proposition 1 We want to show that $\mathbb{E}\left[(y - f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}))^2\right] \leq \mathbb{E}\left[(y - f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}))^2\right]$, where:

- $f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]$ is the optimal predictor after j features have been revealed.
- $\hat{\boldsymbol{x}}^{\sigma,-j} = \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}]$ is the optimal imputation for the missing features.
- $f^\star$ is the optimal predictor trained on the full feature set, i.e., $f^\star(\boldsymbol{x}) = \mathbb{E}[y \mid \boldsymbol{x}]$.

We denote the optimal predictor's residual by $\varepsilon_y = y - \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}]$ and the difference between the optimal and the ITP predictors by $\delta = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] - f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j})$. Then, the expected squared error of the optimal predictor is:

$$\mathbb{E}\left[(y - f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}))^2\right] = \mathbb{E}\left[\varepsilon_y^2\right].$$

Similarly, the expected squared error of the ITP predictor is:

$$\mathbb{E}\left[(y - f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}))^2\right] = \mathbb{E}\left[(\varepsilon_y + \delta)^2\right] = \mathbb{E}\left[\varepsilon_y^2\right] + 2\mathbb{E}\left[\varepsilon_y \delta\right] + \mathbb{E}\left[\delta^2\right]$$

Therefore, we need to show that $\Delta = \mathbb{E}\left[(y - f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j}))^2\right] - \mathbb{E}\left[(y - f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}))^2\right] = 2\mathbb{E}\left[\varepsilon_y \delta\right] + \mathbb{E}\left[\delta^2\right] \geq 0$. Since δ is a function only of $\boldsymbol{x}^{\sigma,j}$ and the residual ε_y is uncorrelated with any function of $\boldsymbol{x}^{\sigma,j}$, the cross-term $\mathbb{E}\left[\varepsilon_y \delta\right] = 0$. Therefore, $\Delta = \mathbb{E}\left[\delta^2\right] \geq 0$. Equality holds if and only if $\delta = 0$ almost surely, i.e., when $\mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}] = f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{\sigma,-j})$. □

Proof of Corollary 1 We aim to prove that for a linear-Gaussian setting, the optimal predictor $f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j})$ based on observed features $[j]$ is equivalent to the ITP predictor, which first optimally imputes the missing features $\boldsymbol{x}^{\sigma,-j}$ and then applies the predictor trained on the full feature set. We assume a linear data generating process $y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon$ with Gaussian features $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu} \in \mathbb{R}^p$ is the mean vector, $\boldsymbol{\Sigma} \in \mathbb{R}^{p\times p}$ is the covariance matrix, $\boldsymbol{\beta} \in \mathbb{R}^p$, and $\epsilon \sim \mathcal{N}(0, \sigma^2_\epsilon)$ is independent of $\boldsymbol{x}$. We prove the result in three steps.

Step 1: We first study the conditional distribution of $\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}$. Since $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the conditional distribution of $\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}$ is Gaussian. Let:

$$\boldsymbol{x}^{\sigma,j} \sim \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_{jj}), \quad \boldsymbol{x}^{\sigma,-j} \sim \mathcal{N}(\boldsymbol{\mu}_{-j}, \boldsymbol{\Sigma}_{-j,-j}),$$

and $\boldsymbol{\Sigma}_{-j,j}$ denote the cross-covariance matrix. Then:

$$\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j} \sim \mathcal{N}(\mathbf{A}\boldsymbol{x}^{\sigma,j} + \mathbf{b}, \boldsymbol{\Sigma}_{-j,-j|j}),$$

where:

$$\mathbf{A} = \boldsymbol{\Sigma}_{-j,j}\boldsymbol{\Sigma}^{-1}_{jj}, \quad \mathbf{b} = \boldsymbol{\mu}_{-j} - \mathbf{A}\boldsymbol{\mu}_j, \quad \boldsymbol{\Sigma}_{-j,-j|j} = \boldsymbol{\Sigma}_{-j,-j} - \boldsymbol{\Sigma}_{-j,j}\boldsymbol{\Sigma}^{-1}_{jj}\boldsymbol{\Sigma}_{j,-j}.$$

The conditional mean is:

$$\hat{\boldsymbol{x}}^{-j} = \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}] = \mathbf{A}\boldsymbol{x}^{\sigma,j} + \mathbf{b}.$$

Step 2: The optimal predictor based on observed features j is:

$$f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}].$$

Substituting $y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon$:

$$f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \mathbb{E}[\boldsymbol{x}^{\sigma,j\top}\boldsymbol{\beta}^j + \boldsymbol{x}^{\sigma,-j\top}\boldsymbol{\beta}^{-j} + \epsilon \mid \boldsymbol{x}^{\sigma,j}].$$

Since ϵ is independent of $\boldsymbol{x}$ with $\mathbb{E}[\epsilon] = 0$:

$$f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \boldsymbol{x}^{\sigma,j\top}\boldsymbol{\beta}^j + \mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}]^\top \boldsymbol{\beta}^{-j}.$$

Substituting $\mathbb{E}[\boldsymbol{x}^{\sigma,-j} \mid \boldsymbol{x}^{\sigma,j}] = \mathbf{A}\boldsymbol{x}^{\sigma,j} + \mathbf{b}$:

$$f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \boldsymbol{x}^{\sigma,j\top}\boldsymbol{\beta}^j + (\mathbf{A}\boldsymbol{x}^{\sigma,j} + \mathbf{b})^\top \boldsymbol{\beta}^{-j}.$$

Expanding:

$$f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \boldsymbol{x}^{\sigma,j\top}\boldsymbol{\beta}^j + \boldsymbol{x}^{\sigma,j\top}\mathbf{A}^\top\boldsymbol{\beta}^{-j} + \mathbf{b}^\top\boldsymbol{\beta}^{-j}.$$

Step 3: The ITP approach computes:

$$f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{-j}) = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{-j}].$$

Substituting $y = \boldsymbol{x}^\top\boldsymbol{\beta} + \epsilon$ and using $\hat{\boldsymbol{x}}^{-j} = \mathbf{A}\boldsymbol{x}^{\sigma,j} + \mathbf{b}$:

$$f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{-j}) = \boldsymbol{x}^{\sigma,j\top}\boldsymbol{\beta}^j + \hat{\boldsymbol{x}}^{-j\top}\boldsymbol{\beta}^{-j}.$$

Expanding $\hat{\boldsymbol{x}}^{-j}$:

$$\hat{\boldsymbol{x}}^{-j\top}\boldsymbol{\beta}^{-j} = (\mathbf{A}\boldsymbol{x}^{\sigma,j} + \mathbf{b})^\top\boldsymbol{\beta}^{-j} = \boldsymbol{x}^{\sigma,j\top}\mathbf{A}^\top\boldsymbol{\beta}^{-j} + \mathbf{b}^\top\boldsymbol{\beta}^{-j}.$$

Thus:

$$f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{-j}) = \boldsymbol{x}^{\sigma,j\top}\boldsymbol{\beta}^j + \boldsymbol{x}^{\sigma,j\top}\mathbf{A}^\top\boldsymbol{\beta}^{-j} + \mathbf{b}^\top\boldsymbol{\beta}^{-j}.$$

This is identical to $f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j})$:

$$f^\star(\boldsymbol{x}^{\sigma,j}, \hat{\boldsymbol{x}}^{-j}) = f^*_{\sigma,j}(\boldsymbol{x}^{\sigma,j}).$$

□

Proof of Proposition 2 We split the argument into two directions: necessity and sufficiency.

Definitions and setup. Throughout, let

$$f^{\star}_{\sigma,j}(\mathbf{X}^{\sigma,j}) = \mathbb{E}\big[Y \mid \mathbf{X}^{\sigma,j}\big] \quad \text{and} \quad f^{\star}\big(\mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j}\big) = \mathbb{E}\Big[Y \,\Big|\, \mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j}\Big],$$

where

$$\hat{\mathbf{X}}^{\sigma,-j} = \mathbb{E}\big[\mathbf{X}^{\sigma,-j} \,\big|\, \mathbf{X}^{\sigma,j}\big].$$

We compare their mean squared errors (MSEs).

(A) Necessity. Assume

$$\mathbb{E}\Big[\big(Y - f^{\star}_{\sigma,j}(\mathbf{X}^{\sigma,j})\big)^2\Big] < \mathbb{E}\Big[\big(Y - f^{\star}(\mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j})\big)^2\Big].$$

We claim that *(i)* and *(ii)* must both hold.

(i) Non-degeneracy: Suppose, for contradiction, that

$$\mathrm{Var}\big(\mathbf{X}^{\sigma,-j} \,\big|\, \mathbf{X}^{\sigma,j}\big) = 0 \quad \text{a.s.}$$

Then $\mathbf{X}^{\sigma,-j}$ is essentially a constant given $\mathbf{X}^{\sigma,j}$. Hence knowing $\mathbf{X}^{\sigma,j}$ is equivalent (almost surely) to knowing *all* of $\mathbf{X}$. In that degenerate setting, we have

$$f^{\star}_{\sigma,j}(\mathbf{X}^{\sigma,j}) = \mathbb{E}[Y \mid \mathbf{X}], \quad f^{\star}(\mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j}) = \mathbb{E}[Y \mid \mathbf{X}],$$

so both predictors coincide and their MSEs are equal, contradicting the strict inequality. Therefore, $\mathrm{Var}(\mathbf{X}^{\sigma,-j} \mid \mathbf{X}^{\sigma,j}) > 0$.

(ii) Nonlinearity: Next, suppose for contradiction that

$$\mathbb{E}\big[Y \mid \mathbf{X}^{\sigma,j}, \mathbf{X}^{\sigma,-j}\big] \text{ is } \textit{linear in } \mathbf{X}^{\sigma,-j}.$$

Concretely, for each fixed value $\mathbf{x}^{\sigma,j}$, there exist scalars $a(\mathbf{x}^{\sigma,j}), b(\mathbf{x}^{\sigma,j})$ such that

$$\mathbb{E}\big[Y \mid \mathbf{X}^{\sigma,j} = \mathbf{x}^{\sigma,j}, \mathbf{X}^{\sigma,-j} = \mathbf{x}^{\sigma,-j}\big] = a(\mathbf{x}^{\sigma,j}) + b(\mathbf{x}^{\sigma,j})\, \mathbf{x}^{\sigma,-j}.$$

Then, for the imputed value $\hat{\mathbf{x}}^{\sigma,-j} = \mathbb{E}[\mathbf{X}^{\sigma,-j} \mid \mathbf{X}^{\sigma,j} = \mathbf{x}^{\sigma,j}]$, we get

$$\mathbb{E}\big[Y \mid \mathbf{X}^{\sigma,j} = \mathbf{x}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j} = \hat{\mathbf{x}}^{\sigma,-j}\big] = a(\mathbf{x}^{\sigma,j}) + b(\mathbf{x}^{\sigma,j})\, \hat{\mathbf{x}}^{\sigma,-j}.$$

By the law of total expectation,

$$\mathbb{E}[Y \mid \mathbf{X}^{\sigma,j} = \mathbf{x}^{\sigma,j}] = \int \Big(a(\mathbf{x}^{\sigma,j}) + b(\mathbf{x}^{\sigma,j})\,\mathbf{x}^{\sigma,-j}\Big)\, dp\big(\mathbf{x}^{\sigma,-j} \mid \mathbf{x}^{\sigma,j}\big) = a(\mathbf{x}^{\sigma,j}) + b(\mathbf{x}^{\sigma,j})\,\hat{\mathbf{x}}^{\sigma,-j}.$$

Hence

$$f^\star_{\sigma,j}(\mathbf{x}^{\sigma,j}) = \mathbb{E}[Y \mid \mathbf{x}^{\sigma,j}] = a(\mathbf{x}^{\sigma,j}) + b(\mathbf{x}^{\sigma,j})\,\hat{\mathbf{x}}^{\sigma,-j} = \mathbb{E}\big[Y \mid \mathbf{x}^{\sigma,j}, \hat{\mathbf{x}}^{\sigma,-j}\big] = f^\star(\mathbf{x}^{\sigma,j}, \hat{\mathbf{x}}^{\sigma,-j}).$$

Thus, on every sample point, the two predictors *coincide*, so their MSEs are equal—contradicting the strict inequality. Therefore the conditional mean cannot be linear in $\mathbf{X}^{\sigma,-j}$, establishing (i).

We have shown that if the strict MSE gap holds, then (i) and (ii) *must* hold. Hence the necessity part is complete.

(B) Sufficiency. We now prove that conditions (i) and (ii) *imply* the strict inequality in MSE.

Step 1. *$f^\star_{\sigma,j}$ is the unique MSE-minimizer among all $\mathbf{X}^{\sigma,j}$-measurable functions.*

Because squared-error loss makes the conditional mean the regression minimizer, we know:

$$f^\star_{\sigma,j}(\mathbf{X}^{\sigma,j}) = \mathbb{E}[Y \mid \mathbf{X}^{\sigma,j}]$$

uniquely minimizes $\mathbb{E}[(Y - Z)^2]$ over all Z measurable w.r.t. $\mathbf{X}^{\sigma,j}$. Note that

$$f^\star\big(\mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j}\big)$$

is *also* a function of $\mathbf{X}^{\sigma,j}$ alone (because $\hat{\mathbf{X}}^{\sigma,-j} = \mathbb{E}[\mathbf{X}^{\sigma,-j} \mid \mathbf{X}^{\sigma,j}]$). Thus it is in the same comparison class of $\mathbf{X}^{\sigma,j}$-measurable functions. Therefore:

$$\mathbb{E}\Big[\big(Y - f^\star_{\sigma,j}(\mathbf{X}^{\sigma,j})\big)^2\Big] \le \mathbb{E}\Big[\big(Y - f^\star(\mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j})\big)^2\Big].$$

So at least the partial predictor is *no worse* in terms of MSE.

Step 2. *Under (i)–(ii), the two predictors differ with positive probability, so the MSE gap is strict.*

By (ii), for each fixed $\mathbf{x}^{\sigma,j}$, the variable $\mathbf{X}^{\sigma,-j}$ still "moves" with nonzero variance. By (i), $\mathbb{E}[Y \mid \mathbf{X}^{\sigma,j} = \mathbf{x}^{\sigma,j}, \mathbf{X}^{\sigma,-j} = \mathbf{x}^{\sigma,-j}]$ is *not* a linear (or affine) function of $\mathbf{x}^{\sigma,-j}$. Hence

$$\mathbb{E}[Y \mid \mathbf{x}^{\sigma,j}] = \int \mathbb{E}\big[Y \mid \mathbf{x}^{\sigma,j}, \mathbf{x}^{\sigma,-j}\big]\, dp(\mathbf{x}^{\sigma,-j} \mid \mathbf{x}^{\sigma,j}) \neq \mathbb{E}\Big[Y \mid \mathbf{x}^{\sigma,j}, \hat{\mathbf{x}}^{\sigma,-j}\Big]$$

on a set of $\mathbf{x}^{\sigma,j}$ of *positive probability*. Thus

$$f^\star_{\sigma,j}(\mathbf{X}^{\sigma,j}) = \mathbb{E}[Y \mid \mathbf{X}^{\sigma,j}] \neq \mathbb{E}[Y \mid \mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j}] = f^\star(\mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j})$$

with strictly positive probability, and due to the uniqueness of the regression minimizer, the MSE must be *strictly* lower for $f^\star_{\sigma,j}$ there.

Consequently,

$$\mathbb{E}\Big[\big(Y - f^\star_{\sigma,j}(\mathbf{X}^{\sigma,j})\big)^2\Big] < \mathbb{E}\Big[\big(Y - f^\star(\mathbf{X}^{\sigma,j}, \hat{\mathbf{X}}^{\sigma,-j})\big)^2\Big],$$

completing the proof of sufficiency. □

Appendix B: Technical Proofs for Section 4: Convergence of Algorithm 1

Throughout the appendix we invoke Assumptions 1–4. For convenience we re-introduce two key objects:

$$g_{\sigma_0}(f) := \max_{\sigma \in \mathcal{B}_B(\sigma_0)} \mathcal{L}(f, \sigma), \qquad \mathcal{R}(f) := \min_{\sigma_0 \in \mathcal{S}} g_{\sigma_0}(f).$$

Recall L is the smoothness constant in Assumption 1 and G is the gradient bound in Assumption 2.

We first prove the following auxiliary lemmas:

LEMMA 3. *Let $\{f_j : \mathbb{R}^d \to \mathbb{R}\}_{j=1}^m$ be L-Lipschitz functions. Then the pointwise maximum*

$$g(x) := \max_{1 \le j \le m} f_j(x)$$

is also L-Lipschitz.

Proof of Lemma 3 Fix any $x, y \in \mathbb{R}^d$. Let $j^* \in \arg\max_j f_j(x)$. Then:

$$g(x) - g(y) = f_{j^*}(x) - \max_j f_j(y) \le f_{j^*}(x) - f_{j^*}(y) \le L\|x - y\|.$$

Similarly, reversing the roles of x and y:

$$g(y) - g(x) \le L\|x - y\|.$$

Hence $|g(x) - g(y)| \le L\|x - y\|$.

LEMMA 4. *Let $\{f_j : \mathbb{R}^d \to \mathbb{R}\}_{j=1}^m$ be L-Lipschitz functions. Then the pointwise minimum*

$$g(x) := \min_{1 \le j \le m} f_j(x)$$

is also L-Lipschitz.

Proof of Lemma 4 Fix any $x, y \in \mathbb{R}^d$. Then:

$$|g(x) - g(y)| = \left| \min_j f_j(x) - \min_j f_j(y) \right| \le \max_j |f_j(x) - f_j(y)| \le L\|x - y\|.$$

The first inequality follows from the fact that for any real-valued functions a_j, b_j,

$$\left| \min_j a_j - \min_j b_j \right| \le \max_j |a_j - b_j|.$$

Proof of Theorem 1 **Step 1: convex and smooth surrogates.** For fixed σ_0 the map g_{σ_0} is the pointwise maximum of $|B_L(\sigma_0)| < \infty$ convex L-smooth functions $f \mapsto \mathcal{L}(f, \sigma)$; hence g_{σ_0} itself is convex and L-smooth.

Step 2: Danskin derivative. By the finite-max Danskin theorem

$$\nabla g_{\sigma_0}(f) \;=\; \nabla_f \mathcal{L}\big(f, \sigma^\star\big), \quad \sigma^\star \in \arg\max_{\sigma \in \mathcal{B}_B(\sigma_0)} \mathcal{L}(f, \sigma).$$

Step 3: subgradient used by the algorithm. At iteration k the algorithm first selects $\sigma_0^{(k)} \in \arg\min_{\sigma_0} g_{\sigma_0}(f^{(k)})$ and then $\sigma^{(k)} \in \arg\max_{\sigma \in \mathcal{B}_B(\sigma_0^{(k)})} \mathcal{L}(f^{(k)}, \sigma)$. Define $d^{(k)} := \nabla_f \mathcal{L}(f^{(k)}, \sigma^{(k)})$. By Step 2 $d^{(k)} = \nabla g_{\sigma_0^{(k)}}(f^{(k)})$.

Because $\mathcal{R}(f) = \min_{\sigma_0} g_{\sigma_0}(f)$, $\sigma_0^{(k)}$ is an *active index* ($g_{\sigma_0^{(k)}}(f^{(k)}) = \mathcal{R}(f^{(k)})$). For a finite minimum of smooth functions the Clarke subdifferential is the convex hull of active gradients (Clarke 1990, Thm. 2.5.1). Hence

$$d^{(k)} \;\in\; \partial^C \mathcal{R}\big(f^{(k)}\big). \tag{A.1}$$

Step 4: one-step decrease in $\mathcal{R}$. Because $g_{\sigma_0^{(k)}}$ is L-smooth, for any u, v

$$g_{\sigma_0^{(k)}}(v) \le g_{\sigma_0^{(k)}}(u) + \nabla g_{\sigma_0^{(k)}}(u)^\top (v - u) + \tfrac{L}{2}\|v - u\|^2. \tag{A.2}$$

Choose $u = f^{(k)}, v = f^{(k+1)} = f^{(k)} - \eta d^{(k)}$. Using (A.1) and $\|d^{(k)}\| \le G$:

$$g_{\sigma_0^{(k)}}(f^{(k+1)}) \le g_{\sigma_0^{(k)}}(f^{(k)}) - \big(\eta - \tfrac{L\eta^2}{2}\big)\|d^{(k)}\|^2.$$

Let $c := \eta - \frac{L\eta^2}{2} > 0$ (Assumption 4). Noting $\mathcal{R} \le g_{\sigma_0^{(k)}}$ for every σ_0:

$$\boxed{\mathcal{R}(f^{(k+1)}) \le \mathcal{R}(f^{(k)}) - c\,\|d^{(k)}\|^2.} \tag{A.3}$$

Step 5: summability of $\|d^{(k)}\|^2$. (A.3) shows $\{\mathcal{R}(f^{(k)})\}$ is monotonically decreasing and bounded below, hence convergent. Summing (A.3) from $k = 0$ to $N - 1$ and letting $N \to \infty$ gives $\sum_{k=0}^{\infty} \|d^{(k)}\|^2 < \infty$; thus $\|d^{(k)}\| \to 0$.

Step 6: convergence to stationarity. By (A.1) $\operatorname{dist}(0, \partial^C \mathcal{R}(f^{(k)})) \le \|d^{(k)}\| \to 0$, proving the first claim. Outer semicontinuity of the Clarke subdifferential for locally Lipschitz functions ∂^C implies that any limit point $\bar{f}$ satisfies $0 \in \partial^C \mathcal{R}(\bar{f})$, completing the proof. □

Proof of Corollary 2 Algorithm 1 without the base-permutation loop optimizes $\mathcal{J}(f) := \max_{\sigma \in \mathcal{S}} \mathcal{L}(f, \sigma)$, a convex L-smooth function (finite max of convex smooth maps).

(i) $O(1/\sqrt{K})$ **rate.** By Danskin, $g^{(k)} := \nabla_f \mathcal{L}(f^{(k)}, \sigma^{(k)}) = \nabla \mathcal{J}(f^{(k)})$. Applying the smooth descent lemma exactly as in Step 4 but with $\mathcal{J}$ (and noting $\mathcal{J}$ is itself the objective) yields $\mathcal{J}(f^{(k+1)}) \leq \mathcal{J}(f^{(k)}) - c\|g^{(k)}\|^2$. A standard telescoping–plus–averaging argument gives the stated $O(1/\sqrt{K})$ bound on $\mathcal{J}(\hat{f}_K) - \mathcal{J}(f^\star)$.

(ii) Linear rate under strong convexity. If $\mathcal{J}$ is μ-strongly convex, the quadratic decrease $\|f^{(k+1)} - f^\star\|^2 \leq (1 - \eta\mu)\|f^{(k)} - f^\star\|^2$ holds for any $\eta \in (0, 2/L)$, proving the linear rate. $\square$

Appendix C: Simple Numerical Example for Section 3

To practically illustrate the limitations of the ITP predictor, we construct a simple toy example.

EXAMPLE 1. Consider the following nonlinear data-generating process:

$$y = x_1^2 + x_2^2 + \epsilon,$$

where $x_1, x_2 \sim \mathcal{N}(0,1)$ (independent) and $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is independent noise. Suppose the observed feature is x_1 ($j = \{1\}$), and the missing feature x_2 is imputed as $\hat{x}_2 = \mathbb{E}[x_2 \mid x_1] = 0$ (since x_2 is independent of x_1).

Optimal Predictor Based on x_1. The optimal predictor given x_1 is:

$$f^\star_{\sigma,j}(x_1) = \mathbb{E}[y \mid x_1] = x_1^2 + \mathbb{E}[x_2^2] = x_1^2 + 1.$$

The MSE of the optimal predictor is:

$$\mathbb{E}\left[(y - f^\star_{\sigma,j}(x_1))^2\right] = \mathbb{E}\left[(x_1^2 + x_2^2 + \epsilon - (x_1^2 + 1))^2\right] = \sigma_\epsilon^2.$$

Impute-Then-Predict Approach. With imputation $\hat{x}_2 = 0$, the ITP predictor becomes:

$$f^\star(x_1, \hat{x}_2) = \mathbb{E}[y \mid x_1, \hat{x}_2] = \mathbb{E}[x_1^2 + x_2^2 + \epsilon \mid x_1, \hat{x}_2] = x_1^2.$$

The MSE of the ITP predictor is:

$$\mathbb{E}\left[(y - f^\star(x_1, \hat{x}_2))^2\right] = \mathbb{E}\left[(x_1^2 + x_2^2 + \epsilon - x_1^2)^2\right] = \mathbb{E}[x_2^4] + \sigma_\epsilon^2.$$

Since $\mathbb{E}[x_2^4] = 3$ (the fourth moment of a standard normal), we find:

$$\mathbb{E}\left[(y - f^\star(x_1, \hat{x}_2))^2\right] = 3 + \sigma_\epsilon^2.$$

The ITP predictor fails to account for the contribution of x_2^2 to y, resulting in a strictly higher MSE:

$$\mathbb{E}\left[(y - f^\star_{\sigma,j}(x_1))^2\right] = \sigma_\epsilon^2 < 3 + \sigma_\epsilon^2 = \mathbb{E}\left[(y - f^\star(x_1, \hat{x}_2))^2\right].$$

Appendix D: Full Illustrative Example for Section 3

[**Vassilis:** Does this help? Should we keep or remove?]

To practically illustrate the limitations of the ITP predictor, consider the following scenario. A patient visits a doctor with symptoms such as fatigue and dizziness. The doctor uses the following features:

- $\boldsymbol{x}^{\sigma,j}$: Basic features, such as age, gender, and medical history.
- $\hat{\boldsymbol{x}}^{\sigma,-j}$: Hemoglobin (hgb) levels, which are missing but will be revealed later through a blood test.

The goal is to predict the probability of anemia ($y=1$) based on these features. Suppose the relationship between the outcome y and the features is described by:

$$f^\star(\boldsymbol{x}^{\sigma,j},\text{hgb}) := \mathbb{P}(y=1 \mid \boldsymbol{x}^{\sigma,j},\text{hgb}) = \frac{1}{1+\exp(-g(\boldsymbol{x}^{\sigma,j}))} \cdot \begin{cases} 1-\exp(-0.5(\text{hgb}-10)), & \text{if hgb} < 10, \\ 0.2+0.1\cdot(\text{hgb}-10), & \text{if hgb} \geq 10, \end{cases}$$

where $g(\boldsymbol{x}^{\sigma,j}) = 0.5\cdot\text{age}+1.5$, and the conditional distribution of hgb given $\boldsymbol{x}^{\sigma,j}$ is $\text{hgb} \sim \mathcal{N}(12,2)$.

Optimal Predictor. The optimal predictor given only the basic features is:

$$f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \mathbb{E}[y \mid \boldsymbol{x}^{\sigma,j}],$$

which integrates over the uncertainty in hgb:

$$f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \int_{-\infty}^{\infty} \mathbb{P}(y=1 \mid \boldsymbol{x}^{\sigma,j},\text{hgb}) \cdot \mathbb{P}(\text{hgb} \mid \boldsymbol{x}^{\sigma,j})\, d\text{hgb}.$$

Substituting the functional forms:

$$f^\star_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \frac{1}{1+\exp(-g(\boldsymbol{x}^{\sigma,j}))} \cdot \left[\int_{-\infty}^{10} (1-\exp(-0.5(\text{hgb}-10))) \cdot \mathcal{N}(12,2)\, d\text{hgb} \right.$$
$$\left. + \int_{10}^{\infty} (0.2+0.1\cdot(\text{hgb}-10)) \cdot \mathcal{N}(12,2)\, d\text{hgb} \right].$$

For numerical calculations: 1. Region hgb < 10:

$$\int_{-\infty}^{10} (1-\exp(-0.5(\text{hgb}-10))) \cdot \mathcal{N}(12,2)\, d\text{hgb}.$$

Using numerical integration, this evaluates to approximately 0.35.

2. Region hgb ≥ 10:

$$\int_{10}^{\infty} (0.2+0.1\cdot(\text{hgb}-10)) \cdot \mathcal{N}(12,2)\, d\text{hgb}.$$

This evaluates to approximately 0.25.

Thus:

$$f^{\star}_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \frac{1}{1+\exp(-g(\boldsymbol{x}^{\sigma,j}))} \cdot (0.35 + 0.25) = \frac{1}{1+\exp(-g(\boldsymbol{x}^{\sigma,j}))} \cdot 0.6.$$

For $g(\boldsymbol{x}^{\sigma,j}) = 0.5 \cdot 30 + 1.5 = 16.5$ (age = 30), we get:

$$f^{\star}_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) = \frac{1}{1+\exp(-16.5)} \cdot 0.6 \approx 0.6.$$

Impute-then-Predict Predictor. In the ITP approach, the missing value hgb is imputed as its conditional expectation $\hat{\text{hgb}} = \mathbb{E}[\text{hgb} \mid \boldsymbol{x}^{\sigma,j}] = 12$. The prediction is:

$$f^{\star}(\boldsymbol{x}^{\sigma,j}, \hat{\text{hgb}}) = \mathbb{P}(y = 1 \mid \boldsymbol{x}^{\sigma,j}, \hat{\text{hgb}}).$$

For $\text{hgb} = 12$:

$$f^{\star}(\boldsymbol{x}^{\sigma,j}, 12) = \frac{1}{1+\exp(-g(\boldsymbol{x}^{\sigma,j}))} \cdot (0.2 + 0.1 \cdot (12 - 10)) = \frac{1}{1+\exp(-g(\boldsymbol{x}^{\sigma,j}))} \cdot 0.4.$$

For $g(\boldsymbol{x}^{\sigma,j}) = 16.5$, this gives:

$$f^{\star}(\boldsymbol{x}^{\sigma,j}, 12) \approx 0.4.$$

Suboptimality of Impute-then-Predict. The gap between the two predictors is:

$$\Delta = f^{\star}_{\sigma,j}(\boldsymbol{x}^{\sigma,j}) - f^{\star}(\boldsymbol{x}^{\sigma,j}, \hat{\text{hgb}}) = 0.6 - 0.4 = 0.2.$$

This gap arises because the optimal predictor integrates over the full distribution of hgb, capturing the possibility of low values that significantly affect the anemia probability. The ITP predictor collapses this uncertainty into a single point (the mean), failing to account for variability and leading to biased predictions.

Appendix E: Numerical Experiment Details for Section 3

Data Generation. The features $\mathbf{X} \in \mathbb{R}^{n \times p}$ are sampled from a multivariate Gaussian distribution with zero mean and covariance matrix:

$$\Sigma = I_p,$$

where I_p is the $p \times p$ identity matrix. The target variable y is generated according to two processes:

- **Linear Process**: $y = \mathbf{X}\boldsymbol{\beta} + \epsilon$, where $\boldsymbol{\beta} \in \mathbb{R}^p$ is a sparse coefficient vector, with 50% of the entries randomly set to zero. The noise $\epsilon \sim \mathcal{N}(0, 0.1^2)$.
- **Nonlinear Process**: $y = \mathbf{X}\boldsymbol{\beta} + \mathbf{X}^2\boldsymbol{\gamma} + \mathbf{X}_{\text{int}}\boldsymbol{\delta} + \epsilon$, where $\mathbf{X}^2$ includes the squared terms of all features, $\mathbf{X}_{\text{int}}$ includes all pairwise interactions, and $\boldsymbol{\gamma}, \boldsymbol{\delta}$ are random coefficient vectors. The noise $\epsilon \sim \mathcal{N}(0, 0.1^2)$.

Each dataset consists of $n = 5000$ samples and $p = 10$ features.

Feature Arrival Orders. We consider two types of feature arrival orders:

- **Worst-case Order**: Features with the weakest correlation with the target y are revealed first.
- **Random Order**: Features are revealed in a random sequence.

Prediction and Imputation Models.

- **Prediction Model**: We use nonlinear models (boosted trees) with the following hyperparameters:
  - —Number of estimators: 100.
  - —Learning rate: 0.1.
  - —Maximum depth: 3.
- **Imputation Method**: Missing features are imputed using MICE (Multiple Imputation by Chained Equations), which iteratively models each feature as a function of the others. We use the scikit-learn implementation with 10 iterations and default settings.

Evaluation. We evaluate both the optimal predictor $f^*_{\sigma,j}(x^{\sigma,j})$ and the ITP approach $f^*(x^{\sigma,j}, \hat{x}^{\sigma,-j})$ at each step j of the feature reveal process. For both methods, we compute the coefficient of determination (R^2) on the training and test datasets. We repeat each experiment 10 times and report the mean and standard deviation of the results.

Appendix F: Numerical Experiment Details for Section 4

In experiments 2-5, we maintain the same *Data Generation* process and settings as in experiment 1, and in this section, we primarily focus on the *Nonlinear data generation process*. We employ worst-case and random order selection and utilize *boosted trees* as our prediction model along with MICE as the imputation method. Specific details have been thoroughly outlined in Appendix E.

Experiment 2. For a given feature arrival order σ, we want to compare the performance of optimal predictor and ITP with the permutation-aware loss function, i.e., $\min_{f\in\mathcal{F}} \sum_{i=1}^{n}\sum_{j=1}^{p} \ell\left(y_i, f\left(\boldsymbol{x}_i^{\sigma,j}\right)\right)$ and $\min_{f\in\mathcal{F}} \sum_{i=1}^{n}\sum_{j=1}^{p} \ell\left(y_i, f\left(\boldsymbol{x}_i^{\sigma,j} \mid \hat{\boldsymbol{x}}_i^{\sigma,-j}\right)\right)$. We repeat the following steps 10 times and report the mean and standard deviation of the results.

Step 1 - Fix a Feature Arrival Order σ. We have two scenarios: worst-case feature order and random-case feature order.

Step 2 - Baseline: Train Impute-Then-Predict.

- Train on the full feature dataset D_{train}.
- During testing, features are sequentially revealed based on σ, and missing values are imputed using the MICE imputer trained on D_{train}.
- Predict using the full-feature model on the D_{imputed} dataset obtained from the last step.

Step 3 - Proposed: Train with the Permutation-Aware Loss Function.

- Expand the dataset D_{expanded} by generating p replicas for each data point, where p corresponds to the number of features:
  - The first replica includes only the first feature in the order σ.
  - The second replica includes the first two features in σ, and so forth, until the p-th replica contains all features.
- Train a predictive model using dataset D_{expanded}.
- During testing, as features arrive according to σ, impute missing values using the trained imputer and predict using the full-feature model on the imputed dataset D_{imputed}.

Step 4 - Evaluation.

- Evaluate the performance by reporting and plotting the training and testing R^2 values (mean and standard deviation) as a function of the number of features that have arrived.
- Compare the performance of different models (e.g., baseline vs. proposed) under various feature arrival orders (worst-case and random-case).
- Ensure robust evaluation by:
  - Properly splitting the data into training and testing sets.

- Repeating the experiment 10 times to account for variability.

- Analyze the results to determine the effectiveness of the proposed permutation-aware loss function in handling sequential feature arrival and imputation.

Experiment 3. We randomly sample a subset of feasible permutations $\mathcal{S} \subset S_p$ where $|\mathcal{S}| = 20$. Among the permutations in subset $\mathcal{S}$, we select the feature order σ such that

$$\sigma = \arg\min_{\sigma \in \mathcal{S}} \left[\sum_{i=1}^{n} \sum_{j=1}^{p} \ell\left(y_i, f\left(\boldsymbol{x}_i^{\sigma,j}\right)\right) \mid f = \min_{f \in \mathcal{F}} \sum_{i=1}^{n} \sum_{j=1}^{p} \ell\left(y_i, f\left(\boldsymbol{x}_i^{\sigma,j}\right)\right) \right].$$

After obtaining the selected features arrival order, we follow the same steps as in Experiment 2. Additionally, we repeat the entire process 10 times and report the mean and standard deviation of the results.

Experiment 4. $|\mathcal{S}| = 10$, $L = 2$, $n_{\text{sample}} = 1000$.

Experiment 5. It can be considered as the dual form of experiment 3, corresponding to the formulation of Problem (6). We select the worst-case features arrival order among the subset $\mathcal{S}$, i.e.,

$$\sigma = \arg\max_{\sigma \in \mathcal{S}} \left[\sum_{i=1}^{n} \sum_{j=1}^{p} \ell\left(y_i, f\left(\boldsymbol{x}_i^{\sigma,j}\right)\right) \mid f = \min_{f \in \mathcal{F}} \sum_{i=1}^{n} \sum_{j=1}^{p} \ell\left(y_i, f\left(\boldsymbol{x}_i^{\sigma,j}\right)\right) \right].$$

The remaining steps are identical to those in Experiment 3.

Appendix G: An Open-Source Library for Learning over Streaming Feature Spaces

To facilitate adoption by practitioners, we implement the proposed framework as an open-source Python library, which we call `sfs-learn` (for "streaming feature space learning"). `dfs-learn` is compatible with the standard Python ML library, `scikit-learn` (Pedregosa et al. 2011).

To use `dfs-learn`, the user must specify the following input parameters:

- Function class and hyperparameters: This parameter corresponds to the function class $\mathcal{F}$ under consideration and the corresponding hyperparameters. The function class is defined according to the `scikit-learn` naming conventions, e.g., `LinearRegression` for linear regression and `DecisionTreeClassifier` for classification trees. The hyperparameters are provided as a Python dictionary (if none is given, they are set to their defaults).
- Feasible permutations: This parameter defines the set of permutations $\mathcal{S}$ under consideration (feasible permutations) and is given as a set of vectors (each describing a permutation). The parameter defaults to "all" meaning that features can arrive in any order.
- Formulation: This parameter decides the type of formulation to be used as per Section **??**. In particular, the options are:

  — `min-min`: Finds the optimal combination of model and permutation (from within the feasible set of permutations) assuming there will be no uncertainty in the feature arrival order (it will precisely follow the prescribed permutation).

  — `min-min-max`: Finds the optimal combination of model and permutation (from within the feasible set of permutations) assuming there will be up to l-mis-orderings in the feature arrival order. l is an additional input parameter and defaults to $l = 1$.

  — `min-max`: Finds the optimal model under the worst-case permutation from within the feasible set, assuming the feature arrival order is unknown.
- Technical parameters relating to the optimization algorithm:

  — Optimization over function space: standard, enhanced, max-enhanced [**Vassilis:** need to rename/clarify those, and also consider any necessary technical parameters, e.g., temperature in simulated annealing]

  — Optimization over permutation space: exhaustive, gradient, simulated annealing, [**Vassilis:** same as above]

  — Imputation method

For reproducibility purposes, every experiment described in this paper is given as a separate Jupyter notebook written in Python.

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OPTIMIZING LYAPUNOV CERTIFICATES VIA STABILITY-PRESERVING QUADRATIZATION FOR POLYNOMIAL SYSTEMS

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Abstract. Region-of-attraction certificates for polynomial systems become expensive as the state dimension and polynomial degree grow, because direct sum-of-squares formulations require a monomial basis whose size grows combinatorially. Quadratization sidesteps this cost by representing a polynomial vector field exactly on an invariant manifold of a quadratic system, so that a quadratic Lyapunov function can certify the region of attraction directly. Even once the lifting map is fixed, this exactness constrains the lifted dynamics only on the manifold, leaving freedom in how they extend off it. Stabilizer gains shape this off-manifold extension and change the transverse dynamics, while representation gauges leave the vector field unchanged but change its matrix representation. Both choices affect the resulting spectral-norm certificate, yet prior work has treated them separately, fixing the gain with a feasibility heuristic and optimizing the gauge only afterward, for that fixed pair.

We formulate optimal dissipative quadratization (ODQ), which designs the gain and the gauge together for a fixed monomial lift, reference extension, and stabilizer factorization, with the Lyapunov weight fixed to the identity. ODQ selects stabilizer gains within a prescribed compact Hurwitz box and, for each candidate gain, optimizes the representation gauge by an exact semidefinite program that attains the global minimum of the spectral-norm bound at that gain. Residual-aware bounds turn this spectral-norm bound into a certified closed Lyapunov sublevel set, accounting for the floating-point residual of the underlying Lyapunov solve. Under our stated assumptions, every accumulation point of the idealized outer search is box-Clarke stationary. A finite run instead returns the best candidate it can independently verify, and we do not claim global optimality for the gain search.

On a planar quintic system, optimizing the gain increases the certified area by a factor of 2.238 over a matched zero-gain gauge. Across a benchmark of 16 heterogeneous polynomial systems with stabilizer freedom, ODQ improves on both fixed-gain lifted baselines. All 36 ODQ runs on the relay benchmark complete, and ODQ is favored in all 27 repeat-level comparisons across the nine fully paired relay cases against a sum-of-squares baseline with a fixed quadratic Lyapunov function, in both the fixed-direction proxy and construction time. Broader comparisons with direct sum-of-squares methods remain mixed.

Key words. polynomial systems, quadratization, region of attraction, Lyapunov functions, semidefinite programming

MSC codes. 93D05, 93D20, 90C22, 93D30

1. Introduction.

The region of attraction (ROA) of an asymptotically stable equilibrium is the set of initial states from which trajectories return to it. Exact quadratization represents a polynomial system on an invariant lifting manifold of a quadratic system. Even with the lifting map fixed, any vector-valued polynomial of total degree at most two that vanishes on this manifold may be added to the lifted right-hand side [7, Ex. 2]. These additions preserve the original trajectories from compatible lifted initial conditions, while they can change transverse stability and the resulting Lyapunov certificate. We treat this off-manifold extension as a design variable for ROA estimation.

ROA estimates are required for post-fault screening in power systems [9] and safe exploration in learning-based control [3]. A closed-form description of the ROA is generally unavailable for nonlinear systems, and the standard substitute is a certified

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inner approximation: a sublevel set of a Lyapunov function, together with a proof that this sublevel set is contained in the ROA. For polynomial vector fields this construction has been carried out by convex optimization for two decades, through sum-of-squares (SOS) programming [25, 27], the V–s iterations and shape-function enlargements built on it [14, 31, 33], and cheaper relaxations of the SOS cone [1].

These programs work over a monomial basis whose size grows combinatorially in the state dimension and the polynomial degree, and the effect is visible in the size of the published examples: many illustrative ROA studies for polynomial systems use two state variables [20, 26, 29], occasionally three [24, 36]. *Quadratization* trades that degree for dimension. Adjoining auxiliary variables $\mathbf{y} = \mathbf{g}(\mathbf{x})$ makes the augmented state $\mathbf{w} = (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^N$ obey an exactly quadratic system $\dot{\mathbf{w}} = A\mathbf{w} + H(\mathbf{w} \otimes \mathbf{w})$. Such a lift always exists [8, 15], modern algorithms produce monomial lifts with provably few auxiliary variables [6], and for a quadratic system with a Hurwitz linear part, a quadratic Lyapunov function yields an ellipsoidal certificate from a closed-form radius or a single semidefinite program [16, 32]. The same quadratic form recurs across otherwise unrelated settings: it is the working class of lift-and-learn [19, 30] and of operator inference [17, 28], it is the entry point of Carleman linearization for reachability analysis [13], and bimolecular mass-action chemical kinetics is quadratic outright [12].

Quadratizations of the same system differ in structure and in how many variables they introduce [2]. Every polynomial system with a dissipative equilibrium admits a *dissipative* quadratization, obtained by doubling a stabilizer gain until the lifted origin passes a Routh–Hurwitz test [7, Thm. 1, Alg. 2, Prop. 3]. There is also a representation gauge: adding a matrix K satisfying $K(\mathbf{w} \otimes \mathbf{w}) = 0$ leaves the quadratic vector field unchanged but can alter the spectral-norm bound. This gauge has been optimized for a fixed pair (A, H) [16]. The two choices are coupled at the certificate level, since the stabilizer gain changes A and H, while A determines the Lyapunov metric in which the gauge is evaluated.

We characterize the full affine space of admissible quadratic extensions for a fixed embedding and restrict the optimization to the stabilizer family. For a fixed monomial inner-quadratic lift, reference extension, prescribed stabilizer factorizations, and Lyapunov weight $Q = I_N$, we optimize the diagonal stabilizer gains Λ and the Kronecker representation coefficients μ. The gains determine the lifted dynamics, while the representation coefficients determine the spectral-norm bound used in the certificate. Over a prescribed compact box B contained in the Hurwitz set, we formulate optimal dissipative quadratization (ODQ) with optimal value $\phi_B := \min_{\Lambda \in B} \min_{\mu \in \mathbb{R}^d} \|J(\Lambda, \mu)\|$, where $d = N^2(N-1)/2$. Its reciprocal $\rho_B := \phi_B^{-1}$ is the optimal Lyapunov-level threshold within this family and box, with $1/0 := +\infty$. For fixed Λ, the gauge problem is an exact convex semidefinite program (SDP); the outer value function is generally nonsmooth and nonconvex.

Contributions and organization. Within this fixed family, we first derive a tangent–transverse spectral splitting that reduces Hurwitz feasibility to the auxiliary block. We prove that the outer value is locally Lipschitz and attains its minimum on every admissible box, and give an example for which the unrestricted threshold is unbounded. We then derive a residual-aware rule for reporting a strict closed certificate and show that, when $\phi_B > 0$ and $Q = I_N$, threshold maximization orders the leading coefficients of scaled small-radius pullback volumes. For computation, we remove gauge directions invisible to the certificate and obtain an exact fixed-gain SDP with $q = N(N^2-1)/3$ effective gauge coordinates. Analytic sensitivities at regular points are combined with

a box-constrained gradient-sampling L-BFGS method. Under the stated assumptions, every accumulation point of the idealized sequence is box-Clarke stationary; finite runs instead return the best independently screened candidate encountered, and we do not claim global optimality of the outer gain search.

The numerical study comprises a planar quintic, a heterogeneous suite of 19 polynomial systems, and a 12-system single-link relay grid. On the planar example, gain optimization increases the certified area by a factor of 2.238 over the matched zero-gain gauge; on all 16 heterogeneous systems with nontrivial stabilizer freedom, ODQ improves the reported size metric over both fixed-gain lifted baselines. Direct sum-of-squares (SOS) methods with quadratic or quartic Lyapunov functions give a structure-dependent size–time tradeoff rather than uniform dominance. All 36 ODQ relay runs completed, and each of the nine fully paired SOS-2 cells favors ODQ in both proxy and time. Within the tested dimensions, the favorable cases are higher-degree, higher-dimensional systems whose sparse nonlinear structure admits a compact quadratic lift. Section 2 gives the required background, Section 3 develops the certificate problem, Section 4 presents the algorithm, and Section 5 reports the experiments.

2. Preliminaries.

2.1. Notation. Vectors are set in bold and their scalar entries in plain type: $\mathbf{x} = (x_1, \dots, x_n)$, and the origin of $\mathbb{R}^n$ is $\mathbf{0}$. Matrices are plain capitals (A, H, P), and scalars are plain lowercase. We write $\mathbb{R}$ for the real numbers and $\mathbb{R}[\mathbf{x}]$ for the ring of polynomials in $\mathbf{x}$, with $\mathbb{R}[\mathbf{x}]^n$ its n-vectors. For a matrix M, $M^\top$ is the transpose, $\|M\|$ the spectral norm (its largest singular value), and $\mathrm{eig}(M)$ is the spectrum; for a vector, $\|\cdot\|$ is the Euclidean norm. We index eigenvalues by decreasing real part, $\mathrm{Re}\,\lambda_1(M) \geq \cdots \geq \mathrm{Re}\,\lambda_N(M)$, so M is *Hurwitz* precisely when $\mathrm{Re}\,\lambda_1(M) < 0$; the quantity $-\mathrm{Re}\,\lambda_1(M) > 0$ is then its *stability margin*. $\mathbb{S}^N$ denotes the symmetric $N \times N$ matrices, $P \succ 0$ ($P \succeq 0$) positive (semi)definiteness, and $\preceq$ the Loewner order. The Jacobian of a differentiable map $\mathbf{f}$ is $J_x(\mathbf{f}) = [\partial f_i / \partial x_j]$. We use $\otimes$ for the Kronecker product and write $\mathbf{w}^{\otimes j} = \mathbf{w} \otimes \cdots \otimes \mathbf{w}$ (j factors) for Kronecker powers; I_k is the $k \times k$ identity and $\mathbf{e}_k$ the k-th standard basis vector. A homogeneous quadratic vector field on $\mathbb{R}^N$ is written $H(\mathbf{w} \otimes \mathbf{w})$ with $H \in \mathbb{R}^{N \times N^2}$, so that its r-th component is the quadratic form $\mathbf{w}^\top H_r \, \mathbf{w}$, where $H_r \in \mathbb{R}^{N \times N}$ is the r-th row of H reshaped to a matrix.

2.2. Polynomial systems and dissipative equilibria. We study polynomial systems of ordinary differential equations (ODEs),

$$\dot{\mathbf{x}} = \mathbf{p}(\mathbf{x}), \qquad \mathbf{x}(t) \in \mathbb{R}^n, \quad \mathbf{p} = (p_1, \dots, p_n) \in \mathbb{R}[\mathbf{x}]^n. \tag{2.1}$$

Each component $p_i \in \mathbb{R}[\mathbf{x}]$ is a polynomial in the state $\mathbf{x} = (x_1, \dots, x_n)$. Our interest is in the behaviour of (2.1) near an equilibrium at which it is locally stable.

DEFINITION 2.1 (equilibrium; dissipativity [7]). *A point $\mathbf{x}^* \in \mathbb{R}^n$ is an* equilibrium *of* (2.1) *if $\mathbf{p}(\mathbf{x}^*) = \mathbf{0}$, and* (2.1) *is* dissipative at $\mathbf{x}^*$ *if every eigenvalue of the Jacobian $J_x(\mathbf{p})|_{\mathbf{x}=\mathbf{x}^*}$ has negative real part.*

We analyse one equilibrium at a time and place it at the origin without loss of generality. The substitution $\mathbf{z} = \mathbf{x} - \mathbf{x}^*$ takes (2.1) to $\dot{\mathbf{z}} = \mathbf{p}(\mathbf{z} + \mathbf{x}^*)$, which is again polynomial, has an equilibrium at $\mathbf{0}$, and has the same Jacobian there. We retain the notation $\mathbf{x}$ and $\mathbf{p}$ for the translated coordinates and vector field, so $\mathbf{p}(\mathbf{0}) = \mathbf{0}$.

DEFINITION 2.2 (dissipativity margin). *The* dissipativity margin *of the origin is*

$$\alpha_p \;:=\; -\operatorname{Re}\lambda_1\big(J_x(\mathbf{p})|_{\mathbf{0}}\big), \tag{2.2}$$

so that (2.1) *is dissipative at* $\mathbf{0}$ *precisely when* $\alpha_p > 0$.

Our analysis assumes that the original system (2.1) is dissipative at the origin, equivalently, $\alpha_p > 0$, since exact quadratization preserves the original dynamics and cannot alter the local stability of this equilibrium. By Lyapunov's indirect method, for every rate $a < \alpha_p$ there exist a neighbourhood of the origin and a constant $C \geq 1$ such that $\|\mathbf{x}(t)\| \leq C\, e^{-at}\|\mathbf{x}(0)\|$ for all $t \geq 0$. Trajectories therefore relax on the asymptotic time scale $1/\alpha_p$ and the constant C absorbs nonnormal transient amplification, which α_p does not bound. Geometrically, $-\alpha_p$ is the spectral abscissa of $J_x(\mathbf{p})|_{\mathbf{0}}$, so the spectrum lies at least α_p to the left of the imaginary axis and α_p is its distance to the stability boundary. Thus, the margin is intrinsic to (2.1) and fixed before any lift is chosen.

2.3. Quadratization, stabilizers, and the quadratic form. A quadratic right-hand side permits closed-form Lyapunov bounds and convex certificate optimization. For a fixed Hurwitz linear part and a fixed positive definite Lyapunov weight, the analytic ellipsoidal ROA radius scales as $1/\|H\|$ when $H \neq 0$ and is inexpensive to evaluate once the Lyapunov equation is solved [16, Prop. 3.1 and §4]. The nonlinear term is represented by a fixed tensor whose projection can be precomputed, allowing exact evaluation of the projected nonlinearity in reduced coordinates without interpolation [18, 30].

Reducing the polynomial degree requires auxiliary state variables. Any polynomial system admits a finite quadratic extension [7, Prop. 1], with freedom in both the lifting map and the off-manifold dynamics. Algorithms for *optimal monomial quadratization* minimize the number of auxiliary variables [5, 6], whereas *dissipative quadratization* also requires the lifted equilibrium to have a Hurwitz Jacobian [7]. We describe these freedoms below and formulate the gain and gauge optimization in Section 3.

DEFINITION 2.3 (quadratization [7]). *A* quadratization *of* (2.1) *consists of new variables* $y_i = g_i(\mathbf{x})$, $i = 1, \ldots, m$, *together with polynomial vectors* $\mathbf{q}_1(\mathbf{x}, \mathbf{y}) \in \mathbb{R}[\mathbf{x}, \mathbf{y}]^n$ *and* $\mathbf{q}_2(\mathbf{x}, \mathbf{y}) \in \mathbb{R}[\mathbf{x}, \mathbf{y}]^m$ *of total degree at most two such that*

$$\dot{\mathbf{x}} = \mathbf{q}_1(\mathbf{x}, \mathbf{y}), \qquad \dot{\mathbf{y}} = \mathbf{q}_2(\mathbf{x}, \mathbf{y}) \tag{2.3}$$

holds along $\mathbf{y} = \mathbf{g}(\mathbf{x})$. *It is* monomial quadratization *if every* g_i *is a monomial, and its* lifting manifold *is* $\mathcal{M} = \{(\mathbf{x}, \mathbf{g}(\mathbf{x})) : \mathbf{x} \in \mathbb{R}^n\} \subset \mathbb{R}^N$ *with* $N = n + m$.

We restrict attention to *monomial* quadratizations with nonconstant auxiliary monomials, so $\mathbf{g}(\mathbf{0}) = \mathbf{0}$ and exactness places the lifted equilibrium at the origin. With $\mathbf{w} = (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^N$, collecting the linear and quadratic terms in (2.3) gives the lifted vector field

$$\dot{\mathbf{w}} = \mathbf{F}(\mathbf{w}) := A\mathbf{w} + H(\mathbf{w} \otimes \mathbf{w}), \qquad A \in \mathbb{R}^{N \times N}, \quad H \in \mathbb{R}^{N \times N^2}. \tag{2.4}$$

Here $A = J_w(\mathbf{F})|_{\mathbf{0}}$ is the linearization at the lifted equilibrium, and H is a Kronecker representation of the homogeneous quadratic term.

Reading off the coefficients of $\mathbf{q}_1, \mathbf{q}_2$ gives one such pair, written (A_0, H_0) and called the *zero-gain lift*. The corresponding field $\mathbf{F}_0(\mathbf{w}) := A_0\mathbf{w} + H_0(\mathbf{w} \otimes \mathbf{w})$ serves as the reference extension below. The original dynamics are exactly the restriction of (2.4) to the invariant manifold $\mathcal{M}$. Off $\mathcal{M}$, equation (2.4) is an *extension* of those

dynamics, and the quadratization is free to shape that extension without touching the system it represents.

DEFINITION 2.4 (admissible quadratic extensions). *Fix the polynomial embedding* $\Phi : \mathbb{R}^n \to \mathbb{R}^N$, $\Phi(\mathbf{x}) = (\mathbf{x}, \mathbf{g}(\mathbf{x}))$, *with image* $\mathcal{M}$ *and* $N = n + m$. *Let* $\mathcal{P}_2 := \mathbb{R}[\mathbf{w}]_{\leq 2}$ *denote the scalar polynomials of total degree at most two, and define the* vanishing ideal

$$I(\mathcal{M}) := \{r \in \mathbb{R}[\mathbf{w}] : r(\Phi(\mathbf{x})) = 0 \text{ for all } \mathbf{x} \in \mathbb{R}^n\}.$$

The admissible quadratic extension space *of* $\mathbf{p}$ *for the fixed embedding is*

$$\mathcal{E}_\Phi(\mathbf{p}) := \{\mathbf{F} \in \mathcal{P}_2^N : \mathbf{F}(\Phi(\mathbf{x})) = J_x(\Phi)(\mathbf{x})\,\mathbf{p}(\mathbf{x}) \text{ for all } \mathbf{x} \in \mathbb{R}^n\}. \tag{2.5}$$

Given a reference extension $\mathbf{F}_0 \in \mathcal{E}_\Phi(\mathbf{p})$, *this is the affine space*

$$\mathcal{E}_\Phi(\mathbf{p}) = \mathbf{F}_0 + \mathcal{V}_\Phi, \qquad \mathcal{V}_\Phi := I(\mathcal{M})^N \cap \mathcal{P}_2^N, \tag{2.6}$$

where $I(\mathcal{M})^N$ *denotes the Cartesian product of* N *copies of* $I(\mathcal{M})$.

Indeed, for $\mathbf{F} \in \mathcal{P}_2^N$ and the fixed reference $\mathbf{F}_0$, exactness in (2.5) is equivalent to $\mathbf{F} - \mathbf{F}_0$ vanishing componentwise on $\mathcal{M}$. A basis of $I(\mathcal{M}) \cap \mathcal{P}_2$ can be computed by substituting $\mathbf{w} = \Phi(\mathbf{x})$ into a general linear combination of $1, w_i, w_i w_j$ $(1 \leq i \leq j \leq N)$ and taking the null space of the resulting coefficient equations. If $r_1, \ldots, r_s$ is such a basis, then $\{\mathbf{e}_k r_\ell : 1 \leq k \leq N,\ 1 \leq \ell \leq s\}$ is a basis of $\mathcal{V}_\Phi$.

Example 2.5 (Admissible extensions of a scalar cubic system). Consider $\dot{x} = p(x) := -x - x^3$ and fix $\Phi(x) = (x, x^2)$, so $\mathcal{M} = \{(x, y) : y = x^2\}$. The quadratic vector field

$$\mathbf{F}_0(x, y) = \begin{pmatrix} -x - xy \\ -2y - 2y^2 \end{pmatrix}$$

is an admissible extension because $\mathbf{F}_0(x, x^2) = (p(x), 2xp(x))^\top = J_x(\Phi)(x)p(x)$.

To find all such extensions, write $r = c_0 + c_1 x + c_2 y + c_3 x^2 + c_4 xy + c_5 y^2 \in \mathcal{P}_2$. Substitution gives

$$r(x, x^2) = c_0 + c_1 x + (c_2 + c_3)x^2 + c_4 x^3 + c_5 x^4.$$

This vanishes identically if and only if $c_0 = c_1 = c_4 = c_5 = 0$ and $c_3 = -c_2$. Thus $I(\mathcal{M}) \cap \mathcal{P}_2 = \operatorname{span}\{h\}$, where $h(x, y) = y - x^2$, and

$$\mathcal{V}_\Phi = \operatorname{span}\left\{\begin{pmatrix} h \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ h \end{pmatrix}\right\}, \qquad \mathcal{E}_\Phi(p) = \left\{\mathbf{F}_0 + \begin{pmatrix} ah \\ bh \end{pmatrix} : a, b \in \mathbb{R}\right\}.$$

The two parameters independently modify the lifted equations away from $\mathcal{M}$ while preserving the dynamics on $\mathcal{M}$. The choice $a = 0$, $b = -\lambda$ gives the family in Example 3.11.

DEFINITION 2.6 (inner-quadratic lift; stabilizers [7]). *A quadratization is* inner-quadratic *if each* g_i *factors as* $g_i = a_i b_i$ *with* $a_i, b_i \in \{x_1, \ldots, x_n, y_1, \ldots, y_{i-1}\}$. *Its* i*-th* stabilizer *is the quadratic polynomial*

$$h_i(\mathbf{x}, \mathbf{y}) \;:=\; y_i - a_i b_i, \tag{2.7}$$

which vanishes identically on $\mathcal{M}$ *and therefore belongs to* $I(\mathcal{M}) \cap \mathcal{P}_2$ *in Definition 2.4.*

Every polynomial system admits an inner-quadratic monomial quadratization [7, Prop. 1], computable by branch and bound [6]. Since $\mathbf{e}_k h_i \in \mathcal{V}_\Phi$ for every $k = 1, \dots, N$, adding $-\lambda_i h_i$ to the k-th lifted equation, with $\lambda_i \in \mathbb{R}$, gives another element of $\mathcal{E}_\Phi(\mathbf{p})$: its restriction to $\mathcal{M}$ is unchanged, while its off-manifold dynamics may change. Those transverse dynamics determine the stability of the lifted equilibrium, which makes the stabilizers the natural design variables. Dissipative quadratization [7] exploits exactly this: it replaces $\mathbf{q}_2$ by $\mathbf{q}_2 - \lambda \mathbf{h}$ with $\mathbf{h} = (h_1, \dots, h_m)$ and raises λ through $1, 2, 4, \dots$ until the lifted origin is dissipative, a search that always terminates [7, Alg. 2, Prop. 3]. It returns the first gain that works.

2.4. Ellipsoidal Lyapunov certificates. Suppose that A in (2.4) is Hurwitz. For a prescribed $Q \succ 0$, let $P \succ 0$ be the unique Lyapunov solution and define V by

$$A^\top P + PA = -Q, \quad V(\mathbf{w}) = \mathbf{w}^\top P \mathbf{w}, \quad \dot{V}(\mathbf{w}) = -\mathbf{w}^\top Q \mathbf{w} + 2\mathbf{w}^\top P H(\mathbf{w} \otimes \mathbf{w}). \tag{2.8}$$

Write $\mathbf{q}(\mathbf{w}) := H(\mathbf{w} \otimes \mathbf{w})$ for the homogeneous quadratic term. For symmetric positive definite P, Q, define its *intrinsic normalized cubic growth* by

$$\beta(P, Q, \mathbf{q}) := \sup_{\mathbf{w} \neq \mathbf{0}} \frac{2|\mathbf{w}^\top P \mathbf{q}(\mathbf{w})|}{\sqrt{\mathbf{w}^\top P \mathbf{w}}\, \mathbf{w}^\top Q \mathbf{w}}. \tag{2.9}$$

This quantity depends on the quadratic map, independently of its matrix representation. For the exact Lyapunov solution in (2.8), set $c(\mathbf{u}) := \mathbf{u}^\top P \mathbf{q}(\mathbf{u})$ and define

$$\rho_{\mathrm{rad}} := \inf_{\substack{\mathbf{u}^\top P \mathbf{u} = 1 \\ c(\mathbf{u}) > 0}} \frac{\mathbf{u}^\top Q \mathbf{u}}{2\, c(\mathbf{u})}, \quad \inf \varnothing := +\infty. \tag{2.10}$$

Proposition 2.7 (Ellipsoidal Lyapunov certificate). *For the exact Lyapunov solution in* (2.8), $\rho_{\mathrm{rad}} = 1/\beta(P, Q, \mathbf{q})$ *is the supremal* $r > 0$ *for which* $\dot{V}(\mathbf{w}) < 0$ *whenever* $0 < V(\mathbf{w}) < r^2$, *with* $1/0 := +\infty$. *For every* $0 < \rho < \rho_{\mathrm{rad}}$, *the closed ellipsoid* $\mathcal{D}(P, \rho) := \{\mathbf{w} \in \mathbb{R}^N : V(\mathbf{w}) \le \rho^2\}$ *is positively invariant and contained in the ROA of the origin. For the chosen Kronecker representation* H,

$$\rho_{\mathrm{an}}(A, H, Q) := \frac{\lambda_{\min}(Q)}{2\|H\|\sqrt{\|P\|}} \le \rho_{\mathrm{rad}}, \qquad 1/0 := +\infty. \tag{2.11}$$

Thus $\mathcal{D}(P, \rho)$ *is certified for every* $0 < \rho < \rho_{\mathrm{an}}(A, H, Q)$.

More generally, let $\widehat{P} = \widehat{P}^\top \succ 0$ *be any computed matrix, put* $\widehat{V}(\mathbf{w}) := \mathbf{w}^\top \widehat{P} \mathbf{w}$, *and set*

$$R := A^\top \widehat{P} + \widehat{P} A + Q, \qquad \delta := \|Q^{-1/2} R Q^{-1/2}\|. \tag{2.12}$$

If $\delta < 1$ *and* $\gamma \ge \beta(\widehat{P}, Q, \mathbf{q})$, *then*

$$\dot{\widehat{V}}(\mathbf{w}) \le -\big(1 - \delta - \gamma \sqrt{\widehat{V}(\mathbf{w})}\big) \mathbf{w}^\top Q \mathbf{w}. \tag{2.13}$$

For $\gamma > 0$ *and* $\theta \in (0, 1)$, *the closed ellipsoid*

$$\mathcal{D}\left(\widehat{P}, \theta \frac{1 - \delta}{\gamma}\right) = \left\{ \mathbf{w} : \widehat{V}(\mathbf{w}) \le \theta^2 \left(\frac{1 - \delta}{\gamma}\right)^2 \right\} \tag{2.14}$$

is positively invariant, has strict Lyapunov decrease away from the origin, and is contained in the region of attraction of the origin. If $\gamma = 0$, $\widehat{V}$ *is a global Lyapunov function. Verified upper bounds may replace* δ *and* γ, *provided the residual bound is below one.*

Proof. See the proof of Proposition 2.7 in Appendix A (p. 28). ∎

The bound (2.11) is the $E = I_N$ case of [16, Prop. 3.1]. For fixed P and Q, ρ_{rad} depends only on the quadratic vector field, whereas ρ_{an} depends on its Kronecker representation; the gauge-optimized bound is introduced after the dynamics-preserving gauge.

3. Problem formulation. Under the standing assumptions of Section 2, fix an inner-quadratic monomial lift $\mathbf{g}$ with $\deg g_i \geq 2$. The original system $\mathbf{p}$, embedding $\Phi(\mathbf{x}) = (\mathbf{x}, \mathbf{g}(\mathbf{x}))$, and lifting manifold $\mathcal{M}$ remain fixed throughout the optimization. Every admissible extension has the form (2.4), with a homogeneous quadratic term and no constant term.

For this fixed embedding, we first select an off-manifold extension and then a matrix representation of its quadratic term. The design variables are the stabilizer gains $\Lambda \in \mathbb{R}^m$ and the representation coefficients $\mu \in \mathbb{R}^d$, with $d = N^2(N-1)/2$. Fixing the lift keeps the state dimension and manifold unchanged; fixing the reference extension and stabilizer factorizations specifies the affine family below. We also fix the Lyapunov weight $Q = I_N$ to isolate gain and representation design. For each extension with $A(\Lambda)$ Hurwitz, the certificate matrix $P(\Lambda) \in \mathbb{S}^N$, $P(\Lambda) \succ 0$, is determined by the Lyapunov equation, rather than optimized independently. Figure 4 in Appendix B shows how these fixed data and the two design variables determine the certificate and the optimization problem.

3.1. The affine family of dissipative lifts. Fix the reference extension $\mathbf{F}_0 = (\mathbf{q}_1, \mathbf{q}_2)$ represented by (A_0, H_0) and one factorization $g_i = a_i b_i$ for each stabilizer $h_i = y_i - a_i b_i$. These data are computed once using the implementation of DQBEE [7]. With these data fixed, Definition 2.4 gives the admissible extension family $\mathcal{E}_\Phi(\mathbf{p}) = \mathbf{F}_0 + \mathcal{V}_\Phi$, where $\mathcal{V}_\Phi = I(\mathcal{M})^N \cap \mathcal{P}_2^N$.

We restrict the extension freedom to the stabilizer subspace

$$\mathcal{S}_h := \operatorname{span}\{\mathbf{e}_{n+i} h_i : 1 \leq i \leq m\} \subseteq \mathcal{V}_\Phi, \qquad \mathbf{F}_\Lambda := \mathbf{F}_0 - \sum_{i=1}^{m} \lambda_i \mathbf{e}_{n+i} h_i. \tag{3.1}$$

Thus $\{\mathbf{F}_\Lambda : \Lambda \in \mathbb{R}^m\} = \mathbf{F}_0 + \mathcal{S}_h$ is an m-dimensional affine subspace of $\mathcal{E}_\Phi(\mathbf{p})$. The restriction gives one gain per auxiliary equation and the transverse matrix $D(\Lambda) = W - \operatorname{diag}(\Lambda)$ established in Lemma 3.1. Directions in $\mathcal{V}_\Phi \setminus \mathcal{S}_h$, including those that modify the physical equations off $\mathcal{M}$, are excluded from the present family. Appendix B.1 (p. 29) gives the coefficient-matching construction of a basis of $\mathcal{V}_\Phi$ and verifies the inclusion and dimension stated above.

Writing $\Lambda = (\lambda_1, \ldots, \lambda_m)$, the selected extension is

$$\dot{\mathbf{x}} = \mathbf{q}_1(\mathbf{x}, \mathbf{y}), \qquad \dot{y}_i = q_{2,i}(\mathbf{x}, \mathbf{y}) - \lambda_i \, h_i(\mathbf{x}, \mathbf{y}), \quad i = 1, \ldots, m. \tag{3.2}$$

Since $h_i \equiv 0$ on $\mathcal{M}$, system (3.2) is a valid quadratization of (2.1) for every Λ. Split the stabilizer (2.7) into linear and quadratic parts, $-\lambda_i h_i = -\lambda_i y_i + \lambda_i a_i b_i$. The term $-\lambda_i y_i$ enters the linear part and $+\lambda_i a_i b_i$ the quadratic part, so

$$A(\Lambda) \;=\; A_0 - \sum_{i=1}^{m} \lambda_i \, \mathbf{e}_{n+i} \mathbf{e}_{n+i}^{\top}, \qquad H(\Lambda) \;=\; H_0 + \sum_{i=1}^{m} \lambda_i \, \mathbf{e}_{n+i} \, s_i^{\top}, \tag{3.3}$$

with $s_i \in \mathbb{R}^{N^2}$ the Kronecker encoding of $a_i b_i$. Both maps are affine in Λ. Each gain subtracts λ_i from one auxiliary diagonal entry of A and modifies one row of H.

The chart $\mathbf{z} = \mathbf{y} - \mathbf{g}(\mathbf{x})$ straightens $\mathcal{M}$ to $\{\mathbf{z} = \mathbf{0}\}$. Since every auxiliary monomial has degree at least two, $J_x(\mathbf{g})|_{\mathbf{0}} = 0$, so this change of coordinates has identity derivative at the origin. The resulting tangent–transverse decomposition is the vector-gain specialization of [7, Eq. (9)].

LEMMA 3.1 (tangent–transverse spectrum splitting). *Set* $J_p := J_x(\mathbf{p})|_{\mathbf{0}} \in \mathbb{R}^{n\times n}$, $R := J_y(\mathbf{q}_1)|_{(\mathbf{0},\mathbf{0})} \in \mathbb{R}^{n\times m}$, *and* $W := J_y(\mathbf{q}_2)|_{(\mathbf{0},\mathbf{0})} \in \mathbb{R}^{m\times m}$. *Then, for every* $\Lambda \in \mathbb{R}^m$,

$$A(\Lambda) = \begin{bmatrix} J_p & R \\ 0 & D(\Lambda) \end{bmatrix}, \qquad D(\Lambda) := W - \operatorname{diag}(\Lambda). \tag{3.4}$$

Consequently,

$$\operatorname{eig}(A(\Lambda)) = \operatorname{eig}(J_p) \cup \operatorname{eig}(D(\Lambda)), \tag{3.5}$$

with algebraic multiplicities. Under the standing assumption $\alpha_p > 0$, $A(\Lambda)$ *is Hurwitz if and only if* $D(\Lambda)$ *is Hurwitz. Moreover, its spectral margin satisfies*

$$\alpha(\Lambda) := -\operatorname{Re}\lambda_1(A(\Lambda)) = \min\{\alpha_p, \alpha_{\mathrm{aux}}(\Lambda)\}, \qquad \alpha_{\mathrm{aux}}(\Lambda) := -\operatorname{Re}\lambda_1(D(\Lambda)). \tag{3.6}$$

When $m = 0$, $A = J_p$ *and we set* $\alpha_{\mathrm{aux}} = +\infty$. *Thus the gains leave the original spectrum fixed and cannot raise the lifted margin above* α_p.

Proof. See the proof of Lemma 3.1 in Appendix B.2 (p. 30). ∎

COROLLARY 3.2 (closed-form transverse feasibility and saturation). *Assume* $m \geq 1$ *and let* $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^m$. *For every* $\Lambda \in \mathbb{R}^m$ *and* $c \in \mathbb{R}$,

$$D(\Lambda + c\mathbf{1}) = D(\Lambda) - cI_m, \qquad \alpha_{\mathrm{aux}}(\Lambda + c\mathbf{1}) = \alpha_{\mathrm{aux}}(\Lambda) + c. \tag{3.7}$$

In particular, $\alpha_{\mathrm{aux}}(\lambda\mathbf{1}) = \lambda - \operatorname{Re}\lambda_1(W)$, *so for every* $\varepsilon > 0$,

$$\alpha_{\mathrm{aux}}(\lambda\mathbf{1}) \geq \varepsilon \quad \Longleftrightarrow \quad \lambda \geq \operatorname{Re}\lambda_1(W) + \varepsilon. \tag{3.8}$$

The full lifted margin satisfies $\alpha(\lambda\mathbf{1}) \geq \varepsilon$ *if and only if both* $\lambda \geq \operatorname{Re}\lambda_1(W) + \varepsilon$ *and* $\alpha_p \geq \varepsilon$ *hold. Under the standing assumption* $\alpha_p > 0$, *the unrestricted set* $\{\Lambda \in \mathbb{R}^m : A(\Lambda)$ *is Hurwitz*$\}$ *is therefore nonempty, and*

$$\alpha(\lambda\mathbf{1}) = \alpha_p \qquad \text{whenever} \qquad \lambda \geq \operatorname{Re}\lambda_1(W) + \alpha_p. \tag{3.9}$$

Let $\Delta H := \sum_{i=1}^m \mathbf{e}_{n+i} s_i^\top$. *Then* $H(\lambda\mathbf{1}) = H_0 + \lambda\Delta H$ *and* $\|H(\lambda\mathbf{1})\| = O(\lambda)$ *as* $\lambda \to \infty$; *if* $\Delta H \neq 0$, *then* $\|H(\lambda\mathbf{1})\|/\lambda \to \|\Delta H\|$.

Proof. See the proof of Corollary 3.2 in Appendix B.2 (p. 31). ∎

The bound (3.8) provides an explicit uniform-gain initializer whenever the resulting vector lies in the prescribed gain box. Although the unrestricted family is feasible, its intersection with a prescribed box can be empty.

Thus feasibility alone does not select a useful gain: once the lifted margin reaches α_p, increasing λ continues to change the quadratic term without improving the linear margin.

Remark 3.3 (Factorization freedom). The stabilizers of a fixed monomial lift need not be unique. Different factorizations into earlier coordinates give stabilizers with the same linear part, all vanishing on $\mathcal{M}$. For example, for $\Phi(x) = (x, x^2, x^3, x^4)$, both $h_3 = y_3 - y_1^2$ and $\widetilde{h}_3 = y_3 - xy_2$ are valid, with $h_3 - \widetilde{h}_3 = xy_2 - y_1^2$. With the reference extension and Λ fixed, changing the factorization can alter $H(\Lambda)$ while leaving $A(\Lambda)$ unchanged. When $A(\Lambda)$ is Hurwitz, its Lyapunov solution is therefore unchanged as well. We keep the stabilizer factorizations fixed throughout the optimization.

After fixing Λ, a separate freedom remains in representing the same quadratic map. Let $\mathcal{K} \subset \mathbb{R}^{N\times N^2}$ be the space of matrices satisfying $K(\mathbf{w}\otimes\mathbf{w}) = 0$ for every $\mathbf{w}\in\mathbb{R}^N$. Two matrices represent the same quadratic map if and only if their difference belongs to $\mathcal{K}$. Thus $H(\Lambda)+\mathcal{K}$ contains all Kronecker representations of the quadratic term of $\mathbf{F}_\Lambda$. The space has dimension $d = \frac{1}{2}N^2(N-1)$. See the representation-space derivation in Appendix B.3 (p. 31). For a fixed basis $\{K_k\}_{k=1}^d$, write

$$H(\Lambda,\mu) := H(\Lambda) + K(\mu), \qquad K(\mu) := \sum_{k=1}^{d} \mu_k K_k, \qquad \mu\in\mathbb{R}^d. \tag{3.10}$$

Unlike the extension directions in $\mathcal{V}_\Phi$, which need only vanish on $\mathcal{M}$, the gauge contributes the zero vector field on all of $\mathbb{R}^N$. Varying μ therefore leaves $\mathbf{F}_\Lambda$, $A(\Lambda)$, and every trajectory unchanged, while it can change the matrix norm used to bound Lyapunov growth. We optimize over all $\mu\in\mathbb{R}^d$ in this exact representation class. The certificate construction below specifies which norm is minimized.

3.2. The certificate and the optimization problem. Fix an extension $\mathbf{F}_\Lambda$ with $A(\Lambda)$ Hurwitz. Its certificate is constructed from the quadratic Lyapunov function of Section 2.4 and the spectral-norm bound below. The choice $Q = I_N$ fixes the Lyapunov weight while the gains and representation vary. The construction is the $Q = I_N$ specialization of [16, Thm. 4.1 and Eqs. (4.8)–(4.12)], building on [32]. Let $P = P(\Lambda) \succ 0$ solve

$$A(\Lambda)^\top P + PA(\Lambda) = -I_N, \tag{3.11}$$

and factor $P = P_f^\top P_f$. Since A is gauge-independent, P depends on Λ but not on μ.

Remark 3.4 (Choice of Lyapunov weight). We fix $Q = I_N$ to isolate the lift and gauge design. A scalar rescaling is redundant: replacing Q by sQ replaces P by sP and only rescales the level parameter of the same ellipsoids. The normalized shape of Q is a genuine design variable. For $Q = Q_f^\top Q_f$, the general certificate replaces J below by $J_Q = (P_f^\top\otimes Q_f^\top)^{-1} G Q_f^{-1}$; it could be optimized over, for example, $Q\succeq\varepsilon I_N$ and $\operatorname{tr} Q = N$. This adds a nonconvex outer variable and is left open.

Partition $H(\Lambda,\mu)$ into blocks $M_i = M_i(\Lambda,\mu)\in\mathbb{R}^{N\times N}$ defined by $(M_i)_{rj} := (H(\Lambda,\mu))_{r,(i-1)N+j}$, so that $H(\Lambda,\mu)(\mathbf{w}\otimes\mathbf{w}) = \sum_{i=1}^N w_i M_i \mathbf{w}$. With col denoting vertical stacking, define

$$\begin{aligned} G(\Lambda,\mu) &:= \operatorname{col}_{i=1}^N \big(M_i^\top P + PM_i\big) \in \mathbb{R}^{N^2\times N},\\ J(\Lambda,\mu) &:= \big(P_f^\top\otimes I_N\big)^{-1} G(\Lambda,\mu) \in \mathbb{R}^{N^2\times N}. \end{aligned} \tag{3.12}$$

The spectral norm of J is independent of the factor: replacing P_f by UP_f, with U orthogonal, replaces J by $(U\otimes I_N)J$. For $V(\mathbf{w}) = \mathbf{w}^\top P\mathbf{w}$,

$$\dot V(\mathbf{w}) = \mathbf{w}^\top\big[-I_N + (\mathbf{w}^\top P_f^\top\otimes I_N)J(\Lambda,\mu)\big]\mathbf{w} \le \|\mathbf{w}\|^2\left[-1+\sqrt{V(\mathbf{w})}\,\|J(\Lambda,\mu)\|\right] \tag{3.13}$$

Hence $\dot V(\mathbf{w}) < 0$ for every nonzero $\mathbf{w}$ with $V(\mathbf{w})\le\rho^2$ whenever $0<\rho<1/\|J(\Lambda,\mu)\|$, with $1/0 := +\infty$; Proposition 2.7 then certifies the corresponding closed ellipsoid.

PROPOSITION 3.5 (Intrinsic growth and spectral-norm relaxation). *Fix symmetric positive definite P, Q and a homogeneous quadratic map $\mathbf{q}(\mathbf{w}) = H(\mathbf{w} \otimes \mathbf{w})$. For each representation $H + K(\mu)$ in (3.10), form G by (3.12) and J_Q by Remark 3.4. Then*

$$\beta(P, Q, \mathbf{q}) \le \gamma_\star := \min_{\mu \in \mathbb{R}^d} \|J_Q(H + K(\mu))\| \le \|J_Q(H + K(\mu))\| \qquad (\mu \in \mathbb{R}^d). \tag{3.14}$$

The minimum is attained. It is the smallest spectral-norm upper bound in this gauge and matricization class; gauge optimality does not establish equality with β.

Proof. See the proof of Proposition 3.5 in Appendix B.4 (p. 31). ∎

For a fixed extension, β and $P(\Lambda)$ are fixed, so the inner optimization improves the spectral-norm upper bound alone. We define its optimized threshold by

$$\rho^\star(\Lambda) := \left(\min_{\mu \in \mathbb{R}^d} \|J(\Lambda, \mu)\| \right)^{-1}. \tag{3.15}$$

For fixed Λ, the certificate map is affine in μ: $J(\Lambda, \mu) = J(\Lambda, 0) + \sum_{k=1}^d \mu_k J_k(\Lambda)$, which implies $\mu \mapsto \|J(\Lambda, \mu)\|$ is convex. Its minimum is attained because $\operatorname{span}\{J_k(\Lambda)\}_{k=1}^d$ is a closed finite-dimensional subspace. Every μ already provides a valid threshold; minimizing the norm only enlarges it. Consequently, if $A(\Lambda)$ is Hurwitz, then for every $0 < \rho < \rho^\star(\Lambda)$, $\mathcal{D}(P(\Lambda), \rho)$ is contained in the ROA of the lifted system. The value $\rho^\star(\Lambda)$ is the best threshold furnished by this norm bound and satisfies $\rho^\star(\Lambda) \le \rho_{\mathrm{rad}}$; equality need not hold. If the minimum in (3.15) is zero, we set $\rho^\star(\Lambda) = +\infty$; an optimizing gauge then has $J = 0$, and (3.13) gives $\dot{V} = -\|\mathbf{w}\|^2$, so every finite ρ is certified. Although $H(\Lambda)$ is affine in Λ, the dependence across gains is generally nonconvex because $P(\Lambda)$ and $P_f(\Lambda)$ are obtained from the Lyapunov equation (3.11).

The lifted certificate pulls back to the original coordinates through $\Phi(\mathbf{x}) := (\mathbf{x}, \mathbf{g}(\mathbf{x}))$. Define

$$S(\Lambda, \rho) := \left\{ \mathbf{x} \in \mathbb{R}^n : \widetilde{V}_\Lambda(\mathbf{x}) \le \rho^2 \right\}, \quad \widetilde{V}_\Lambda(\mathbf{x}) := \Phi(\mathbf{x})^\top P(\Lambda) \Phi(\mathbf{x}). \tag{3.16}$$

For $0 < \rho < \rho^\star(\Lambda)$, exactness and uniqueness imply that the lifted trajectory from $\Phi(\mathbf{x}_0)$ is $\Phi(\mathbf{x}(t))$ and remains on $\mathcal{M}$. Since the lifted ellipsoid is positively invariant and contained in the lifted ROA, this trajectory is global and converges to the origin. Hence $S(\Lambda, \rho)$ is positively invariant and contained in the ROA of (2.1).

COROLLARY 3.6 (Path-connected pullback). *If $A(\Lambda)$ is Hurwitz and $0 < \rho < \rho^\star(\Lambda)$, then $S(\Lambda, \rho)$ is path-connected; its connected component containing the origin is therefore the entire set.*

Proof. See the proof of Corollary 3.6 in Appendix B.5 (p. 31). ∎

After optimizing the representation at each gain, the remaining choice is the extension $\mathbf{F}_\Lambda$. It changes both $P(\Lambda)$ and $\rho^\star(\Lambda)$, and hence the pullback certificate (3.16), while preserving the original dynamics on $\mathcal{M}$.

By Lemma 3.1, feasibility reduces to the Hurwitz condition on the auxiliary block $D(\Lambda)$. An unrestricted maximizing gain need not exist: Example 3.11 below gives a member of the present stabilizer family for which $\rho^\star(\lambda) \to +\infty$. We therefore include the gain domain, which is also the domain used by the implementation, in the problem data.

DEFINITION 3.7 (admissible gain box). *An* admissible gain box *is a nonempty box $B = [\underline{\Lambda}, \overline{\Lambda}] \subset \mathbb{R}^m$ contained in the Hurwitz set $\mathcal{F} := \{\Lambda : A(\Lambda)$ is Hurwitz$\}$.*

Equivalently, compactness gives a uniform margin $\gamma_B := \min_{\Lambda\in B} -\operatorname{Re}\lambda_1(A(\Lambda)) > 0$. *A common Lyapunov inequality checked at all vertices is a sufficient, implementation-verifiable certificate that* $B \subset \mathcal{F}$ *because* A *is affine in* Λ. *Since* $\mathcal{F}$ *is open and nonempty, every* $\Lambda^0 \in \mathcal{F}$ *lies in the interior of some nondegenerate closed box* $B \Subset \mathcal{F}$.

PROBLEM 3.8 (box-constrained certificate optimization). *Given the fixed polynomial system* $\mathbf{p}$, *embedding* Φ, *weight* $Q = I_N$, *and lift data* $(A_0, H_0, \{h_i\}_{i=1}^m)$, *the derived quantities* $(W, \{s_i\}_{i=1}^m, \{K_k\}_{k=1}^d)$, *and an admissible gain box* B, *solve*

$$\boxed{\phi_B := \min_{\Lambda\in B} \phi(\Lambda), \qquad \phi(\Lambda) := \min_{\mu\in\mathbb{R}^d} \|J(\Lambda,\mu)\|, \qquad \rho_B := \phi_B^{-1},} \tag{3.17}$$

with the convention $1/0 := +\infty$. *The map* $J(\Lambda,\mu)$ *is assembled from* (3.3), (3.10), (3.11), *and* (3.12). *Equivalently, using the definitions above, an expanded formulation of* (3.17) *is*

$$\begin{aligned}
\min_{\substack{\Lambda\in B,\ \mu\in\mathbb{R}^d,\\ P\in\mathbb{S}^N,\ t\in\mathbb{R}}} \quad & t \\
\text{s.t.} \quad & \begin{bmatrix} tI_{N^2} & J \\ J^\top & tI_N \end{bmatrix} \succeq 0, \\
& J = \left(P_f^\top \otimes I_N\right)^{-1} G(\Lambda,\mu), \qquad P = P_f^\top P_f, \\
& A(\Lambda)^\top P + PA(\Lambda) = -I_N, \qquad P \succ 0, \\
& A(\Lambda) = A_0 - \sum_{i=1}^m \lambda_i\, \mathbf{e}_{n+i}\mathbf{e}_{n+i}^\top, \\
& H(\Lambda,\mu) = H_0 + \sum_{i=1}^m \lambda_i\, \mathbf{e}_{n+i} s_i^\top + \sum_{k=1}^d \mu_k K_k.
\end{aligned} \tag{3.18}$$

Its optimal value is $t_B = \phi_B$. *Proposition 3.10 below justifies attainment. A closed certified set is reported with the safe radius of Corollary 3.12, never with the open threshold* ρ_B *itself.*

See the derivation of the box-constrained formulation in Appendix B.6 (p. 32), including the exact spectral-norm epigraph.

Remark 3.9. The optimization variables are $\Lambda \in B$ and $\mu \in \mathbb{R}^d$. The embedding, reference extension, stabilizer factorizations, Lyapunov weight, and gain box are fixed problem data. Thus optimality in Problem 3.8 refers to the stated spectral-norm certificate within this family and box. The admissible-box condition guarantees a unique symmetric solution $P(\Lambda) \succ 0$ of (3.11) at every outer point. Thus P is not a free design variable; it appears in (3.18) only to make the dependence on Λ explicit. After eliminating P, the outer dependence of $J(\Lambda,\mu)$ on Λ combines the affine map $H(\Lambda)$ with the nonlinear Lyapunov–Cholesky map $\Lambda \mapsto P(\Lambda) \mapsto P_f(\Lambda)$. For fixed Λ, the problem in (μ, t) is an exact convex SDP; the box-constrained outer problem is generally nonsmooth and nonconvex. The objective optimizes the Lyapunov level threshold. Since $P(\Lambda)$ also changes with the gain, it does not by itself order the volumes or set inclusions of pullbacks obtained at different gains.

The next result removes gauge directions invisible to the certificate before applying Weierstrass' theorem.

PROPOSITION 3.10 (well-posed box value). *The value function ϕ is locally Lipschitz on $\mathcal{F}$. Consequently, it attains a global minimum on every admissible gain box B. If $\phi_B > 0$, then $\rho^\star = 1/\phi$ attains its box maximum at exactly the same gains. If $\phi_B = 0$, an optimizing pair satisfies $J = 0$ and the corresponding quadratic Lyapunov function is global.*

Proof. See the proof of Proposition 3.10 in Appendix B.7 (p. 32). ∎

Example 3.11 (unbounded spectral-norm threshold). For $\dot{x} = -x - x^3$ with lift $y = x^2$, consider the stabilizer family

$$\dot{x} = -x - xy, \qquad \dot{y} = -(2+\lambda)y + \lambda x^2 - 2y^2, \qquad \lambda > -2. \tag{3.19}$$

Put $s := \lambda + 2 > 0$. Then $P = \mathrm{diag}(1/2, 1/(2s))$ and $\dot{V} = -(x^2+y^2)(1+2y/s)$. A gauge whose two blocks in G are 0 and $-2I_2/s$ gives $\phi(\lambda) \le 2\sqrt{2/s}$. Taking $x = 0$ and $y < 0$, the cubic contribution is $2|y|^3/s$, while (3.13) bounds it above by $|y|^3\|J\|/\sqrt{2s}$. Hence every gauge satisfies $\|J\| \ge 2\sqrt{2/s}$, so $\phi(\lambda) = 2\sqrt{2/s}$ and $\rho^\star(\lambda) = \sqrt{s}/(2\sqrt{2}) \to +\infty$. Thus the unrestricted problem need not have a maximizing gain; a boundary box optimum should be checked by widening B.

COROLLARY 3.12 (Numerical certificate). *Fix a gain, a gauge satisfying $K(\mathbf{w} \otimes \mathbf{w}) \equiv 0$, and $\widehat{P} = \widehat{P}^\top \succ 0$. Assemble $\widehat{J}$ from $\widehat{P}$ by* (3.12). *Suppose verified bounds satisfy*

$$\|A(\Lambda)^\top \widehat{P} + \widehat{P}A(\Lambda) + I_N\| \le \widehat{\delta} < 1, \qquad \|\widehat{J}\| \le \widehat{j}.$$

For $\eta \in (0,1)$ and $\widehat{j} > 0$, the ellipsoid $\mathcal{D}(\widehat{P}, \rho_{\mathrm{safe}})$ with $\rho_{\mathrm{safe}} := (1-\eta)(1-\widehat{\delta})/\widehat{j}$ is positively invariant and contained in the lifted ROA; its pullback under Φ is contained in the ROA of (2.1). *If $\widehat{j} = 0$ is verified exactly, $\mathbf{w}^\top \widehat{P} \mathbf{w}$ is a global Lyapunov function.*

Proof. Proposition 3.5 gives $\beta(\widehat{P}, I_N, \mathbf{q}) \le \|\widehat{J}\| \le \widehat{j}$. Apply Proposition 2.7 with $Q = I_N$, $\theta = 1 - \eta$, and the verified upper bounds $\widehat{\delta}, \widehat{j}$. Exactness of the lift gives the pullback claim. ∎

For numerical use, symmetrize the returned P, verify positive definiteness, and bound the normalized residual and the norm of the reassembled matrix. The SDP epigraph variable alone does not bound the returned matrix. Check the gauge identity coefficientwise; any nonzero defect must be included in the nonlinear growth bound using (B.3) in Appendix B.8. Verified bounds must account for rounding errors in assembly and norm evaluation. Certificate validity requires these bounds, independently of solver optimality.

Problem 3.8 reduces to the fixed-pair convex construction in [16, §4] when $m = 0$ or Λ is fixed, since only the gauge μ remains. By contrast, the doubling rule in [7, Alg. 2] finds a feasible scalar gain but optimizes neither the gauge nor the certificate over B.

Remark 3.13 (Multiple equilibria). To preserve dissipativity at several equilibria $x_1^*, \dots, x_\ell^*$ simultaneously, as in [7], impose $D_j(\Lambda)$ Hurwitz for every j, where

$$D_j(\Lambda) = W_j - \mathrm{diag}(\Lambda) L_j, \qquad W_j = J_y \mathbf{q}_2(x_j^*, \mathbf{g}(x_j^*)) - J_x \mathbf{g}(x_j^*) J_y \mathbf{q}_1(x_j^*, \mathbf{g}(x_j^*)),$$

and $L_j = J_y \mathbf{h}(x_j^*, \mathbf{g}(x_j^*))$. The second term in W_j vanishes at the standing origin because $J_x \mathbf{g}(\mathbf{0}) = 0$, but it must be retained at a shifted equilibrium. The certificate itself remains anchored at one equilibrium.

3.3. The certified volume at leading order. Problem 3.8 maximizes the lifted level radius $\rho^\star(\Lambda)$, whereas the geometric quantity of interest is the volume of its pullback $S(\Lambda,\rho)$. These objectives need not agree a priori because the metric $P(\Lambda)$ varies with the gain and Φ is nonlinear. They do agree to leading order near the origin. Partition P conformally with $\mathbf{w}=(\mathbf{x},\mathbf{y})$ as

$$P(\Lambda)=\begin{bmatrix} P_{xx}(\Lambda) & P_{xy}(\Lambda)\\ P_{xy}(\Lambda)^\top & P_{yy}(\Lambda)\end{bmatrix}, \qquad P_{xx}(\Lambda)\in\mathbb{S}^n. \tag{3.20}$$

Since every auxiliary monomial has degree at least two, $\Phi(\mathbf{x})=(\mathbf{x},O(\|\mathbf{x}\|^2))$, and hence

$$\widetilde{V}_\Lambda(\mathbf{x})=\mathbf{x}^\top P_{xx}(\Lambda)\mathbf{x}+O(\|\mathbf{x}\|^3). \tag{3.21}$$

Proposition 3.14 (Leading-order certified volume). *For every feasible Λ, as $\rho\downarrow 0$,*

$$\operatorname{vol} S(\Lambda,\rho)=\kappa_n\frac{\rho^n}{\sqrt{\det P_{xx}(\Lambda)}}+o(\rho^n), \qquad \kappa_n:=\frac{\pi^{n/2}}{\Gamma(n/2+1)}. \tag{3.22}$$

Proof. A two-sided quadratic sandwich gives the result; see the proof of Proposition 3.14 in Appendix B.9 (p. 33). ∎

If $\phi_B>0$, all thresholds on B are finite, and the logarithm of the leading coefficient at the scaled level $\rho=\varepsilon\rho^\star(\Lambda)$ suggests the score

$$n\log\rho^\star(\Lambda)-\tfrac{1}{2}\log\det P_{xx}(\Lambda). \tag{3.23}$$

The apparent correction is constant across gains.

Proposition 3.15 (Gain invariance of the quadratic germ). *Fix $Q\in\mathbb{S}^N$, $Q\succ 0$, and let $P_Q(\Lambda)\succ 0$ solve $A(\Lambda)^\top P_Q+P_QA(\Lambda)=-Q$. Then, for every feasible Λ, its physical block satisfies*

$$P_{Q,xx}(\Lambda)=\bar{P}_Q, \qquad J_p^\top\bar{P}_Q+\bar{P}_QJ_p=-Q_{xx},$$

where Q_{xx} is the physical block of Q. Thus $\bar{P}_Q$ is the unique solution and is independent of Λ; in particular, P_{xx} in (3.20) equals the gain-invariant matrix $\bar{P}_I$ obtained for $Q=I_N$.

Proof. See the proof of Proposition 3.15 in Appendix B.10 (p. 33). ∎

Corollary 3.16 (Radius and leading-order volume). *If $\phi_B>0$, then, for every $\Lambda\in B$ and as $\varepsilon\downarrow 0$,*

$$\operatorname{vol} S\big(\Lambda,\varepsilon\rho^\star(\Lambda)\big)=\frac{\kappa_n\big(\rho^\star(\Lambda)\big)^n}{\sqrt{\det\bar{P}_I}}\,\varepsilon^n+o(\varepsilon^n).$$

Hence Problem 3.8 ranks gains exactly by the leading coefficients of these scaled certified volumes. If $\phi_B=0$, Proposition 3.10 instead gives an optimizing gain with a global quadratic Lyapunov certificate.

Thus, for the fixed weight $Q=I_N$, the gain affects the leading-order volume only through the achievable radius, not through the local shape factor. This is a local statement: finite-radius volumes and set inclusions can still depend on the higher-order lift and on P_{xy} and P_{yy}. For $Q=I_N$, $\bar{P}_I$ is precisely the Lyapunov matrix of the plant linearization, so all lifts share the plant's quadratic germ. Within this family, changing that germ requires varying the normalized weight Q_{xx}, as anticipated in Remark 3.4.

4. Solving the fixed-family certificate problem. Problem 3.8 has one convex and one nonconvex level. For fixed Λ, the Lyapunov equation fixes $P(\Lambda)$ and the gauge is optimized by an exact SDP; the remaining box-constrained value function ϕ is locally Lipschitz but generally nonsmooth and nonconvex. This section develops the corresponding value-and-derivative oracle and a gradient-sampling L-BFGS method. The lift, stabilizer factorizations, and admissible box are fixed throughout.

4.1. The exact reduced inner oracle. Fix $\Lambda \in \mathcal{F}$ and put $J_0 := J(\Lambda, 0)$. The linear certificate gauge map and its image are $\mathcal{T}_\Lambda \mu := J(\Lambda, \mu) - J_0$ and $\mathcal{W}_\Lambda := \operatorname{im} \mathcal{T}_\Lambda \subset \mathbb{R}^{N^2 \times N}$. Write $\langle U, V \rangle_F := \operatorname{tr}(U^\top V)$ and let $\|Z\|_*$ denote the nuclear norm, the sum of the singular values of Z.

PROPOSITION 4.1 (Gauge-image distance and duality). *For every fixed $\Lambda \in \mathcal{F}$,*

$$\phi(\Lambda) = \min_{W \in \mathcal{W}_\Lambda} \|J_0 + W\| = \operatorname{dist}_{\|\cdot\|}(J_0, \mathcal{W}_\Lambda), \tag{4.1}$$

where the distance is induced by the spectral norm. Its dual is

$$\phi(\Lambda) = \max_{\substack{Z \in \mathbb{R}^{N^2 \times N} \\ \|Z\|_* \le 1,\ Z \in \mathcal{W}_\Lambda^\perp}} \langle Z, J_0 \rangle_F. \tag{4.2}$$

Here $Z \in \mathcal{W}_\Lambda^\perp$ is equivalent to $\langle Z, \mathcal{T}_\Lambda \mathbf{e}_k \rangle_F = 0$ for $k = 1, \dots, d$. Both optima are attained. A matrix $J^\star \in J_0 + \mathcal{W}_\Lambda$ is optimal if and only if there exists $Z^\star \in \mathcal{W}_\Lambda^\perp$ with $\|Z^\star\|_* \le 1$ and $\langle Z^\star, J^\star \rangle_F = \|J^\star\|$; equivalently, $\partial \|\cdot\|(J^\star) \cap \mathcal{W}_\Lambda^\perp \neq \varnothing$. For any gauge μ and any dual-feasible Z,

$$0 \le \|J(\Lambda, \mu)\| - \phi(\Lambda) \le \|J(\Lambda, \mu)\| - \langle Z, J_0 \rangle_F. \tag{4.3}$$

Thus a zero primal–dual gap certifies global inner optimality.

Proof. See the proof of Proposition 4.1 in Appendix C.1 (p. 34). ∎

This formulation depends on the gauge image, not its basis; it still optimizes a spectral-norm upper bound on β. The distance dual and gap bound are standard convex norm duality. The quadratization-specific step is identifying $\mathcal{W}_\Lambda$ and removing its redundant gauge coordinates.

The proof of Proposition 3.10 identifies the P-independent space $\mathcal{C}$ seen by the certificate. Fix a basis $(B_1^{(\ell)}, \dots, B_N^{(\ell)})$, $\ell = 1, \dots, q$, of $\mathcal{C}$, where $q = N(N^2 - 1)/3$, and define

$$\begin{aligned} \mathcal{B}_\ell &:= \operatorname{col}_{r=1}^N B_r^{(\ell)}, & X(\Lambda) &:= P_f(\Lambda)^\top \otimes I_N, \\ G_{\mathrm{r}}(\Lambda, \nu) &:= G(\Lambda, 0) + \sum_{\ell=1}^q \nu_\ell \mathcal{B}_\ell, & J_{\mathrm{r}}(\Lambda, \nu) &:= X(\Lambda)^{-1} G_{\mathrm{r}}(\Lambda, \nu). \end{aligned} \tag{4.4}$$

The basis matrices $\mathcal{B}_\ell$ are independent of Λ and are computed once. Their images $J_\ell(\Lambda) := X(\Lambda)^{-1} \mathcal{B}_\ell$ form a basis of $\mathcal{W}_\Lambda$, so the reduced coordinates describe the same distance problem.

PROPOSITION 4.2 (Exact reduced inner oracle). *For every $\Lambda \in \mathcal{F}$,*

$$\boxed{\begin{aligned} \phi(\Lambda)^2 = \tau^\star(\Lambda) := \min_{\nu \in \mathbb{R}^q,\, \tau \in \mathbb{R}} \ & \tau \\ \text{s.t.} \quad & F_\Lambda(\nu, \tau) := \begin{bmatrix} P(\Lambda) \otimes I_N & G_{\mathrm{r}}(\Lambda, \nu) \\ G_{\mathrm{r}}(\Lambda, \nu)^\top & \tau I_N \end{bmatrix} \succeq 0. \end{aligned}} \tag{4.5}$$

The problem is strictly feasible and attains its optimum. If $\nu^\star$ is returned, a raw gauge representative is recovered from the minimum-norm linear problem

$$\mu^\star = \operatorname*{arg\,min}_{\mu\in\mathbb{R}^d} \ \tfrac{1}{2}\|\mu\|^2 \quad \text{s.t.} \quad \operatorname{col}_{r=1}^{N}\big(S_r(\mu)^\top P + PS_r(\mu)\big) = \sum_{\ell=1}^{q} \nu_\ell^\star \mathcal{B}_\ell, \tag{4.6}$$

where $S_r(\mu)$ are the slices of $K(\mu)$ in (3.10)*. This reconstruction leaves the quadratic vector field and the value $\phi(\Lambda)$ unchanged.*

Proof. Since $P\otimes I_N = XX^\top$, the Schur complement of (4.5) is $G_{\mathrm{r}}^\top (P\otimes I_N)^{-1} G_{\mathrm{r}} \preceq \tau I_N$, equivalently $J_{\mathrm{r}}^\top J_{\mathrm{r}} \preceq \tau I_N$. Thus the reduced generalized-Schur formulation is exactly the fixed-Λ spectral-norm problem, without placing X^{-1} in the cone model. For the full argument, including attainment and reconstruction, see the proof of Proposition 4.2 in Appendix C.2 (p. 34). ∎

4.2. Inner optimality and value-function sensitivities. Let $(\nu^\star, \tau^\star)$ solve (4.5) and put $J^\star := J_{\mathrm{r}}(\Lambda, \nu^\star)$. If $J^\star \neq 0$ and U_1, V_1 contain paired orthonormal leading singular vectors, so that $J^\star V_1 = \phi(\Lambda) U_1$, then [34]

$$\partial\|\cdot\|(J^\star) = \left\{U_1 \Theta V_1^\top : \Theta \succeq 0,\ \operatorname{tr}\Theta = 1\right\}; \tag{4.7}$$

at $J^\star = 0$ it is the nuclear-norm unit ball. Convex optimality of the inner problem is therefore equivalent to the existence of $Z^\star \in \partial\|\cdot\|(J^\star)$ satisfying

$$\langle Z^\star, J_\ell(\Lambda)\rangle_F = 0, \qquad \ell = 1, \dots, q, \tag{4.8}$$

with the condition vacuous when $q = 0$.

The gain derivatives can be evaluated without finite differences. Put $E_i := \mathbf{e}_{n+i}\mathbf{e}_{n+i}^\top$ and use a dot for ∂_{λ_i} while holding $\nu = \nu^\star$ fixed. From (3.3) and (3.11),

$$\begin{aligned} \dot{A}_i &= -E_i, & \dot{H}_i &= \mathbf{e}_{n+i} s_i^\top, \\ A(\Lambda)^\top \dot{P}_i + \dot{P}_i A(\Lambda) &= E_i P + P E_i, & \dot{P}_i &= \dot{P}_{f,i}^\top P_f + P_f^\top \dot{P}_{f,i}. \end{aligned} \tag{4.9}$$

Let M_r^0 be the rth slice of $H(\Lambda)$ and $\dot{M}_{r,i}$ the corresponding slice of $\dot{H}_i$. Differentiating (4.4) gives

$$\begin{aligned} \dot{G}_i^0 &= \operatorname{col}_{r=1}^{N}\left(\dot{M}_{r,i}^\top P + (M_r^0)^\top \dot{P}_i + \dot{P}_i M_r^0 + P\dot{M}_{r,i}\right), \\ \dot{J}_i &= X^{-1}\left[\dot{G}_i^0 - (\dot{P}_{f,i}^\top \otimes I_N) J^\star\right]. \end{aligned} \tag{4.10}$$

The superscript in G_i^0 emphasizes that the fixed reduced directions $\mathcal{B}_\ell$ have zero gain derivative.

Proposition 4.3 (Regularity and envelope sensitivities). *Let $\mathcal{U} \subset \mathcal{F}$ be open and let A, H be the affine maps in* (3.3)*. For any fixed $Q \in \mathbb{S}^N$, $Q \succ 0$, the solution $P_Q(\Lambda) \succ 0$ of $A^\top P_Q + P_Q A = -Q$ is real analytic on $\mathcal{U}$, and*

$$A^\top P_{Q,i} + P_{Q,i} A = -\left(A_i^\top P_Q + P_Q A_i\right), \qquad P_{Q,i} := \partial_{\lambda_i} P_Q, \quad A_i := \partial_{\lambda_i} A. \tag{4.11}$$

For the fixed weight $Q = I_N$ in Problem 3.8, $P = P_{I_N}$, its positive-diagonal Cholesky factor P_f, and J_{r} depend real analytically on the gain. The value ϕ is locally Lipschitz, hence continuous, on $\mathcal{U}$, and its reduced inner minimizers are locally uniformly bounded.

Fix $\Lambda \in \mathcal{U}$, let $(\nu^\star, \tau^\star)$ be a primal optimum and $Y^\star \succeq 0$ a dual optimum of (4.5), and put $J^\star = J_{\mathrm{r}}(\Lambda, \nu^\star)$. Partition $Y^\star = (Y^\star_{ab})_{a,b=1}^2$ conformally with F_Λ. The KKT conditions include

$$\operatorname{tr} Y^\star_{22} = 1, \qquad \langle Y^\star_{12}, \mathcal{B}_\ell \rangle_F = 0 \ (\ell = 1, \dots, q), \qquad \langle Y^\star, F_\Lambda(\nu^\star, \tau^\star) \rangle_F = 0. \tag{4.12}$$

If $\phi(\Lambda) > 0$, then

$$Z^\star := -\frac{X^\top Y^\star_{12}}{\phi(\Lambda)} \in \partial \|\cdot\|(J^\star), \qquad \|Z^\star\|_* = 1, \tag{4.13}$$

and $Z^\star$ solves (4.2) and satisfies (4.8). The vector $g = (g_1, \dots, g_m)$ with

$$g_i = \left\langle Z^\star, \dot{J}_i \right\rangle_F = -\frac{\left\langle Y^\star_{11}, \dot{P}_i \otimes I_N \right\rangle_F + 2 \left\langle Y^\star_{12}, \dot{G}^0_i \right\rangle_F}{2\phi(\Lambda)} \tag{4.14}$$

belongs to $\partial_C \phi(\Lambda)$ and equals $\nabla \phi(\Lambda)$ wherever ϕ is differentiable. If, in addition, the reduced minimizer $\nu^\star$ is unique and the largest singular value of $J^\star$ is simple, with paired unit singular vectors u, v, then ϕ is differentiable at Λ and

$$\partial_{\lambda_i} \phi(\Lambda) = g_i = u^\top \dot{J}_i v, \qquad i = 1, \dots, m. \tag{4.15}$$

Proof. See the proof of Proposition 4.3 in Appendix C.3 (p. 35). ∎

The differentiability criterion concerns uniqueness of the reduced minimizer ν; raw gauge representatives can remain nonunique. The partial derivative in (4.15) holds $\nu = \nu^\star$ fixed. The Clarke inclusion uses the fixed reduced basis $\mathcal{B}_\ell$, which makes the conic dual feasible set independent of Λ; it requires neither a unique inner minimizer nor a simple leading singular value. At $\phi(\Lambda) = 0$, $Z = 0$ is a distance-dual optimum, but the normalization (4.13) is undefined. Approximate numerical multipliers remain sensitivity candidates unless the stated optimality conditions are established.

A single Clarke subgradient does not generally certify descent, so the outer method samples nearby ordinary gradients. Let $\mathcal{R} \subset \mathcal{U}$ be the set where ϕ is differentiable. For $B \Subset \mathcal{U} \Subset \mathcal{F}$, define

$$\partial_\epsilon \phi(\Lambda) := \operatorname{cl} \operatorname{co} \{\nabla \phi(z) : z \in \mathbb{B}_\epsilon(\Lambda) \cap \mathcal{U} \cap \mathcal{R}\}, \qquad \partial_C \phi(\Lambda) = \bigcap_{\epsilon > 0} \partial_\epsilon \phi(\Lambda). \tag{4.16}$$

The equality follows from local Lipschitz continuity and is the target approximated by a finite bundle of nearby ordinary gradients.

4.3. Box-constrained GS–L-BFGS-B. Fixed gain coordinates can be removed, so we assume here that B is nondegenerate. For conditioning, the implementation uses $D_B := \operatorname{diag}(\overline{\Lambda} - \underline{\Lambda})$, $\theta := D_B^{-1}(\Lambda - \underline{\Lambda}) \in C := [0,1]^m$, and $\psi(\theta) := \phi(\underline{\Lambda} + D_B \theta)$, so $\nabla \psi(\theta) = D_B^\top \nabla \phi(\Lambda)$. Let $\mathcal{U}_B := D_B^{-1}(\mathcal{U} - \underline{\Lambda})$. At θ^k, continuously sample

$$z_{kj} = \theta^k + \epsilon_k \xi_{kj}, \qquad \xi_{kj} \sim \mathrm{Unif}(\mathbb{B}_1), \qquad \mathcal{G}_k := \{g_{kj} = \nabla \psi(z_{kj})\}_{j=1}^{p_k}, \qquad p_k \geq m+1, \tag{4.17}$$

retaining only samples in $\mathcal{U}_B$ for which the derivative oracle is accepted. In the ideal algorithm below these are ordinary gradients; the finite implementation uses dual-envelope candidates that pass its numerical consistency checks. The points z_{kj} need not lie in C, and the deterministic center θ^k is not a bundle atom.

The L-BFGS memory supplies a Hessian metric. For a bundle-derived secant pair (s, r) and $M \succ 0$, Powell damping with $\kappa \in (0, 1)$ uses

$$\begin{aligned} \vartheta &= \begin{cases} 1, & s^\top r \geq \kappa s^\top M s, \\ \dfrac{(1-\kappa) s^\top M s}{s^\top M s - s^\top r}, & s^\top r < \kappa s^\top M s, \end{cases} \\ \widetilde{r} &= \vartheta r + (1-\vartheta) M s, \\ M^+ &= M - \frac{M s s^\top M}{s^\top M s} + \frac{\widetilde{r}\widetilde{r}^\top}{s^\top \widetilde{r}}. \end{aligned} \tag{4.18}$$

Here s is the most recent accepted displacement and r is the change from the preceding QP-dual aggregate to the next bundle mean. Only the most recent pairs are retained. Spectral clipping of the rebuilt matrix enforces $0 < \underline{b} I \preceq M_k \preceq \bar{b} I$.

For $x \in C$, use the convex-analysis convention $N_C(x) := \{n : n^\top (y - x) \leq 0 \text{ for all } y \in C\}$. Given $\mathcal{G}_k$ and M_k, the finite-displacement model is

$$\begin{aligned} \min_{d \in \mathbb{R}^m,\, \zeta \in \mathbb{R}} \quad & \zeta + \tfrac{1}{2} d^\top M_k d \\ \text{s.t.} \quad & g_{kj}^\top d \leq \zeta, \quad j = 1, \ldots, p_k, \\ & -\theta^k \leq d \leq \mathbf{1} - \theta^k. \end{aligned} \tag{4.19}$$

Lemma 4.4 (Bundle model and residual). *Problem* (4.19) *has a unique displacement* d_k. *There exist* $\alpha_k \in \mathbb{R}^{p_k}_+$, $\mathbf{1}^\top \alpha_k = 1$, $\bar{g}_k := \sum_j \alpha_{kj} g_{kj}$, *and* $n_k \in N_C(\theta^k + d_k)$ *such that*

$$M_k d_k + \bar{g}_k + n_k = 0, \qquad \zeta_k = \max_j g_{kj}^\top d_k = \bar{g}_k^\top d_k = -d_k^\top M_k d_k - n_k^\top d_k \leq -d_k^\top M_k d_k. \tag{4.20}$$

Consequently, with

$$\chi_k := \|d_k\|_{M_k} := (d_k^\top M_k d_k)^{1/2}, \tag{4.21}$$

one has $d_k = 0$ *if and only if* $0 \in \operatorname{co} \mathcal{G}_k + N_C(\theta^k)$; *if* $d_k \neq 0$, *every bundle atom satisfies* $g_{kj}^\top d_k \leq -\chi_k^2 < 0$.

Proof. Strong convexity gives uniqueness of d_k. The KKT multiplier of the epigraph inequalities lies in the simplex, complementarity gives $\zeta_k = \bar{g}_k^\top d_k$, and the box multipliers form n_k. Since the zero displacement is feasible, the normal-cone inequality gives $n_k^\top d_k \geq 0$, proving (4.20); setting $d_k = 0$ gives the stated equivalence. ∎

For $\chi_k > \omega_k$, define the first acceptable backtracking index and the next iterate by

$$\begin{aligned} \ell_k &:= \min\Big\{ \ell \in \{0, \ldots, L_k - 1\} : \psi(\theta^k + \beta^\ell d_k) \leq \psi(\theta^k) - c\beta^\ell \chi_k^2 \Big\}, \\ \theta^{k+1} &= \begin{cases} \theta^k + \beta^{\ell_k} d_k, & \ell_k \text{ exists}, \\ \theta^k, & \text{otherwise}. \end{cases} \end{aligned} \tag{4.22}$$

The endpoint bounds in (4.19) make every trial feasible. A missing ℓ_k is a null step, not a stationarity test.

Algorithm 4.1 Box-constrained GS–L-BFGS-B

Require: $\theta^0 \in C$, $c, \beta, \kappa \in (0,1)$, $p_k \geq m+1$, radii ϵ_k, tolerances ω_k, and caps L_k
Require: $0 < \underline{b} \leq \overline{b}$ and a finite L-BFGS memory cap
Evaluate the inner oracle at θ^0 and initialize the set of screened records $\mathcal{E}$.
for $k = 0, 1, \ldots$ **do**
  if $\psi(\theta^k) = 0$ is verified exactly **then**
    Return the corresponding global Lyapunov certificate.
  end if
  Sample (4.17) until $p_k \geq m+1$ accepted derivatives are obtained; retain every residual-valid in-box value record in $\mathcal{E}$.
  Update and safeguard M_k from the most recent serious-step secant pair; solve (4.19) and compute χ_k.
  if $\chi_k \leq \omega_k$ **then**
    Set $\theta^{k+1} \leftarrow \theta^k$, reduce (ϵ_k, ω_k), and reset the metric.
  else
    Apply (4.22); after a serious step retain its secant data, and after a null step reduce ϵ_k and reset the metric.
  end if
  Add every residual-valid in-box line-search record to $\mathcal{E}$.
end for
return the element of $\mathcal{E}$ with smallest recomputed spectral norm.

To make the anytime return rule precise, let $\mathcal{A}_K$ contain every completed value-oracle record through iteration K, including sampled points and rejected Armijo trials, and define

$$\mathcal{E}_K := \{a \in \mathcal{A}_K : \Lambda(a) \in B,\ a \text{ passes Corollary 3.12}\}, \quad \widehat{a}_K \in \operatorname*{arg\,min}_{a \in \mathcal{E}_K} \widehat{j}(a), \tag{4.23}$$

where $\widehat{j}(a)$ is the independently recomputed norm of J stored with the record. Writing $(\widehat{\Lambda}, \widehat{P}) := (\Lambda(\widehat{a}_K), P(\widehat{a}_K))$, the reported set is

$$S_{\mathrm{safe}} := \left\{ \mathbf{x} : \Phi(\mathbf{x})^\top \widehat{P} \Phi(\mathbf{x}) \leq \rho_{\mathrm{safe}}(\widehat{a}_K)^2 \right\}. \tag{4.24}$$

The record $\widehat{a}_K$ may differ from the iterate at which χ_k is reported; a sample with an unusable derivative can still enter $\mathcal{E}_K$ when its value and certificate pass their independent checks. If $\mathcal{E}_K$ is empty, the finite run reports failure rather than a certificate.

4.4. Cost and conditional guarantee. An uncached generic value-oracle call requires one $N \times N$ Lyapunov solve, one Cholesky factorization, and the SDP (4.5), which has $q+1 = O(N^3)$ scalar variables and a positive-semidefinite block of order $N^2 + N$. A direct full sensitivity evaluation adds m Lyapunov solves with the same coefficient operator. The bundle problem has $m+1$ variables and $p_k + 2m$ linear constraints. If an iteration dispatches n_k^{samp} gradient-sample attempts and performs n_k^{ls} line-search evaluations, it uses $n_k^{\mathrm{samp}} + n_k^{\mathrm{ls}}$ inner-oracle calls before caching, with $p_k \leq n_k^{\mathrm{samp}}$ because rejected derivatives still consume an evaluation.

THEOREM 4.5 (Conditional box-Clarke stationarity). *Consider the ideal sequence generated by Algorithm 4.1. Assume exact inner values and ordinary gradients,* $\epsilon_k, \omega_k \downarrow$

0, $L_k \to \infty$, and $0 < \underline{b} I \preceq M_k \preceq \bar{b} I$. Suppose the finite hulls $\widehat{\partial}_k := \operatorname{co} \mathcal{G}_k$ are gradient-consistent: along every subsequence $\theta^{k_r} \to \bar{\theta}$,

$$\operatorname{dist}_{\mathrm{H}}\Big(\widehat{\partial}_{k_r}, \partial_C \psi(\bar{\theta})\Big) \longrightarrow 0. \tag{4.25}$$

Here $\operatorname{dist}_{\mathrm{H}}$ *is the Hausdorff distance. Then $\theta^k \in C$, $\psi(\theta^k)$ is nonincreasing, and every accumulation point satisfies*

$$0 \in \partial_C \psi(\bar{\theta}) + N_C(\bar{\theta}). \tag{4.26}$$

Equivalently, for $\bar{\Lambda} = \underline{\Lambda} + D_B \bar{\theta}$, $0 \in \partial_C \phi(\bar{\Lambda}) + N_B(\bar{\Lambda})$. If $\phi_B > 0$, this is also Clarke stationarity for maximizing $\rho^\star = 1/\phi$ on B.

Proof. See the proof of Theorem 4.5 in Appendix C.4 (p. 37). ∎

Condition (4.25) is an explicit asymptotic assumption; a fixed bundle of $m+1$ samples does not imply it [4]. Hence the theorem gives neither global optimality nor a convergence rate. With finite bundles, solver tolerances, and line-search caps, χ_k is only a sampled-bundle diagnostic; validity of the returned closed set follows instead from the independent screen in Corollary 3.12.

5. Numerical experiments.

Computational environment. All experiments were run in CPython 3.11.12 under macOS 26.6.1 (arm64) on an Apple M2 Pro with ten CPU cores and 16 GB of RAM. The implementation used DQBEE 0.3.0 [7], NumPy 1.26.4, SciPy 1.17.1, and SymPy 1.14.0. The conic programs in our optimal dissipative quadratization (ODQ) method were modeled in CVXPY 1.7.5 and solved numerically by MOSEK 11.2.1 [11, 23]. The SOS baselines, applied directly in the original coordinates, used SUMOFSQUARES.PY 1.3.1 [35] through PICOS 2.6.2 and the same MOSEK backend.

Compared methods and implementation. For each system, DQBEE generated one monomial lift and its inner-quadratic factorizations, which were fixed for all lifted methods with $Q = I_N$. ODQ solves Problem 3.8 using Algorithm 4.1, searching over the stabilizer gain Λ with a convex gauge solve at each gain. In the benchmark suite, *Base* uses the DQBEE feasibility gain Λ_{feas} with $\mu = 0$, whereas *Gauge* fixes Λ_{feas} and optimizes only μ; the first experiment instead uses a matched *Gauge* arm at $\Lambda = 0$. Thus all lifted comparisons share the same lift and Q, with no search over either.

The direct SOS baselines act on the original system. SOS-2 fixes the quadratic Lyapunov function from the linearization and maximizes only its certified level. SOS-4(r) searches a quartic Lyapunov function using at most r alternations between V and the SOS multiplier s, followed by a final level search. Hence 2 and 4 denote $\deg V$, while r is a round budget, not a radius or ratio; a larger r need not give a nested or larger set. We compare the audited closed sets in the original coordinates, not the raw values of ρ_{safe} or the SOS level c, whose scalings are method dependent.

Each ODQ oracle call used the reduced generalized-Schur program (4.5). Gain-independent sparse operators were cached, and a parameterized CVXPY–MOSEK model was reused and warm-started across gains. The common outer settings were $m+1$ full-ball samples, five L-BFGS pairs, metric eigenvalues in $[10^{-3}, 10^{3}]$, and halving updates of the sampling radius ϵ and stationarity tolerance ω from $(\epsilon_0, \omega_0) = (5 \times 10^{-2}, 10^{-2})$ to $(\epsilon_{\min}, \omega_{\min}) = (10^{-4}, 10^{-4})$. Armijo backtracking used $(c, \beta) = (10^{-4}, 1/2)$ and at most 20 trials; the scaled bundle-QP feasibility and KKT tolerance was 2×10^{-5}. Gain boxes and computational budgets are specified with each experiment.

Verification and timing. For $m > 0$, the gain box and its initially 5%-padded sampling neighborhood passed a common-Lyapunov vertex test with requested margin 10^{-7}; the padding was halved if necessary. Lifted certificates used Corollary 3.12 with $\eta = 10^{-6}$, and the nominal open threshold was diagnostic only. Each certificate was independently reassembled, with the floating-point estimates of $\|J\|$ and the normalized Lyapunov residual moved upward by one unit in the last place. These are residual-aware floating-point checks, not interval or rational proofs.

The direct SOS programs used $\varepsilon_{\mathrm{SOS}} = 10^{-4}$ and the smallest even multiplier degree at least $\max\{2, \deg f - 1\}$. A reported set $\{V \le \alpha^2 c\}$ had to pass an independent audit of solver status, coefficient residuals, and Gram-matrix semidefiniteness at tolerance 10^{-7}; we used the largest passing $\alpha \in \{1 - 10^{-k} : k = 3, \dots, 6\} \cup \{0.995, 0.99, 0.98\}$. Construction time includes method-specific construction and auditing, DQBEE for lifted methods, and the stable-box test for ODQ. The planar experiment uses one untimed warm-up per solver path; in the heterogeneous suite, each fresh-process execution group performs the corresponding untimed warm-up. Reported construction times exclude process startup, warm-up, geometry, and serialization, whereas the 600-second group deadline includes them. The relay protocol instead reports launch-to-serialization wall time. Monte Carlo checks, box widening, and plotting are excluded throughout.

Experimental questions. The planar-quintic experiment isolates gain selection by comparing ODQ with the matched $\Lambda = 0$ *Gauge* arm, and then compares both with SOS-2 and SOS-4(r) in certified area and construction time. The second experiment compares Base, Gauge, and ODQ on a heterogeneous 19-system suite and tests the resulting structural hypothesis on a separately locked 12-system relay family. It includes $m = 0$ controls and direct-SOS references. The table retains separate cohort blocks because the timing boundaries differ, while the discussion considers the evidence jointly.

5.1. The pipeline end to end: a planar quintic. Consider

$$\dot{x}_1 = -x_1 + x_2, \qquad \dot{x}_2 = -2x_2 - x_1^5. \tag{5.1}$$

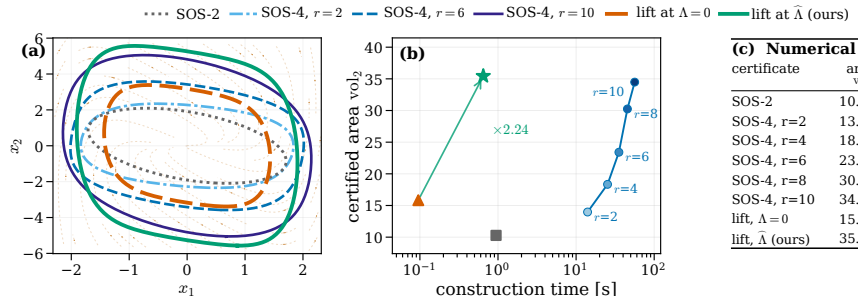
Its linearization has eigenvalues -1 and -2, while $V_0 = x_1^6 + 3x_2^2$ satisfies $\dot{V}_0 = -6x_1^6 - 12x_2^2$. Thus the origin is globally asymptotically stable, and the finite areas below measure certificate conservatism rather than the true basin.

DQBEE selects $y_1 = x_1^2$ and $y_2 = x_1^3$, giving, before stabilizer injection,

$$\dot{x}_1 = -x_1 + x_2, \quad \dot{x}_2 = -2x_2 - y_1 y_2, \quad \dot{y}_1 = -2y_1 + 2x_1 x_2, \quad \dot{y}_2 = -3y_2 + 3x_2 y_1. \tag{5.2}$$

The stabilizers are $h_1 = y_1 - x_1^2$ and $h_2 = y_2 - x_1 y_1$; hence $N = 4$, $m = 2$, and $q = 20$. We solve Problem 3.8 on $B = [0, 12]^2$, with caps of 60 outer iterations and 300 oracle calls, and compare with the matched $\Lambda = 0$ Gauge arm, SOS-2, and SOS-4(r) for $r \in \{2, 4, 6, 8, 10\}$. The padded box passed the common stability screen. Areas are those of the full origin-connected pullbacks; an 8001-slice integration agreed with 10^6-sample Monte Carlo estimates to within 0.081%.

Figure 1 summarizes the results. ODQ terminates after 21 outer iterations and 135 oracle calls at $\widehat{\Lambda} = (0.7337, 1.5003)$, with $\rho_{\mathrm{safe}} = 3.040679$, area 35.44, and total time 0.65 s. At $\Lambda = 0$, separate gauge optimization gives $\rho_{\mathrm{safe}} = 1.832534$, area 15.83, and time 0.10 s. Gain optimization therefore increases the certified area by a factor of 2.238 with the lift and Q fixed. The largest direct certificate tested, SOS-4(10), has area 34.48 and takes 56.27 s. In this single post-warm-up execution, ODQ is thus 2.8% larger and about 87 times faster.



(c) Numerical summary

certificate	area vol_2	DQbee [ms]	total [s]
SOS-2	10.28	--	0.94
SOS-4, r=2	13.97	--	14.03
SOS-4, r=4	18.34	--	25.28
SOS-4, r=6	23.41	--	35.33
SOS-4, r=8	30.25	--	45.41
SOS-4, r=10	34.48	--	56.27
lift, $\Lambda = 0$	15.83	83	0.10
lift, $\bar{\Lambda}$ (ours)	35.44	83	0.65

FIG. 1. *Planar-quintic comparison. (a) Certified boundaries in the original coordinates; representative SOS-4 budgets are shown. (b) Certified area versus construction time for all eight runs. (c) Numerical summary. Panel (a) uses independent axis scales for readability; all areas are for the full origin-connected components, and all times follow the common convention above.*

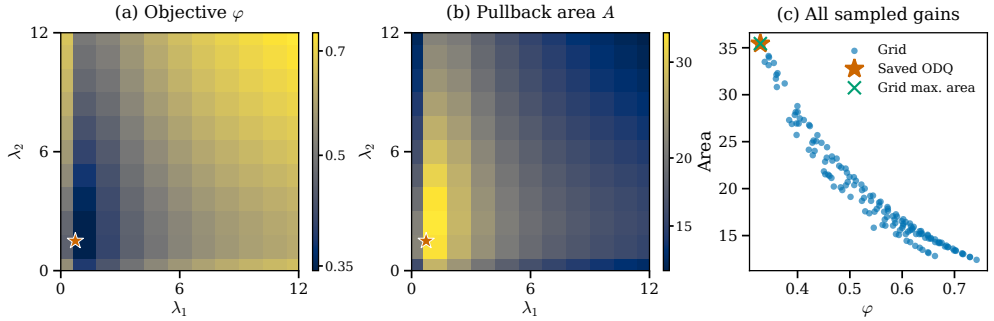


FIG. 2. *Quintic objective and original-coordinate area. Panels (a)–(b) show values on the* 11×11 *full-box grid; panel (c) includes the 25 local-grid points, with the shared origin counted once. Stars mark the saved ODQ gain, and the cross marks the largest sampled area. Each point uses the residual-aware safe radius; colors in (a)–(b) use logarithmic scales. These are sampled values, not continuous bounds between grid points.*

Repeating the search on $[0, 24]^2$ and $[0, 48]^2$ again returns an interior gain and changes ρ_{safe} by at most 6.8×10^{-8}. This indicates insensitivity to the chosen upper endpoint for this instance, without asserting an unrestricted optimum.

Figure 2 compares the objective with area at 145 fixed gains: an 11×11 grid on $[0, 12]^2$ and a 5×5 grid on $[0, 3]^2$, sharing the origin. The rank correlation between $1/\phi$ and area is 0.981. The sampled gain $(0.75, 1.5)$ has area 35.4478, slightly above the saved ODQ area 35.4405, although its objective is 0.010% larger. This finite-radius difference is consistent with the leading-order result in Section 3.3: minimizing ϕ need not maximize area at the reported level. The grid establishes neither optimum over the box.

5.2. Systematic comparison across a benchmark suite.

Benchmark design. We use two complementary, source-locked cohorts with three repeats each: a heterogeneous suite tests dependence on system structure, while a controlled relay grid probes dimensional scaling. The first contains 19 polynomial systems with $1 \le n \le 10$, $2 \le \deg f \le 7$, and $2 \le N \le 10$: standard and literature-derived models, synthetic degree and coupling probes, three already-quadratic controls

with $m = 0$, and one system with known basin area. Exact vector fields and provenance appear in Appendix D.

We use an origin-connected physical measure only when every complete arm occupies multiple grid cells without boundary clipping, giving three length, seven area, and four volume rows. The remaining five use the same fixed-direction first-exit proxy R for all methods in the row, avoiding comparisons of unresolved or truncated measures. The quantity R is a numerical size score, not a certified inradius.

To prevent a slow direct-SOS solve from suppressing completed methods, each system is split into one lifted group (Base, Gauge, and ODQ) and three direct groups, one for each SOS arm. Every group runs in a fresh process with an independent 600-second deadline; at most four groups run concurrently, with each child restricted to one numerical-solver thread. This common cap bounds the high-dimensional attempts while making every missing value an observed computational or audit outcome. A cell is displayed only when all three repeats return an audited certificate.

For $m > 0$, both cohorts stop ODQ when the sampled residual in (4.21) satisfies $\chi_k \le \omega_k$ at $(\epsilon_k, \omega_k) = (\epsilon_{\min}, \omega_{\min})$, or at a cap of 40 outer iterations or 200 oracle calls. The caps bound repeated SDP work, whereas the residual test measures finite-sample stationarity at the prescribed resolution; neither a budget exit nor a failed line search certifies stationarity. The best screened candidate is returned as in (4.23). All lifted and SOS-2 groups complete. The two quartic arms time out in every repeat on `duffing-chain-3` and `liu-10d`; SOS-4(8) additionally passes the audit in only 1/3 repeats on `cubic-ring-3` and 2/3 on `shear-cubic-a1`. Appendix D records the execution and completeness checks.

The relay grid is $(p, n) \in \{3, 5, 7\} \times \{2, 4, 6, 8\}$, with $m = (p-1)/2$ and $N = n+m$. Thus increasing n at fixed p leaves the number of stabilizer gains unchanged. All rows use R to avoid dimension-dependent volume grids. Each method runs in an isolated process with a predeclared 600-second deadline, bounding its cost independently of other methods. A timeout or completed budget crossing retires that method only at larger n within the same degree and repeat; numerical failures alone do not trigger retirement. These resource-limited outcomes are distinguished from completed comparisons in Appendix D.2.

Results and analysis. Table 1 gives the full comparison, while Figure 3 isolates the complete direct-SOS pairs. Across the 16 heterogeneous systems with $m > 0$, ODQ improves the reported metric over both Base and Gauge in every row, by median factors of 2.892 and 1.652, respectively; the median ODQ/Gauge time factor is 7.172. All returned sets pass the independent audit. Six outer searches reach the oracle cap, so those entries are best audited box candidates rather than stationary points. Against SOS-2, ODQ is larger on 6/19 systems, faster on 11/19, and both on three, with median size ratio $M_{\mathrm{ODQ}}/M_{\mathrm{sos}-2} = 0.951$. It is faster on all 17 and 15 repeat-complete quartic cells for two and eight rounds, respectively; the corresponding median SOS/ODQ time factors are 12.794 and 45.814, and ODQ is larger on five cells at each budget. On the relay grid, all 36 ODQ runs complete, and the 12 cell-median times remain between 1.69 and 13.08 seconds for $N \le 11$. All 27 repeat-level comparisons in the nine fully paired SOS-2 cells favor ODQ in both proxy and time, with geometric-mean ODQ/SOS-2 proxy and SOS-2/ODQ time factors of 1.162 and 4.305. In the five repeat-complete relay quartic cells, the proxy is 1.03–1.69 times larger, but construction is 7.41–109 times slower; the missing high-dimensional endpoints rule out a family-wide quartic comparison.

The favorable relay scaling is consistent with its one-link structure. For fixed odd degree p, DQBEE introduces only $m = (p-1)/2$ auxiliary powers, so $N = n + m$,

TABLE 1

Relative results (three-repeat medians, two significant digits). For ODQ, the table reports the absolute pair (M/T), where $M \in \{L, A, V, R\}$ denotes length, area, volume, or the fixed-direction proxy and T is the construction time in seconds. Each complete comparator is reported as (R_M/R_T), with $R_M = M_{\text{method}}/M_{\text{ODQ}}$ and $R_T = T_{\text{method}}/T_{\text{ODQ}}$; $R_M < 1$ and $R_T > 1$ favor ODQ. Bold marks strict joint ODQ dominance over every complete direct-SOS arm. Missing cells: $--^{R}$ retired by the relay protocol, $--^{A}_{k/3}$ attempted but not repeat-complete (k of three repeats), and TO three hard timeouts. Absolute pair results appear in Table 3.

			Proposed	Lifted baselines		Direct SOS		
System or (p, n)	$n/N/m$	M	ODQ	Base	Gauge	SOS–2	SOS–4(2)	SOS–4(8)
Reference systems and scalar degree probes								
`vanderpol`	2/3/1	A	(2.7/0.3)	(0.33/0.38)	(0.73/0.41)	(2.4/1.9)	(2.5/19)	(2.5/51)
`planar-quintic`	2/4/2	A	**(35/0.72)**	(0.21/0.18)	(0.45/0.19)	(0.29/1.7)	(0.39/21)	(0.85/65)
`cubic`	1/2/1	L	(1.9/0.2)	(0.67/0.59)	(0.67/0.62)	(1/1.2)	(1/8.3)	(1/21)
`quintic`	1/3/2	L	(1.3/0.74)	(0.7/0.17)	(0.72/0.19)	(1.5/0.46)	(1.5/3.8)	(1.5/4)
`bistable`	1/2/1	R	(0.051/0.4)	(0.017/0.27)	(0.052/0.29)	(2/0.67)	(2/3.5)	(2/4.2)
`duffing`	2/3/1	A	(3.2/0.19)	(0.36/0.51)	(0.64/0.55)	(1.1/2.8)	(1.2/31)	(1.2/36)
Quadratic controls ($m = 0$)								
`quad-cross`	2/2/0	A	(8.9/0.0098)	(0.67/0.35)	(1/0.95)	(1.2/49)	(2.1/570)	(1.8/1500)
`quad-2d`	2/2/0	A	(3.9/0.0096)	(0.99/0.4)	(1/0.99)	(1.3/49)	(2.4/580)	(2.1/1100)
Synthetic degree and cascade probes								
`cubic-3d`	3/7/4	V	(36/4.3)	(0.24/0.036)	(0.53/0.049)	(0.73/0.21)	(0.92/5.7)	(0.8/18)
`cascade-3d`	3/6/3	V	(32/2.3)	(0.3/0.056)	(0.54/0.071)	(0.69/0.39)	(0.76/10)	(0.94/33)
`septic`	1/4/3	L	(1.1/1.1)	(0.69/0.12)	(0.71/0.13)	(1.8/0.38)	(1.8/3)	(1.8/3.3)
`septic-2d`	2/5/3	A	**(18/1.6)**	(0.32/0.093)	(0.57/0.11)	(0.42/1.8)	(0.57/18)	(0.9/59)
Semidiscrete, chain, and ring models								
`chafee-3`	3/6/3	V	(6.4/1.8)	(0.44/0.11)	(0.45/0.12)	(1.1/0.67)	(0.93/13)	(0.93/46)
`duffing-chain-2`	4/6/2	R	(0.98/1.1)	(0.59/0.13)	(0.72/0.17)	(1.1/3.4)	(1.2/73)	(1.3/260)
`duffing-chain-3`	6/9/3	R	(0.95/6.4)	(0.58/0.023)	(0.73/0.043)	(1/2.1)	TO	TO
`cubic-ring-3`	3/9/6	V	(33/11)	(0.24/0.021)	(0.58/0.033)	(0.81/0.088)	(1.4/2)	$--^{A}_{1/3}$
Literature-derived network models								
`networked-vdp-2`	4/6/2	R	(0.48/1.1)	(0.62/0.12)	(0.93/0.15)	(1.8/1.8)	(2/71)	(2/110)
`liu-10d`	10/10/0	R	**(19/0.29)**	(0.74/0.26)	(1/1.1)	(0.53/160)	TO	TO
Prescribed-basin geometry test								
`shear-cubic-a1`	2/6/4	A	(0.63/3.4)	(0.11/0.12)	(0.25/0.13)	(1.7/0.89)	(3.1/9.2)	$--^{A}_{2/3}$
Single-link relay family (D.3)*; instances are indexed by* (p, n)								
$(3, 2)$	2/3/1	R	(1.2/1.7)	(0.75/1)	(0.98/0.94)	(0.87/1.1)	$--^{A}_{2/3}$	(1/10)
$(3, 4)$	4/5/1	R	**(0.89/2.3)**	(0.51/0.72)	(0.98/0.77)	(0.77/2)	$--^{A}_{0/3}$	$--^{A}_{0/3}$
$(3, 6)$	6/7/1	R	**(0.7/3.5)**	(0.49/0.43)	(0.99/0.47)	(0.79/3.6)	$--^{A}_{0/3}$	$--^{A}_{0/3}$
$(3, 8)$	8/9/1	R	**(0.53/5.7)**	(0.42/0.3)	(1/0.28)	(0.85/5.5)	TO	$--^{A}_{0/3}$
$(5, 2)$	2/4/2	R	(1/1.9)	(0.7/0.92)	(0.91/0.93)	(0.94/1.3)	(1.1/7.4)	$--^{A}_{0/3}$
$(5, 4)$	4/6/2	R	(0.8/2.8)	(0.45/0.63)	(0.96/0.68)	(0.82/6.2)	(1.7/110)	TO
$(5, 6)$	6/8/2	R	**(0.62/2.9)**	(0.45/0.62)	(0.99/0.58)	(0.89/51)	TO	$--^{R}$
$(5, 8)$	8/10/2	R	(0.48/7.7)	(0.37/0.21)	(0.99/0.24)	TO	$--^{R}$	$--^{R}$
$(7, 2)$	2/5/3	R	(0.94/2.2)	(0.64/0.71)	(0.91/0.64)	(0.99/1.6)	(1.2/11)	(1.3/26)
$(7, 4)$	4/7/3	R	**(0.74/3.8)**	(0.41/0.51)	(0.96/0.42)	(0.85/20)	TO	TO
$(7, 6)$	6/9/3	R	(0.57/5.6)	(0.41/0.29)	(0.98/0.31)	TO	$--^{R}$	$--^{R}$
$(7, 8)$	8/11/3	R	(0.45/13)	(0.33/0.13)	(0.98/0.15)	$--^{R}$	$--^{R}$	$--^{R}$

whereas the gain dimension and the $m+1$-sample bundle do not grow with n. Increasing n lengthens only the stable linear chain without adding nonlinear monomials. The cached reduced-Schur oracle therefore uses a sparse quadratic lift and a cone of order $N^2 + N$, while the largest direct-SOS decrease Gram matrices have orders $\binom{n+(p+1)/2}{(p+1)/2}$ for SOS-2 and $\binom{n+(p+3)/2}{(p+3)/2}$ for SOS-4. At $(p, n) = (7, 8)$, these orders are 495 and 1287, compared with 132 for the corresponding ODQ cone. The lifted comparison shows that Gauge closes most of the gap from Base to ODQ: $M_{\text{Gauge}}/M_{\text{ODQ}}$ ranges from 0.907 to 0.995, whereas $M_{\text{Base}}/M_{\text{ODQ}}$ ranges from 0.331 to 0.751. Gain optimization adds a smaller but consistent improvement. The relay results suggest that ODQ may be especially useful for higher-degree, higher-dimensional nonlinear systems whose sparse structure admits a compact quadratic lift. This conclusion is empirical rather

than asymptotic, and the advantage is not universal: SOS-2 is both cheaper and larger on five low-dimensional heterogeneous systems.

Numerical diagnostics, failures, and the timing breakdown appear in Appendix D.4.

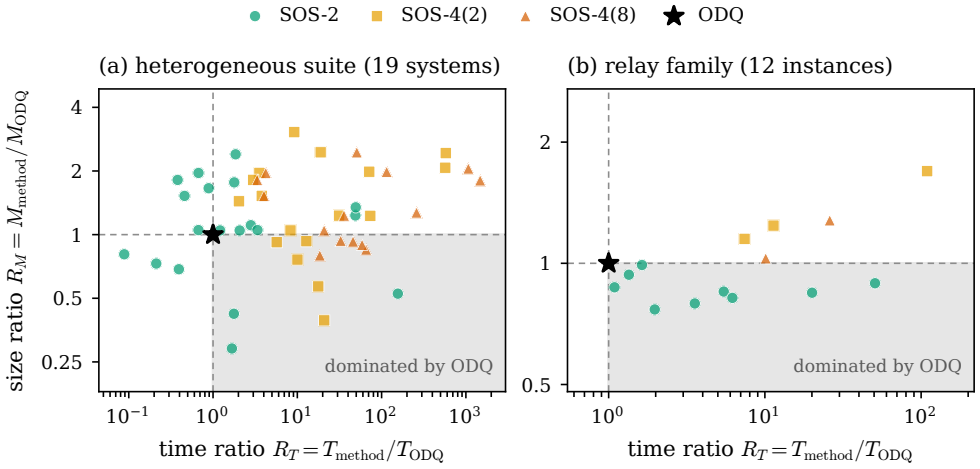


FIG. 3. *Certified size versus construction time for the direct-SOS comparators, relative to ODQ. Each colored marker represents one complete direct-SOS cell of Table 1, placed at* $(R_T, R_M) = (T_{\text{method}}/T_{\text{ODQ}},\ M_{\text{method}}/M_{\text{ODQ}})$*; the star marks ODQ at* $(1, 1)$*. Points in the shaded quadrant are dominated by ODQ; axes are logarithmic and differ between panels.*

6. Conclusion and Future Work. This paper separates the choices that enter dissipative quadratization and its ROA analysis. These are the lifting map, the off-manifold extension, the representation of its quadratic term, and the Lyapunov certificate itself. For a fixed embedding, we characterize the full affine space of admissible quadratic extensions and identify the stabilizer family within it as a structured affine subspace. Extension choices can change transverse stability and the resulting Lyapunov metric, while representation gauges preserve the vector field but can change the spectral-norm bound used in the certificate. Fixing a monomial lift, a reference extension, stabilizer factorizations, the weight $Q = I_N$, and an admissible gain box, we optimize the diagonal stabilizer gains together with the Kronecker gauge.

The tangent–transverse splitting reduces feasibility to the auxiliary block, and the fixed-gain gauge problem is an exact reduced SDP whose global optimum minimizes the stated spectral-norm bound, though equality with the intrinsic cubic growth is not guaranteed. We prove attainment on the prescribed box, give a residual-aware closed certificate, and show that the level threshold ranks the leading-order volumes of scaled pullbacks when $\phi_B > 0$. An explicit unbounded-ray example shows why the box must be part of the problem data. The resulting algorithm combines the reduced oracle with a box-constrained nonsmooth search, and under the hypotheses of Theorem 4.5, every accumulation point of the idealized sequence is box-Clarke stationary. Its finite implementation instead returns the best independently screened candidate encountered, and we do not claim global optimality.

The experiments show both the benefit and the limits of this design. On the planar quintic, gain optimization increases the certified area from 15.83 to 35.44 with the lift and Q fixed, and ODQ improves on both fixed-gain lifted baselines for every

heterogeneous system with $m > 0$. Direct SOS nonetheless remains competitive, since SOS-2 is both larger and cheaper on five low-dimensional heterogeneous systems. In all nine fully paired single-link relay cells, ODQ has a larger fixed-direction proxy and is faster than SOS-2, and the relay's sparse nonlinear structure admits a compact quadratic lift as the physical dimension grows. We expect the method to extend to similarly structured higher-degree, higher-dimensional systems, but these finite experiments do not establish an asymptotic advantage or uniform dominance over direct SOS.

Three directions would relax the present formulation. Optimizing over the full admissible extension space, allowing normalized metric design, and bounding the intrinsic cubic growth more tightly would each remove one of its current restrictions. Global gain optimization and interval or rational residual bounds remain separate questions, of search quality and of certificate validity respectively. Beyond ROA estimation, extension design could also be used to seek tighter truncation-error bounds and reachable-set enclosures in Carleman linearization [13], which would require finite-horizon error estimates and consistent lifting of initial sets that the present Lyapunov results alone do not provide. Barrier certificates and feedback synthesis offer further settings in which to study how extension choices affect safety and controlled invariance. The off-manifold extension of an exact lift should therefore be chosen together with the certificate used for the original system.

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Appendix contents

Appendix A. Proof of the ellipsoidal Lyapunov certificate 28

Appendix B. Problem formulation: overview and proofs 29
B.1 Extension space 29
B.2 Transverse feasibility 30
B.3 Representation space 31
B.4 Intrinsic growth and spectral-norm relaxation 31
B.5 Pullback connectedness 31
B.6 Derivation of the box-constrained formulation 32
B.7 Well-posedness of the box value 32
B.8 Gauge-defect correction 33
B.9 Leading-order certified volume 33
B.10 Gain invariance of the quadratic germ 33

Appendix C. Inner-oracle and outer-convergence details 34
C.1 The gauge-image distance and its dual 34
C.2 The reduced oracle and its dual 34
C.3 Proof of the sensitivity formula 35
C.4 Proof of conditional Clarke stationarity 37

Appendix D. Benchmark definitions and supplementary results 38
D.1 Heterogeneous suite 38
D.2 Single-link relay family 39
D.3 Supplementary geometry 40
D.4 Numerical diagnostics, failures, and timing 41

Appendix A. Proof of the ellipsoidal Lyapunov certificate.

Proof of Proposition 2.7. For any symmetric positive definite P, Q, the quotient in (2.9) is unchanged under positive rescaling of $\mathbf{w}$. It can therefore be restricted to the compact ellipsoid $\{\mathbf{u} : \mathbf{u}^\top P\mathbf{u} = 1\}$. Its denominator is positive there, so the supremum is finite and attained. Since $\mathbf{q}$ is homogeneous quadratic, $\mathbf{q}(-\mathbf{u}) = \mathbf{q}(\mathbf{u})$, whereas $\mathbf{u}^\top P\mathbf{q}(\mathbf{u})$ changes sign. Consequently,

$$\beta(P, Q, \mathbf{q}) = \max_{\mathbf{u}^\top P\mathbf{u}=1} \frac{2|\mathbf{u}^\top P\mathbf{q}(\mathbf{u})|}{\mathbf{u}^\top Q\mathbf{u}} = \max_{\mathbf{u}^\top P\mathbf{u}=1} \frac{2\mathbf{u}^\top P\mathbf{q}(\mathbf{u})}{\mathbf{u}^\top Q\mathbf{u}}. \tag{A.1}$$

For the exact Lyapunov solution, write each nonzero $\mathbf{w} = t\mathbf{u}$, where $t = \sqrt{V(\mathbf{w})} > 0$ and $\mathbf{u}^\top P\mathbf{u} = 1$. Homogeneity gives

$$\dot{V}(t\mathbf{u}) = t^2\mathbf{u}^\top Q\mathbf{u}\left[-1 + t\frac{2\mathbf{u}^\top P\mathbf{q}(\mathbf{u})}{\mathbf{u}^\top Q\mathbf{u}}\right]. \tag{A.2}$$

If $\beta > 0$, this is negative for $0 < t < 1/\beta$. At a maximizing direction in (A.1), it equals zero at $t = 1/\beta$; hence no larger radius has strict decrease throughout its punctured interior. Taking reciprocals over the positive directions in (A.1) gives (2.10) and $\rho_{\mathrm{rad}} = 1/\beta$. If $\beta = 0$, the cubic contribution vanishes identically and $\dot{V} = -\mathbf{w}^\top Q\mathbf{w} < 0$ for every nonzero $\mathbf{w}$, so $\rho_{\mathrm{rad}} = +\infty$.

For any representation $\mathbf{q}(\mathbf{w}) = H(\mathbf{w} \otimes \mathbf{w})$, Cauchy–Schwarz in the P inner product and $\|\mathbf{w} \otimes \mathbf{w}\| = \|\mathbf{w}\|^2$ yield

$$2|\mathbf{w}^\top P\mathbf{q}(\mathbf{w})| \le 2\sqrt{V(\mathbf{w})}\sqrt{\|P\|}\,\|H\|\,\|\mathbf{w}\|^2.$$

Using $\mathbf{w}^\top Q\mathbf{w} \geq \lambda_{\min}(Q)\|\mathbf{w}\|^2$ gives

$$\beta(P, Q, \mathbf{q}) \leq \frac{2\sqrt{\|P\|}\,\|H\|}{\lambda_{\min}(Q)}. \tag{A.3}$$

Taking reciprocals proves (2.11), including $H = 0$ under the stated convention.

Now let $\widehat{P}$ be any symmetric positive definite matrix. The symmetric normalized residual satisfies $-\delta I_N \preceq Q^{-1/2}RQ^{-1/2} \preceq \delta I_N$. Thus $\mathbf{w}^\top R\mathbf{w} \leq \delta \mathbf{w}^\top Q\mathbf{w}$, and

$$\begin{aligned}\dot{\widehat{V}}(\mathbf{w}) &= \mathbf{w}^\top(-Q + R)\mathbf{w} + 2\mathbf{w}^\top \widehat{P}\mathbf{q}(\mathbf{w}) \\ &\leq -(1-\delta)\mathbf{w}^\top Q\mathbf{w} + \gamma\sqrt{\widehat{V}(\mathbf{w})}\,\mathbf{w}^\top Q\mathbf{w},\end{aligned}$$

which proves (2.13). On (2.14), $\gamma\sqrt{\widehat{V}} \leq \theta(1-\delta)$. With $\kappa := \lambda_{\min}(\widehat{P}^{-1/2}Q\widehat{P}^{-1/2}) > 0$, it follows that

$$\dot{\widehat{V}} \leq -(1-\theta)(1-\delta)\mathbf{w}^\top Q\mathbf{w} \leq -(1-\theta)(1-\delta)\kappa\widehat{V}. \tag{A.4}$$

The derivative is strictly negative on the boundary, so a trajectory cannot leave the ellipsoid. The polynomial vector field is locally Lipschitz, and the invariant ellipsoid is compact; hence every trajectory starting there exists for all $t \geq 0$. Integration of (A.4) gives $\widehat{V}(\mathbf{w}(t)) \leq \widehat{V}(\mathbf{w}(0))e^{-(1-\theta)(1-\delta)\kappa t}$ and convergence to the origin. This also proves the exact-solution closed-ellipsoid claims: when $\beta > 0$, take $\widehat{P} = P$, $\delta = 0$, $\gamma = \beta$, and $\theta = \rho\beta < 1$. If $\gamma = 0$, the bound $\dot{\widehat{V}} \leq -(1-\delta)\kappa\widehat{V}$ holds globally. Every finite sublevel set is compact and invariant, so the same argument gives global existence and exponential convergence. This covers the exact case $\beta = 0$ as well. Replacing δ and γ by upper bounds with the residual bound below one preserves all the inequalities. ∎

Appendix B. Problem formulation: overview and proofs.

B.1. Extension space. For the fixed polynomial embedding, substitution defines the linear map

$$T_\Phi : \mathcal{P}_2 \longrightarrow \mathbb{R}[\mathbf{x}], \qquad T_\Phi r = r \circ \Phi, \qquad \ker T_\Phi = I(\mathcal{M}) \cap \mathcal{P}_2.$$

Two degree-at-most-two vector fields agree on $\mathcal{M}$ if and only if their difference belongs componentwise to $\ker T_\Phi$. Since $\mathbf{F}_0$ is exact, this gives $\mathcal{E}_\Phi(\mathbf{p}) = \mathbf{F}_0 + (\ker T_\Phi)^N$. To compute the kernel, write

$$r(\mathbf{w}) = c_0 + \sum_{i=1}^{N} c_i w_i + \sum_{1 \leq i \leq j \leq N} c_{ij} w_i w_j.$$

Substitution and coefficient matching in $\mathbf{x}$ give a finite homogeneous linear system in the $1 + N + N(N+1)/2$ coefficients. Its null space gives a basis $r_1, \ldots, r_s$ of $\ker T_\Phi$, and $\{\mathbf{e}_k r_\ell : 1 \leq k \leq N,\ 1 \leq \ell \leq s\}$ is a basis of $\mathcal{V}_\Phi$; in particular, $\dim \mathcal{V}_\Phi = Ns$.

Each h_i belongs to $\ker T_\Phi$, so every direction $\mathbf{e}_{n+i}h_i$ in (3.1) belongs to $\mathcal{V}_\Phi$. Its linear part is $\mathbf{e}_{n+i}y_i$. These linear parts are linearly independent, which proves $\dim \mathcal{S}_h = m$. Hence $\Lambda \mapsto \mathbf{F}_\Lambda$ parametrizes $\mathbf{F}_0 + \mathcal{S}_h$ injectively. The restriction can exclude both auxiliary and physical equation directions. For example, if $m > 0$, then $\mathbf{e}_1 h_1 \in \mathcal{V}_\Phi \setminus \mathcal{S}_h$ because h_1 is nonzero and all fields in $\mathcal{S}_h$ have zero physical components. Thus the inclusion is strict whenever $m > 0$. Exactness also gives $\mathbf{F}(\mathbf{0}) = J_x\Phi(\mathbf{0})\mathbf{p}(\mathbf{0}) = 0$ for every admissible extension, so its constant term vanishes.

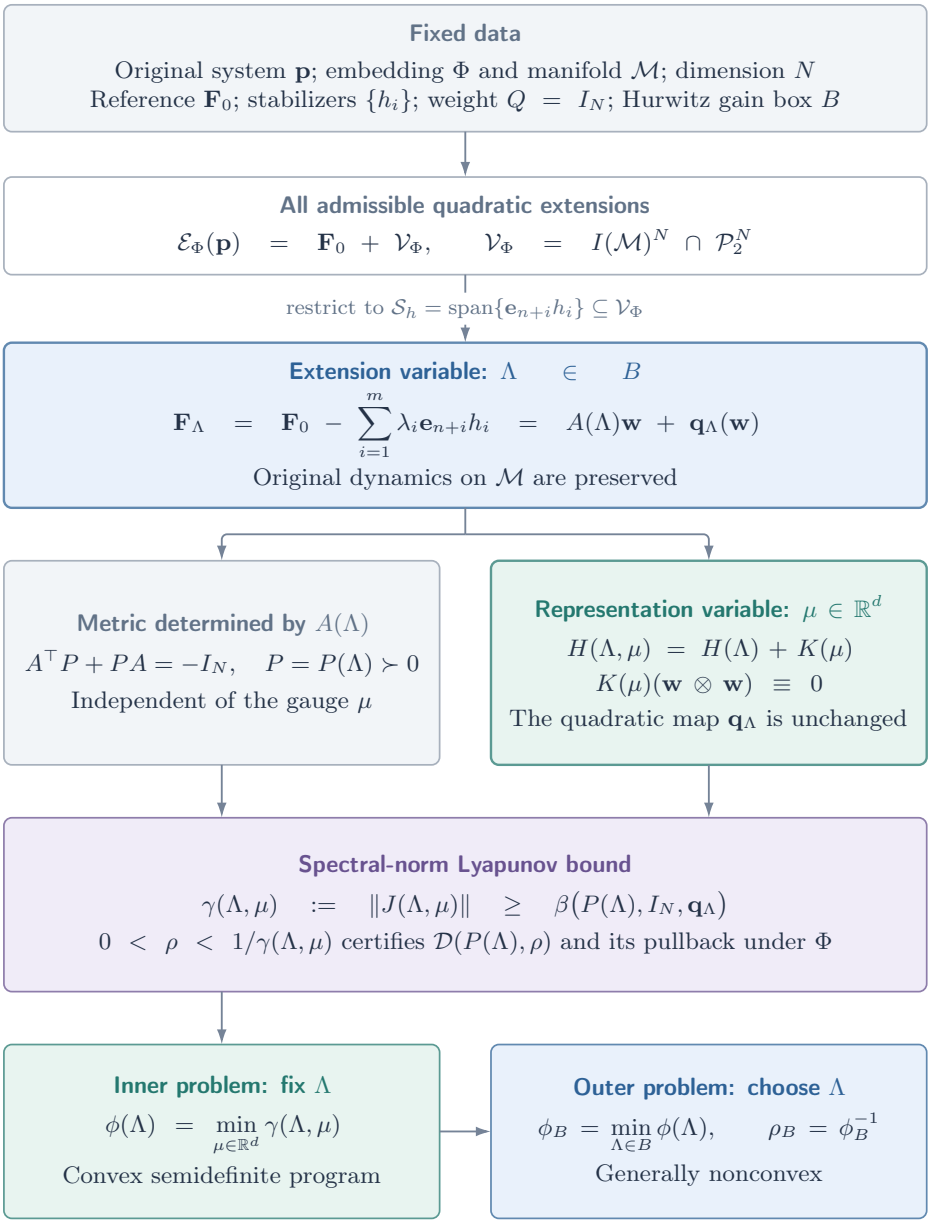


FIG. 4. *Construction of Problem 3.8. The gains select the extension; the gauge changes only its matrix representation. The diagram states the optimization objective; certificates from numerical candidates require the verification in Corollary 3.12.*

B.2. Transverse feasibility.

Proof of Lemma 3.1. Let $\Phi(\mathbf{x}) = (\mathbf{x}, \mathbf{g}(\mathbf{x}))$ and let $\mathbf{F}_\Lambda$ denote the right-hand side of (3.2). Exactness of the lift gives

$$\mathbf{F}_\Lambda(\Phi(\mathbf{x})) = J_x(\Phi)(\mathbf{x})\,\mathbf{p}(\mathbf{x}).$$

Because $\mathbf{p}(\mathbf{0}) = \mathbf{0}$, $\mathbf{g}(\mathbf{0}) = \mathbf{0}$, and $J_x(\mathbf{g})|_{\mathbf{0}} = 0$, differentiation at the origin yields

$$A(\Lambda)\begin{bmatrix} I_n \\ 0 \end{bmatrix} = \begin{bmatrix} I_n \\ 0 \end{bmatrix} J_p;$$

the quadratic part of $\mathbf{F}_\Lambda$ has zero derivative there. Hence the first n columns of $A(\Lambda)$ are $[J_p^\top, 0]^\top$. Moreover, both factors a_i and b_i vanish at the origin, so

$$J_y(\mathbf{h})|_{(\mathbf{0},\mathbf{0})} = I_m, \qquad \mathbf{h} := (h_1, \dots, h_m).$$

The remaining blocks are therefore the gain-independent matrix R and $J_y(\mathbf{q}_2 - \operatorname{diag}(\Lambda)\mathbf{h})|_{(\mathbf{0},\mathbf{0})} = W - \operatorname{diag}(\Lambda)$, proving (3.4). Its characteristic polynomial factors over the two diagonal blocks, which gives (3.5). The Hurwitz equivalence and (3.6) follow by taking spectral abscissae and using $\alpha_p > 0$. ∎

Proof of Corollary 3.2. By (3.4), adding $c\mathbf{1}$ to the gains subtracts cI_m from the transverse matrix and shifts every transverse eigenvalue by $-c$. Taking negative spectral abscissae gives (3.7), and setting $\Lambda = \mathbf{0}$ gives (3.8). The full-margin claims follow from $\alpha(\Lambda) = \min\{\alpha_p, \alpha_{\mathrm{aux}}(\Lambda)\}$ in Lemma 3.1. Finally, (3.3) gives the affine formula for H, and continuity of the spectral norm gives its asymptotic behavior. ∎

B.3. Representation space. For the representation freedom, reshape the r-th row of $K \in \mathbb{R}^{N\times N^2}$ as $K_r \in \mathbb{R}^{N\times N}$. Its contribution to component r is $\mathbf{w}^\top K_r \mathbf{w}$, which vanishes identically if and only if K_r is skew-symmetric: evaluation at $\mathbf{e}_i$ gives $(K_r)_{ii} = 0$, and evaluation at $\mathbf{e}_i + \mathbf{e}_j$ gives $(K_r)_{ij} + (K_r)_{ji} = 0$. There are $N(N-1)/2$ independent entries per component, so $\dim \mathcal{K} = N^2(N-1)/2$. Finally, two matrices H_1, H_2 represent the same homogeneous quadratic map if and only if $(H_1 - H_2)(\mathbf{w} \otimes \mathbf{w}) \equiv 0$. This proves that (3.10) parametrizes all representations of the fixed quadratic map.

B.4. Intrinsic growth and spectral-norm relaxation.

Proof of Proposition 3.5. Factor $P = P_f^\top P_f$ and $Q = Q_f^\top Q_f$. For any exact representation, write $\mathbf{q}(\mathbf{w}) = \sum_i w_i M_i \mathbf{w}$ and $G = \mathrm{col}_i(M_i^\top P + P M_i)$. Then

$$\begin{aligned} 2\mathbf{w}^\top P \mathbf{q}(\mathbf{w}) &= (\mathbf{w} \otimes \mathbf{w})^\top G \mathbf{w} \\ &= (P_f \mathbf{w} \otimes Q_f \mathbf{w})^\top J_Q (Q_f \mathbf{w}), \qquad J_Q = (P_f^\top \otimes Q_f^\top)^{-1} G Q_f^{-1}. \end{aligned}$$

Taking absolute values and applying the spectral-norm inequality gives

$$2|\mathbf{w}^\top P \mathbf{q}(\mathbf{w})| \le \|J_Q\| \, \|P_f \mathbf{w}\| \, \|Q_f \mathbf{w}\|^2 = \|J_Q\| \sqrt{V(\mathbf{w})}\, \mathbf{w}^\top Q \mathbf{w}.$$

Division by the positive denominator for $\mathbf{w} \neq 0$ proves $\beta \le \|J_Q\|$. The gauge identity leaves $\mathbf{q}$, and hence β, unchanged. Since J_Q is affine in μ, its image is a closed affine subspace of a finite-dimensional matrix space. Intersecting this image with the closed norm ball of radius $\|J_Q(H)\|$ gives a nonempty compact set on which the norm attains its minimum. This proves (3.14) and attainment. ∎

B.5. Pullback connectedness.

Proof of Corollary 3.6. Fix $\mathbf{x}_0 \in S(\Lambda, \rho)$, set $P = P(\Lambda)$ and $\mathbf{w}_0 = \Phi(\mathbf{x}_0)$, and let $\mu^\star$ attain the minimum in (3.15). Put $j_\star := \|J(\Lambda, \mu^\star)\|$ and $\delta := 1 - \rho j_\star > 0$. For $\mathbf{w} \in \mathcal{D}(P, \rho)$, (3.13) gives $\dot V(\mathbf{w}) \le -\delta \|\mathbf{w}\|^2 \le -cV(\mathbf{w})$, where $c := \delta/\lambda_{\max}(P) > 0$. A first-exit argument makes $\mathcal{D}(P, \rho)$ positively invariant; its compactness and the

polynomiality of the lifted field imply that the solution $\mathbf{w}(t)$ from $\mathbf{w}_0$ is global. Gronwall's inequality gives $V(\mathbf{w}(t)) \le e^{-ct}V(\mathbf{w}_0)$, hence $\mathbf{w}(t) \to \mathbf{0}$. Exactness, $\mathbf{F}_\Lambda(\Phi(\mathbf{x})) = J_x\Phi(\mathbf{x})\mathbf{p}(\mathbf{x})$, and uniqueness yield $\mathbf{w}(t) = \Phi(\mathbf{x}(t))$; thus $\mathbf{x}(t)$ remains in $S(\Lambda,\rho)$ and converges to $\mathbf{0}$. Consequently, $s \mapsto \mathbf{x}(s/(1-s))$ for $0 \le s < 1$, extended by $\mathbf{0}$ at $s = 1$, is a continuous path in $S(\Lambda,\rho)$ from $\mathbf{x}_0$ to the origin. Every point admits such a path, so any two points can be joined through the origin. ∎

B.6. Derivation of the box-constrained formulation. Equation (3.15) already establishes the fixed-gain identity $\rho^\star(\Lambda) = \phi(\Lambda)^{-1}$, and Proposition 3.10 gives attainment on B. Hence, if $\phi_B > 0$, order reversal under inversion gives

$$\max_{\Lambda\in B} \rho^\star(\Lambda) = \left(\min_{\Lambda\in B} \phi(\Lambda)\right)^{-1} = \rho_B. \tag{B.1}$$

If $\phi_B = 0$, the same proposition supplies an optimizing pair with $J = 0$, so both sides of (B.1) are $+\infty$ under the convention in (3.17).

It remains to introduce the epigraph variable. For $J \in \mathbb{R}^{N^2\times N}$,

$$\|J\| \le t \quad\Longleftrightarrow\quad \begin{bmatrix} tI_{N^2} & J \\ J^\top & tI_N \end{bmatrix} \succeq 0. \tag{B.2}$$

Indeed, the block inequality forces $t \ge 0$. For $t > 0$, the Schur complement gives

$$tI_N - t^{-1}J^\top J \succeq 0 \quad\Longleftrightarrow\quad J^\top J \preceq t^2 I_N \quad\Longleftrightarrow\quad \|J\| \le t.$$

For $t = 0$, its 2×2 principal minors show that the block inequality holds if and only if $J = 0$, so (B.2) also covers the boundary case. Consequently,

$$\phi_B = \min_{\Lambda\in B,\ \mu\in\mathbb{R}^d,\ t\in\mathbb{R}} \left\{ t : \begin{bmatrix} tI_{N^2} & J(\Lambda,\mu) \\ J(\Lambda,\mu)^\top & tI_N \end{bmatrix} \succeq 0 \right\}.$$

Substituting (3.3), (3.10), (3.11), and (3.12) gives precisely (3.18). As noted in Remark 3.9, the Lyapunov equation uniquely fixes $P = P(\Lambda) \succ 0$; P_f is its upper Cholesky factor with positive diagonal, not an independent decision variable. Thus the substitution introduces no additional freedom and preserves the optimal value $t_B = \phi_B$.

B.7. Well-posedness of the box value.

Proof of Proposition 3.10. Write a gauge direction as $S = (S_i)_{i=1}^N$, so $K(\mathbf{w}\otimes\mathbf{w}) = 0$ is equivalent to $(S_i)_{rj} = -(S_j)_{ri}$. For $P = P^\top \succ 0$, let $T_P(S)_i := S_i^\top P + PS_i$. The bijection $U_i = PS_i$ gives $T_P(S)_i = U_i^\top + U_i$. Define

$$\mathcal{C} := \left\{ (B_i)_{i=1}^N \in (\mathbb{S}^N)^N : (B_i)_{jk} + (B_j)_{ki} + (B_k)_{ij} = 0 \text{ for all } i,j,k \right\},$$

which is independent of P. Gauge antisymmetry gives $\operatorname{im} T_P \subseteq \mathcal{C}$, while $\ker T_P$ corresponds under $U_i = PS_i$ to $\wedge^3\mathbb{R}^N$. Thus $\dim\operatorname{im} T_P = d - \binom{N}{3} =: q = N(N^2-1)/3$. The cyclic-sum map has rank $\binom{N+2}{3}$ because its restriction to fully symmetric tensors is multiplication by 3; hence $\dim\mathcal{C} = q$ and $\operatorname{im} T_P = \mathcal{C}$.

Fixing a basis of $\mathcal{C}$ gives a nonredundant parametrization $J(\Lambda,\nu) = J_0(\Lambda) + \mathcal{L}_\Lambda\nu$, $\nu \in \mathbb{R}^q$, with $\mathcal{L}_\Lambda$ injective by (3.12). On $\mathcal{F}$, the Lyapunov solution and its positive-diagonal Cholesky factor depend analytically on Λ; hence J_0 and $\mathcal{L}_\Lambda$ are locally Lipschitz. The case $q = 0$ is immediate. Otherwise, fix a closed ball $K \Subset \mathcal{F}$ and set

$c_K := \min_{\Lambda\in K,\,\|\nu\|=1}\|\mathcal{L}_\Lambda\nu\| > 0$ and $M_K := \max_{\Lambda\in K}\|J_0(\Lambda)\|$. The bound $\|J(\Lambda,\nu)\| \geq c_K\|\nu\| - M_K$ makes every inner problem coercive, hence attained; comparison with $\nu = 0$ places every minimizer in the common ball $\|\nu\| \leq R_K := 2M_K/c_K$. Uniform Lipschitz continuity of J on this set now gives

$$|\phi(\Lambda_1) - \phi(\Lambda_2)| \leq \sup_{\|\nu\|\leq R_K} \|J(\Lambda_1,\nu) - J(\Lambda_2,\nu)\| \leq L_K\|\Lambda_1 - \Lambda_2\|.$$

Thus ϕ is locally Lipschitz and attains its minimum on the compact box B. If $\phi_B > 0$, strict monotonicity of $x \mapsto 1/x$ gives the same optimizing gains; if $\phi_B = 0$, attainment gives an optimizing pair with $J = 0$, and (3.13) makes the corresponding quadratic Lyapunov function global. ∎

B.8. Gauge-defect correction. For a returned representation $\widetilde{H}$ that has a nonzero coefficient defect, choose ΔH such that $\mathbf{q}(\mathbf{w}) - \widetilde{H}(\mathbf{w}\otimes\mathbf{w}) = \Delta H(\mathbf{w}\otimes\mathbf{w})$ identically. Applying (A.3) to the defect gives the valid bound

$$\beta(\widehat{P}, Q, \mathbf{q}) \leq \|J_Q(\widetilde{H})\| + \frac{2\sqrt{\|\widehat{P}\|}\,\|\Delta H\|}{\lambda_{\min}(Q)}, \tag{B.3}$$

where J_Q is assembled using $\widehat{P}$. Thus a nonzero gauge residual requires a correction to the growth bound. If the gauge identity holds exactly, one may take $\Delta H = 0$. In numerical use, both terms in (B.3) and the normalized Lyapunov residual require upper bounds that include rounding errors.

B.9. Leading-order certified volume.

Proof of Proposition 3.14. Fix Λ and write $P = P(\Lambda)$. With the blocks in (3.20), set $q(\mathbf{x}) := \mathbf{x}^\top P_{xx}\mathbf{x}$ and

$$r(\mathbf{x}) := \widetilde{V}_\Lambda(\mathbf{x}) - q(\mathbf{x}) = 2\mathbf{x}^\top P_{xy}\mathbf{g}(\mathbf{x}) + \mathbf{g}(\mathbf{x})^\top P_{yy}\mathbf{g}(\mathbf{x}).$$

Since $\mathbf{g}(\mathbf{x}) = O(\|\mathbf{x}\|^2)$, there is $c > 0$ such that $|r(\mathbf{x})| \leq c\|\mathbf{x}\|^3$ near the origin. Let $\lambda_x := \lambda_{\min}(P_{xx}) > 0$. On $\mathbb{B}_\delta := \{\|\mathbf{x}\| \leq \delta\}$, for all sufficiently small δ,

$$(1-\varepsilon_\delta)q(\mathbf{x}) \leq \widetilde{V}_\Lambda(\mathbf{x}) \leq (1+\varepsilon_\delta)q(\mathbf{x}), \qquad \varepsilon_\delta := \frac{c\delta}{\lambda_x} < 1. \tag{B.4}$$

Let $E(s) := \{\mathbf{x} : q(\mathbf{x}) \leq s^2\}$. Since $P \succ 0$ and Φ contains $\mathbf{x}$, $\widetilde{V}_\Lambda(\mathbf{x}) \geq \lambda_{\min}(P)\|\mathbf{x}\|^2$; hence, for fixed δ and all sufficiently small ρ, both $S(\Lambda,\rho)$ and the inner ellipsoid below lie in $\mathbb{B}_\delta$. Equation (B.4) then gives

$$E\left(\frac{\rho}{\sqrt{1+\varepsilon_\delta}}\right) \subseteq S(\Lambda,\rho) \subseteq E\left(\frac{\rho}{\sqrt{1-\varepsilon_\delta}}\right).$$

Because $\operatorname{vol} E(s) = \kappa_n s^n/\sqrt{\det P_{xx}}$, division by the proposed leading term yields bounds $(1+\varepsilon_\delta)^{-n/2}$ and $(1-\varepsilon_\delta)^{-n/2}$. Letting first $\rho \downarrow 0$ and then $\delta \downarrow 0$ squeezes the ratio to one, proving (3.22). ∎

B.10. Gain invariance of the quadratic germ.

Proof of Proposition 3.15. By Lemma 3.1, $A(\Lambda) = \left[\begin{smallmatrix} J_p & R \\ 0 & D(\Lambda)\end{smallmatrix}\right]$, and only $D(\Lambda)$ depends on Λ. The $(1,1)$ block of the Lyapunov equation is $J_p^\top P_{Q,xx} + P_{Q,xx}J_p = -Q_{xx}$. Since J_p is Hurwitz, this equation has a unique solution, independent of Λ. ∎

Appendix C. Inner-oracle and outer-convergence details.

C.1. The gauge-image distance and its dual.

Proof of Proposition 4.1. Fix Λ and abbreviate $\mathcal{T} = \mathcal{T}_\Lambda$ and $\mathcal{W} = \mathcal{W}_\Lambda$. The attainable matrices are exactly $J_0 + \mathcal{W}$. Since $\mathcal{W} = -\mathcal{W}$, minimizing their spectral norm gives (4.1). The subspace $\mathcal{W}$ is closed and finite-dimensional. The intersection of $J_0 + \mathcal{W}$ with any nonempty bounded norm sublevel set is compact, so a minimizing matrix $J^\star$ exists; any preimage of $J^\star - J_0$ under $\mathcal{T}$ is a minimizing gauge.

Introduce a matrix variable M and impose $M = J_0 + \mathcal{T}\mu$. The Lagrangian is

$$L(M, \mu, Z) = \|M\| + \langle Z, J_0 + \mathcal{T}\mu - M\rangle_F.$$

Its infimum over μ is finite exactly when $\langle Z, \mathcal{T}\mu\rangle_F = 0$ for all μ, or $Z \in \mathcal{W}^\perp$. Spectral–nuclear norm duality gives $\langle Z, M\rangle_F \le \|Z\|_*\|M\|$, with $\sup_{\|M\|\le 1}\langle Z, M\rangle_F = \|Z\|_*$: equality follows by taking $M = UV^\top$ in a compact singular value decomposition $Z = U\Sigma V^\top$. Consequently,

$$\inf_M\{\|M\| - \langle Z, M\rangle_F\} = \begin{cases} 0, & \|Z\|_* \le 1, \\ -\infty, & \|Z\|_* > 1. \end{cases}$$

This yields the dual in (4.2). Its feasible set is nonempty and compact, so its maximum is attained.

For completeness, strong duality follows directly from convex optimality. The spectral norm is continuous everywhere, so the convex Fermat rule on $J_0 + \mathcal{W}$ gives

$$0 \in \partial\|\cdot\|(J^\star) + \mathcal{W}^\perp.$$

Hence some $Z^\star \in \partial\|\cdot\|(J^\star)$ lies in $\mathcal{W}^\perp$. The subgradient inequality for a norm is equivalent to

$$Z \in \partial\|\cdot\|(J) \quad\Longleftrightarrow\quad \|Z\|_* \le 1, \qquad \langle Z, J\rangle_F = \|J\|. \tag{C.1}$$

Indeed, applying that inequality at 0 and $2J$ forces equality at J, after which the inequality for arbitrary matrices is precisely the dual-norm bound. Thus $Z^\star$ is dual feasible and

$$\langle Z^\star, J_0\rangle_F = \langle Z^\star, J^\star\rangle_F = \|J^\star\| = \phi(\Lambda).$$

Conversely, such a $Z^\star$ bounds the norm of every matrix in $J_0 + \mathcal{W}$ from below by $\|J^\star\|$, proving the stated optimality criterion. Finally, weak duality gives $\langle Z, J_0\rangle_F \le \phi(\Lambda) \le \|J(\Lambda, \mu)\|$ for every primal–dual feasible pair, which proves (4.3). ∎

C.2. The reduced oracle and its dual.

Proof of Proposition 4.2. Let $X = P_f^\top \otimes I_N$. Since $P = P_f^\top P_f$, $P \otimes I_N = XX^\top$. Congruence of $F_\Lambda(\nu, \tau)$ by $\mathrm{diag}(X^{-1}, I_N)$, or equivalently its Schur complement, gives

$$F_\Lambda(\nu, \tau) \succeq 0 \quad\Longleftrightarrow\quad \begin{bmatrix} I_{N^2} & J_{\mathrm{r}}(\Lambda, \nu) \\ J_{\mathrm{r}}(\Lambda, \nu)^\top & \tau I_N \end{bmatrix} \succeq 0 \quad\Longleftrightarrow\quad J_{\mathrm{r}}^\top J_{\mathrm{r}} \preceq \tau I_N. \tag{C.2}$$

Minimizing τ therefore gives $\tau^\star = \min_\nu \|J_{\mathrm{r}}(\Lambda, \nu)\|^2 = \phi(\Lambda)^2$. Strict feasibility follows by fixing any ν and taking $\tau > \|J_{\mathrm{r}}(\Lambda, \nu)\|^2$. For $q > 0$, linear independence of the $\mathcal{B}_\ell$ and invertibility of X make $\nu \mapsto X^{-1}\sum_\ell \nu_\ell \mathcal{B}_\ell$ injective, hence bounded below on the unit sphere. The objective is therefore coercive in ν; for $q = 0$ attainment is immediate.

For the reconstruction, Proposition 3.10 proves that the image of the raw gauge under $S \mapsto \mathrm{col}_r(S_r^\top P + P S_r)$ is exactly $\mathcal{C}$. Thus the constraint in (4.6) is feasible. Its strictly convex quadratic objective selects a unique representative, and substitution in (3.12) adds precisely $\sum_\ell \nu_\ell^\star \mathcal{B}_\ell$ to $G(\Lambda, 0)$. Hence it realizes the same J, value, and vector field. ∎

For later use, the dual of (4.5) is

$$\begin{aligned} \max_{Y \succeq 0} \quad & -\langle Y_{11}, P \otimes I_N \rangle_F - 2\langle Y_{12}, G(\Lambda, 0) \rangle_F \\ \text{s.t.} \quad & \operatorname{tr} Y_{22} = 1, \qquad \langle Y_{12}, \mathcal{B}_\ell \rangle_F = 0, \quad \ell = 1, \dots, q, \end{aligned} \tag{C.3}$$

where Y is partitioned conformally with F_Λ. This follows from the Lagrangian $\tau - \langle Y, F_\Lambda \rangle_F$ by stationarity in (τ, ν). Strict feasibility gives strong duality; primal–dual optimality adds the complementarity condition in (4.12).

C.3. Proof of the sensitivity formula. The spectral subdifferential in (4.7) and the affine chain rule give

$$0 \in \partial_\nu \|J_{\mathrm{r}}(\Lambda, \nu^\star)\| \quad \Longleftrightarrow \quad \exists Z^\star \in \partial \|\cdot\|(J^\star) : \langle Z^\star, J_\ell(\Lambda) \rangle_F = 0 \quad (\ell = 1, \dots, q), \tag{C.4}$$

which proves (4.8) by the convex Fermat rule.

We next make the Cholesky derivative in (4.9) explicit. Set $C_i := P_f^{-\top} \dot{P}_i P_f^{-1}$ and let $\mathrm{up}_{1/2}(C_i)$ retain the strict upper triangle of C_i, take half of its diagonal, and set its strict lower triangle to zero. Then

$$\dot{P}_{f,i} = \mathrm{up}_{1/2}(C_i) P_f. \tag{C.5}$$

Indeed, with $R_i := \dot{P}_{f,i} P_f^{-1}$, differentiation of $P = P_f^\top P_f$ gives $C_i = R_i^\top + R_i$; the upper-triangularity of R_i determines it uniquely. Differentiating the block definition of $G(\Lambda, 0)$ gives the first line of (4.10), and differentiating $X J_{\mathrm{r}} = G_{\mathrm{r}}$ gives its second line. The sign in the Lyapunov sensitivity follows from $\dot{A}_i = -E_i$:

$$\dot{A}_i^\top P + A(\Lambda)^\top \dot{P}_i + \dot{P}_i A(\Lambda) + P \dot{A}_i = 0 \quad \Longleftrightarrow \quad A(\Lambda)^\top \dot{P}_i + \dot{P}_i A(\Lambda) = E_i P + P E_i.$$

Proof of Proposition 4.3. For $\Lambda \in \mathcal{U}$, define the Lyapunov operator on $\mathbb{S}^N$ by

$$\mathcal{A}_\Lambda(S) := A(\Lambda)^\top S + S A(\Lambda).$$

Since $A(\Lambda)$ is Hurwitz, this operator is invertible, and

$$P_Q(\Lambda) = -\mathcal{A}_\Lambda^{-1}(Q) = \int_0^\infty e^{A(\Lambda)^\top t} Q e^{A(\Lambda) t}\, dt \succ 0.$$

Indeed, the integral converges and solves the equation by integration of its time derivative. If $\mathcal{A}_\Lambda(S) = 0$, then $e^{A^\top t} S e^{A t}$ is constant in t and tends to zero, so $S = 0$; in finite dimensions this proves invertibility. The coefficients of $\mathcal{A}_\Lambda$ are affine in Λ, and matrix inversion is real analytic on the invertible matrices. Thus P_Q is real analytic on $\mathcal{U}$. Differentiating its defining equation, with Q fixed, gives

$$A_i^\top P_Q + A^\top P_{Q,i} + P_{Q,i} A + P_Q A_i = 0,$$

which proves (4.11); the derivative is unique by the same invertibility argument. These are local assertions and require only that $A(\Lambda)$ be Hurwitz at each point of $\mathcal{U}$.

The positive-diagonal Cholesky factor is real analytic on the positive definite cone, as follows from its recursive formulas using positive square roots and division by positive pivots. Hence P_f, X^{-1}, and J_{r} are real analytic for $Q = I_N$. The fixed linearly independent matrices $\mathcal{B}_\ell$ make the reduced gauge map injective throughout $\mathcal{U}$. For every closed ball $K \Subset \mathcal{U}$, the proof of Proposition 3.10 therefore supplies constants $c_K > 0$ and $M_K \geq 0$ such that

$$\|J_{\mathrm{r}}(\Lambda, \nu)\| \geq c_K \|\nu\| - M_K, \qquad \|\nu^\star(\Lambda)\| \leq 2M_K/c_K \quad (\Lambda \in K)$$

when $q > 0$. Minimization is consequently restricted to a common compact ball in ν, on which J_{r} is uniformly Lipschitz in the gain. Comparing the two objectives at each other's minimizers proves local Lipschitz continuity of ϕ, as detailed in that proof. When $q = 0$, there is no inner variable and the conclusion follows directly from the smoothness of J_{r} and the norm inequality.

The conic KKT conditions follow from (C.3) and strong duality. Write $\phi = \phi(\Lambda) > 0$ and choose paired orthonormal bases U_1, V_1 of the leading singular subspaces of $J^\star$. Solving the block equations for the kernel of $F_\Lambda(\nu^\star, \phi^2)$ shows that it is spanned by the columns of

$$T := \begin{bmatrix} -\phi X^{-\top} U_1 \\ V_1 \end{bmatrix}.$$

Indeed, the first block equation determines the upper block as $-X^{-\top} J^\star v$, and the second becomes $((J^\star)^\top J^\star - \phi^2 I_N)v = 0$. For two positive semidefinite matrices, zero Frobenius inner product implies zero product. Complementarity therefore places the range of $Y^\star$ in this kernel, so

$$Y^\star = T \Theta T^\top, \qquad \Theta \succeq 0, \qquad \operatorname{tr} \Theta = \operatorname{tr} Y_{22}^\star = 1. \tag{C.6}$$

It follows that

$$-\phi^{-1} X^\top Y_{12}^\star = U_1 \Theta V_1^\top =: Z^\star, \qquad \|Z^\star\|_* = \operatorname{tr} \Theta = 1.$$

Equation (4.7) gives $Z^\star \in \partial \|\cdot\|(J^\star)$, and conic stationarity gives

$$\langle Z^\star, J_\ell(\Lambda) \rangle_F = -\phi^{-1} \langle Y_{12}^\star, \mathcal{B}_\ell \rangle_F = 0.$$

Proposition 4.1 now proves that $Z^\star$ is a distance-dual optimum.

To relate the two sensitivity formulas, put $\dot{X}_i = \dot{P}_{f,i}^\top \otimes I_N$. The blocks in (C.6) are

$$Y_{11}^\star = \phi^2 X^{-\top} U_1 \Theta U_1^\top X^{-1}, \qquad Y_{12}^\star = -\phi X^{-\top} Z^\star.$$

Using $\dot{P}_i \otimes I_N = \dot{X}_i X^\top + X \dot{X}_i^\top$ and $J^\star V_1 = \phi U_1$ gives

$$\langle Y_{11}^\star, \dot{P}_i \otimes I_N \rangle_F = 2\phi \langle Z^\star, X^{-1} \dot{X}_i J^\star \rangle_F, \qquad \langle Y_{12}^\star, \dot{G}_i^0 \rangle_F = -\phi \langle Z^\star, X^{-1} \dot{G}_i^0 \rangle_F.$$

Together with (4.10), these identities prove

$$2\phi \langle Z^\star, \dot{J}_i \rangle_F = -\langle Y_{11}^\star, \dot{P}_i \otimes I_N \rangle_F - 2 \langle Y_{12}^\star, \dot{G}_i^0 \rangle_F. \tag{C.7}$$

It remains to establish the outer subgradient claim. The dual feasible set

$$\mathcal{Y} := \{ Y \succeq 0 : \operatorname{tr} Y_{22} = 1, \quad \langle Y_{12}, \mathcal{B}_\ell \rangle_F = 0 \ (\ell = 1, \ldots, q) \}$$

is independent of the gain. Fix the optimum $Y^\star$ at Λ and define

$$d_{Y^\star}(\Lambda') := -\langle Y_{11}^\star, P(\Lambda') \otimes I_N \rangle_F - 2\langle Y_{12}^\star, G(\Lambda', 0) \rangle_F.$$

Weak duality gives $d_{Y^\star}(\Lambda') \le \phi(\Lambda')^2$ for every $\Lambda' \in \mathcal{F}$, with equality at Λ. Since $d_{Y^\star}$ is smooth and $d_{Y^\star}(\Lambda) = \phi^2 > 0$, it is positive in a neighborhood of Λ. There $h := \sqrt{d_{Y^\star}}$ is a smooth lower bound for ϕ, with $h(\Lambda) = \phi(\Lambda)$ and $\partial_{\lambda_i} h(\Lambda) = g_i$ by (C.7). Taylor expansion gives

$$\phi(\Lambda + \Delta) \ge \phi(\Lambda) + g^\top \Delta + o(\|\Delta\|).$$

Thus g is a Fréchet subgradient and, by local Lipschitz continuity, belongs to $\partial_C \phi(\Lambda)$. If ϕ is differentiable at Λ, the same inequality applied in opposite directions gives $g = \nabla\phi(\Lambda)$.

Finally, suppose that $\nu^\star$ is unique and the leading singular value of $J^\star$ is positive and simple. Set $f(\Lambda', \nu) := \|J_{\mathrm{r}}(\Lambda', \nu)\|$. Then f is continuously differentiable near $(\Lambda, \nu^\star)$, and $a := \nabla_\Lambda f(\Lambda, \nu^\star)$ has components $a_i = \langle uv^\top, \dot{J}_i \rangle_F$. For each sufficiently small Δ, choose a minimizer ν_Δ at $\Lambda + \Delta$. Local uniform boundedness and continuity imply that every cluster point of ν_Δ as $\Delta \to 0$ minimizes $f(\Lambda, \cdot)$; uniqueness therefore gives $\nu_\Delta \to \nu^\star$. Optimality at the two gains yields

$$\begin{aligned} f(\Lambda + \Delta, \nu_\Delta) - f(\Lambda, \nu_\Delta) &\le \phi(\Lambda + \Delta) - \phi(\Lambda) \\ &\le f(\Lambda + \Delta, \nu^\star) - f(\Lambda, \nu^\star). \end{aligned}$$

Both bounds equal $a^\top \Delta + o(\|\Delta\|)$ by continuity of $\nabla_\Lambda f$ and $\nu_\Delta \to \nu^\star$. Thus ϕ is differentiable and

$$\partial_{\lambda_i} \phi(\Lambda) = \langle uv^\top, \dot{J}_i \rangle_F = u^\top \dot{J}_i v, \tag{C.8}$$

which proves (4.15). ∎

At a nonsmooth point with $\phi(\Lambda) > 0$, different exact optimal multipliers can select different Clarke subgradients. An individual selection need not determine the nearby-gradient hull in (4.16). The outer convergence theorem therefore retains its assumption that the sampled hulls converge to the Clarke subdifferential [10].

Finally, the damping in (4.18) gives $s^\top \widetilde{r} \ge \kappa s^\top M s > 0$. Thus the BFGS update is positive definite whenever $M \succ 0$ and $s \ne 0$; rebuilding from finitely many such pairs and clipping its eigenvalues gives the metric safeguards assumed in Theorem 4.5.

C.4. Proof of conditional Clarke stationarity. Recall that $\psi^\circ(x; d) := \limsup_{z \to x,\, t \downarrow 0} [\psi(z + td) - \psi(z)]/t$ denotes the Clarke directional derivative.

Proof of Theorem 4.5. The bounds in (4.19) give $\theta^k + d_k \in C$. Convexity of C then makes every backtracked point $\theta^k + a d_k$, $a \in [0, 1]$, feasible. A serious step satisfies (4.22), whereas a null step leaves the iterate unchanged; hence $\psi(\theta^k)$ is nonincreasing. Compactness of C gives accumulation points.

Fix a subsequence $\theta^{k_r} \to \bar{\theta}$ and suppose that $0 \notin \partial_C \psi(\bar{\theta}) + N_C(\bar{\theta})$. We first show that the corresponding residuals are bounded away from zero. Otherwise, after passing to a subsequence, $\chi_{k_r} \to 0$; the metric safeguards imply $d_{k_r} \to 0$. By (4.25), the aggregates $\bar{g}_{k_r} \in \widehat{\partial}_{k_r}$ in Lemma 4.4 are bounded, and a further subsequence converges to some $\bar{g} \in \partial_C \psi(\bar{\theta})$. Equation (4.20) makes $n_{k_r} = -M_{k_r} d_{k_r} - \bar{g}_{k_r}$ bounded. Since the convex normal-cone mapping has a closed graph and $n_{k_r} \in N_C(\theta^{k_r} + d_{k_r})$, its limit satisfies $n \in N_C(\bar{\theta})$ and $\bar{g} + n = 0$, a contradiction. Consequently, for some $\delta > 0$,

$$\chi_{k_r}^2 = d_{k_r}^\top M_{k_r} d_{k_r} \ge \delta \tag{C.9}$$

for all sufficiently large r.

The displacements lie in the compact set $C - C$, and the safeguarded metrics lie in a compact set. Along any further subsequence of the chosen indices, take $d_{k_r} \to \bar{d} \neq 0$ and $M_{k_r} \to \bar{M}$. Hausdorff convergence in (4.25) implies uniform convergence of the support functions on bounded sets. Lemma 4.4 therefore yields

$$\psi^\circ(\bar{\theta}; \bar{d}) = \max_{g \in \partial_C \psi(\bar{\theta})} g^\top \bar{d} = \lim_{r \to \infty} \max_{g \in \widehat{\partial}_{k_r}} g^\top d_{k_r} \leq -\bar{d}^\top \bar{M} \bar{d} < 0. \tag{C.10}$$

Because $c < 1$, joint upper semicontinuity of the Clarke directional derivative and continuity of $d^\top M d$ imply that sufficiently small steps satisfy

$$\psi(\theta + ad) - \psi(\theta) \leq -ca\, d^\top M d. \tag{C.11}$$

The cluster set of (d_{k_r}, M_{k_r}) is compact; a finite-cover argument makes the upper step bound $a_0 > 0$ uniform for all sufficiently large r. Equation (C.9) and $\omega_{k_r} \downarrow 0$ exclude the low-residual branch. Since $L_{k_r} \to \infty$, the grid in (4.22) eventually accepts a step satisfying $\beta^{\ell_{k_r}} \geq \beta \min\{1, a_0\}$. Hence every sufficiently large r decreases ψ by at least $c\beta \min\{1, a_0\}\delta > 0$, contradicting nonnegativity and monotone convergence of the values. Thus every accumulation point satisfies (4.26).

The affine change of variables gives $\partial_C \psi(\theta) = D_B^\top \partial_C \phi(\Lambda)$ and $N_C(\theta) = D_B^\top N_B(\Lambda)$, so stationarity is equivalent in the two coordinates. If $\phi_B > 0$, the gradient-limit characterization applied to the C^1 map $s \mapsto -1/s$ gives $\partial_C(-1/\phi) = \phi^{-2} \partial_C \phi$ on B. Since N_B is a cone, this is precisely the Clarke condition for maximizing $\rho^\star = 1/\phi$. □

Appendix D. Benchmark definitions and supplementary results.

D.1. Heterogeneous suite. The suite was assembled to vary polynomial degree, state and lift dimension, and nonlinear coupling, rather than sampled from a statistical population. It contains standard models and examples used in the stability literature (reversed Van der Pol, Duffing, a Chafee–Infante semidiscretization, and systems drawn or adapted from [7, 21, 22]); scalar degree probes and synthetic cascades, chains, and rings; and the already-quadratic controls `quad-cross`, `quad-2d`, and `liu-10d`. The networked Van der Pol system is the deterministic local variant motivated by the interconnected model in [21]. Table 2 records the exact executed vector fields, dimensions, and construction or literature source.

The retained geometry test, `shear-cubic-a1`, is obtained by pushing forward the base field

$$g(z) = \begin{bmatrix} -z_1 - z_2 + z_1 \|z\|^2 \\ z_1 - z_2 + z_2 \|z\|^2 \end{bmatrix} \tag{D.1}$$

under $T_1(z) = (z_1, z_2 + z_1^2)$. Since $\frac{d}{dt}\|z\|^2 = 2\|z\|^2(\|z\|^2 - 1)$, the base field has the open unit disk as its basin. The identity $\det\big(DT_1(z)\big) = 1$ shows that T_1 preserves area. Its basin is the sheared unit disk

$$T_1\big(\{z : \|z\| < 1\}\big) = \big\{x : x_1^2 + (x_2 - x_1^2)^2 < 1\big\}, \tag{D.2}$$

which has area π and a quartic boundary. This gives a coordinate-shape benchmark with known basin area.

Table 2: Executed vector fields in the heterogeneous suite. The dimension tuple is $(n, \deg f, N, m)$.

System	$(n, \deg f, N, m)$	Dynamics	Provenance
Reference systems and scalar degree probes			
`vanderpol`	$(2, 3, 3, 1)$	$\dot{x_1} = -x_2$; $\dot{x_2} = x_1 + x_2\,(x_1^2 - 1)$	reversed Van der Pol [22, Ex. 6]
`planar-quintic`	$(2, 5, 4, 2)$	$\dot{x_1} = -x_1 + x_2$; $\dot{x_2} = -x_1^5 - 2x_2$	Section 5.1
`cubic`	$(1, 3, 2, 1)$	$\dot{x_1} = x_1^3 - x_1$	[7, Ex. 2]
`quintic`	$(1, 5, 3, 2)$	$\dot{x_1} = x_1^5 - x_1$	scalar degree probe
`bistable`	$(1, 3, 2, 1)$	$\dot{x_1} = -x_1^3 + 0.4x_1^2 - 0.03x_1$	bistable reaction [7, §5.2]
`duffing`	$(2, 3, 3, 1)$	$\dot{x_1} = x_2$; $\dot{x_2} = x_1^3 - x_1 - x_2$	Duffing oscillator [7, §5.1]
Quadratic controls ($m = 0$)			
`quad-cross`	$(2, 2, 2, 0)$	$\dot{x_1} = x_1x_2 - x_1$; $\dot{x_2} = x_1^2 - 2x_2$	synthetic quadratic control
`quad-2d`	$(2, 2, 2, 0)$	$\dot{x_1} = -x_1 + x_2^2$; $\dot{x_2} = x_1x_2 - x_2$	synthetic quadratic control
Synthetic degree and cascade probes			
`cubic-3d`	$(3, 3, 7, 4)$	$\dot{x_1} = -x_1 + x_2^3$; $\dot{x_2} = -2x_2 + x_3^2$; $\dot{x_3} = x_1^2 - 3x_3$	mixed-degree coupling probe
`cascade-3d`	$(3, 3, 6, 3)$	$\dot{x_1} = -x_1 + x_2^3$; $\dot{x_2} = -x_2 + x_3^3$; $\dot{x_3} = -x_3$	triangular cascade
`septic`	$(1, 7, 4, 3)$	$\dot{x_1} = x_1^7 - x_1$	scalar degree probe
`septic-2d`	$(2, 7, 5, 3)$	$\dot{x_1} = -x_1 + x_2$; $\dot{x_2} = -x_1^7 - 2x_2$	degree-seven cascade
Semidiscrete, chain, and ring models			
`chafee-3`	$(3, 3, 6, 3)$	$\dot{x_1} = x_1^3 - 1.4x_1 + 0.2x_2$; $\dot{x_2} = 0.2x_1 + x_2^3 - 1.4x_2 + 0.2x_3$; $\dot{x_3} = 0.2x_2 + x_3^3 - 1.4x_3$	Chafee–Infante-type semidiscretization, $u_t = \nu u_{xx} - u + u^3$ (three nodes)
`duffing-chain-2`	$(4, 3, 6, 2)$	$\dot{q_1} = v_1$; $\dot{v_1} = q_1^3 - 2.0q_1 + 0.5q_2 - 1.0v_1$; $\dot{q_2} = v_2$; $\dot{v_2} = 0.5q_1 + q_2^3 - 2.0q_2 - 1.0v_2$	Duffing-chain variant; cf. [7, §5.3]
`duffing-chain-3`	$(6, 3, 9, 3)$	$\dot{q_1} = v_1$; $\dot{v_1} = q_1^3 - 2.0q_1 + 0.5q_2 - 1.0v_1$; $\dot{q_2} = v_2$; $\dot{v_2} = 0.5q_1 + q_2^3 - 2.0q_2 + 0.5q_3 - 1.0v_2$; $\dot{q_3} = v_3$; $\dot{v_3} = 0.5q_2 + q_3^3 - 2.0q_3 - 1.0v_3$	Duffing-chain variant; cf. [7, §5.3]
`cubic-ring-3`	$(3, 3, 9, 6)$	$\dot{x_1} = -x_1 + x_2^3$; $\dot{x_2} = -2x_2 + x_3^3$; $\dot{x_3} = x_1^3 - 3x_3$	directed cubic ring
Literature-derived network models			
`networked-vdp-2`	$(4, 3, 6, 2)$	$\dot{x_1} = -x_2$; $\dot{x_2} = -\frac{x_1x_4}{10} + x_1 + x_2\left(\frac{4x_1^2}{5} - \frac{4}{5}\right)$; $\dot{x_3} = -x_4$; $\dot{x_4} = -\frac{x_2x_3}{10} + x_3 + x_4\left(\frac{11x_3^2}{5} - \frac{11}{5}\right)$	deterministic network variant [21]
`liu-10d`	$(10, 2, 10, 0)$	$\dot{x_1} = -x_1 + 0.5x_2 - 0.1x_9^2$; $\dot{x_2} = -0.5x_1 - x_2$; $\dot{x_3} = -0.1x_1^2 - x_3 + 0.5x_4$; $\dot{x_4} = -0.5x_3 - x_4$; $\dot{x_5} = -x_5 + 0.5x_6 + 0.1x_7^2$; $\dot{x_6} = -0.5x_5 - x_6$; $\dot{x_7} = -x_7 + 0.5x_8$; $\dot{x_8} = -0.5x_7 - x_8$; $\dot{x_9} = 0.5x_{10} - x_9$; $\dot{x_{10}} = -x_{10} + 0.1x_2^2 - 0.5x_9$	[22, Ex. 8]
Prescribed-basin geometry test			
`shear-cubic-a1`	$(2, 6, 6, 4)$	$\dot{x_1} = x_1^5 - 2x_1^3x_2 + x_1^3 + x_1^2 + x_1x_2^2 - x_1 - x_2$; $\dot{x_2} = x_1^6 - x_1^4x_2 + x_1^4 + 2x_1^3 - x_1^2x_2^2 + x_1^2x_2 - x_1^2 - 2x_1x_2 + x_1 + x_2^3 - x_2$	area-preserving shear of (D.1)

D.2. Single-link relay family. For odd $p \geq 3$ and $n \geq 2$, the relay family is

$$\begin{aligned} \dot{x}_i &= a_i(-x_i + x_{i+1}), \qquad i = 1, \ldots, n-1, \\ \dot{x}_n &= a_n(-x_n + x_1^p), \qquad a_i = \frac{2 + ((i-1) \bmod 3)}{2}. \end{aligned} \tag{D.3}$$

The executed grid is $(p, n) \in \{3, 5, 7\} \times \{2, 4, 6, 8\}$. Its origin Jacobian is upper triangular with diagonal entries $-a_i$, and hence is Hurwitz. Writing $q = (p-1)/2$, exact symbolic reconstruction verifies the DQBEE auxiliaries $x_1^2, \ldots, x_1^{q+1}$, so $m = q$ and $N = n + q$ in every cell. The two nonzero equilibria $\pm\mathbf{1}$ motivate retaining the directions $\pm\mathbf{1}/\sqrt{n}$ in the common geometry set.

The grid, arm definitions, execution order, and retirement rule were frozen before

TABLE 3

Absolute results corresponding to Table 1. Each complete cell gives the three-repeat median (M/T) to two significant digits, with T in seconds and $M \in \{L, A, V, R\}$. Bold and the $--^{R}$, $--^{A}_{k/3}$, and TO conventions follow Table 1.

			Proposed	Lifted baselines		Direct SOS		
System or (p,n)	$n/N/m$	M	ODQ	Base	Gauge	SOS–2	SOS–4(2)	SOS–4(8)
Reference systems and scalar degree probes								
`vanderpol`	2/3/1	A	(2.7/0.3)	(0.9/0.11)	(2/0.12)	(6.5/0.55)	(6.6/5.6)	(6.6/15)
`planar-quintic`	2/4/2	A	**(35/0.72)**	(7.3/0.13)	(16/0.14)	(10/1.2)	(14/15)	(30/47)
`cubic`	1/2/1	L	(1.9/0.2)	(1.3/0.12)	(1.3/0.12)	(2/0.24)	(2/1.6)	(2/4.1)
`quintic`	1/3/2	L	(1.3/0.74)	(0.92/0.13)	(0.94/0.14)	(2/0.34)	(2/2.8)	(2/3)
`bistable`	1/2/1	R	(0.051/0.4)	(0.00089/0.11)	(0.0026/0.12)	(0.1/0.27)	(0.1/1.4)	(0.1/1.7)
`duffing`	2/3/1	A	(3.2/0.19)	(1.2/0.095)	(2.1/0.1)	(3.6/0.53)	(4/5.9)	(4/6.7)
Quadratic controls ($m = 0$)								
`quad-cross`	2/2/0	A	(8.9/0.0098)	(5.9/0.0034)	(8.9/0.0093)	(11/0.48)	(18/5.5)	(16/14)
`quad-2d`	2/2/0	A	(3.9/0.0096)	(3.9/0.0039)	(3.9/0.0096)	(5.3/0.47)	(9.6/5.6)	(8.1/10)
Synthetic degree and cascade probes								
`cubic-3d`	3/7/4	V	(36/4.3)	(8.5/0.15)	(19/0.21)	(26/0.9)	(33/24)	(29/79)
`cascade-3d`	3/6/3	V	(32/2.3)	(9.6/0.13)	(17/0.16)	(22/0.9)	(24/23)	(30/74)
`septic`	1/4/3	L	(1.1/1.1)	(0.76/0.13)	(0.78/0.14)	(2/0.42)	(2/3.3)	(2/3.7)
`septic-2d`	2/5/3	A	**(18/1.6)**	(5.7/0.15)	(10/0.17)	(7.5/2.9)	(10/29)	(16/96)
Semidiscrete, chain, and ring models								
`chafee-3`	3/6/3	V	(6.4/1.8)	(2.8/0.19)	(2.9/0.22)	(6.7/1.2)	(5.9/23)	(5.9/81)
`duffing-chain-2`	4/6/2	R	(0.98/1.1)	(0.58/0.15)	(0.71/0.19)	(1/3.6)	(1.2/79)	(1.2/280)
`duffing-chain-3`	6/9/3	R	(0.95/6.4)	(0.55/0.15)	(0.7/0.28)	(1/13)	TO	TO
`cubic-ring-3`	3/9/6	V	(33/11)	(7.8/0.22)	(19/0.35)	(26/0.94)	(47/22)	$--^{A}_{1/3}$
Literature-derived network models								
`networked-vdp-2`	4/6/2	R	(0.48/1.1)	(0.3/0.13)	(0.45/0.16)	(0.84/1.9)	(0.95/78)	(0.95/120)
`liu-10d`	10/10/0	R	**(19/0.29)**	(14/0.074)	(19/0.32)	(10/45)	TO	TO
Prescribed-basin geometry test								
`shear-cubic-a1`	2/6/4	A	(0.63/3.4)	(0.072/0.4)	(0.16/0.43)	(1/3)	(1.9/31)	$--^{A}_{2/3}$
Single-link relay family (D.3)*; instances are indexed by* (p,n)								
$(3,2)$	2/3/1	R	(1.2/1.7)	(0.87/1.7)	(1.1/1.6)	(1/1.8)	$--^{A}_{2/3}$	(1.2/17)
$(3,4)$	4/5/1	R	**(0.89/2.3)**	(0.45/1.6)	(0.88/1.8)	(0.68/4.5)	$--^{A}_{0/3}$	$--^{A}_{0/3}$
$(3,6)$	6/7/1	R	**(0.7/3.5)**	(0.35/1.5)	(0.7/1.6)	(0.56/13)	$--^{A}_{0/3}$	$--^{A}_{0/3}$
$(3,8)$	8/9/1	R	**(0.53/5.7)**	(0.22/1.7)	(0.53/1.6)	(0.45/31)	TO	$--^{A}_{0/3}$
$(5,2)$	2/4/2	R	(1/1.9)	(0.72/1.7)	(0.94/1.8)	(0.96/2.5)	(1.2/14)	$--^{A}_{0/3}$
$(5,4)$	4/6/2	R	(0.8/2.8)	(0.36/1.8)	(0.77/1.9)	(0.65/17)	(1.3/310)	TO
$(5,6)$	6/8/2	R	**(0.62/2.9)**	(0.28/1.8)	(0.61/1.7)	(0.55/150)	TO	$--^{R}$
$(5,8)$	8/10/2	R	(0.48/7.7)	(0.18/1.6)	(0.47/1.8)	TO	$--^{R}$	$--^{R}$
$(7,2)$	2/5/3	R	(0.94/2.2)	(0.6/1.6)	(0.85/1.4)	(0.93/3.6)	(1.2/25)	(1.2/57)
$(7,4)$	4/7/3	R	**(0.74/3.8)**	(0.3/1.9)	(0.71/1.6)	(0.63/77)	TO	TO
$(7,6)$	6/9/3	R	(0.57/5.6)	(0.23/1.6)	(0.56/1.8)	TO	$--^{R}$	$--^{R}$
$(7,8)$	8/11/3	R	(0.45/13)	(0.15/1.7)	(0.44/2)	$--^{R}$	$--^{R}$	$--^{R}$

this relay cohort was run. Each arm has three isolated repeats and an independent 600-second deadline; a timeout or completed budget crossing retires only the same method at larger n within the same degree and repeat. Geometry has a separate 120-second budget and uses 4096 seeded random directions, the $2n$ signed coordinate directions, and the two equilibrium directions, with the same radial scan and bisection resolution as above.

Table 1 gives the normalized comparison used in the main text, while Table 3 reports the absolute score and construction-time medians for all six methods and both cohorts.

D.3. Supplementary geometry. For the planar quintic, write $x = x_1$, $y = x_2$, let $z(x) = (x, x^2, x^3)^\top$, and partition the metric with respect to (z, y) as $\left[\begin{smallmatrix} C & b \\ b^\top & a \end{smallmatrix}\right]$. Since $a > 0$, completing the square gives

$$V(x,y) = a\big(y + a^{-1}b^\top z(x)\big)^2 + z(x)^\top (C - a^{-1}bb^\top) z(x).$$

TABLE 4

Maximum stored numerical diagnostics at the historical ODQ returns. Columns include every returned certificate in the corresponding cohort; no new optimization was run to produce this table.

Diagnostic	Heterogeneous (57)	Relay (36)
Stored normalized Lyapunov residual bound	8.02×10^{-15}	4.81×10^{-16}
Certificate-map reconstruction residual ($\|\cdot\|_F$)	8.15×10^{-14}	2.92×10^{-14}
Epigraph discrepancy $\lvert\sqrt{\tau}-\|J\|_2\rvert$	2.45×10^{-7}	7.75×10^{-9}
Epigraph discrepancy/ $\max\{1,\|J\|_2\}$	9.87×10^{-8}	7.54×10^{-9}
Gauge KKT residual $\max_k \lvert\langle Z, J_k\rangle\rvert$	4.45×10^{-12}	2.25×10^{-11}
Generalized-Schur dual normalization residual	1.99×10^{-8}	9.64×10^{-9}

Writing $f(x)=\rho_{\text{safe}}^2-z(x)^\top(C-a^{-1}bb^\top)z(x)$, the vertical slice has width $2\sqrt{f(x)/a}$ wherever $f(x)>0$. We isolate all real roots of this degree-six polynomial and integrate over every positive interval. This does not assume a star-shaped set; all sampled quintic sets have one component. Symbolic checks verify the square completion, including examples with disconnected sets and repeated roots. At the saved Gauge and ODQ points, the area agrees with the original 8001-slice estimates to relative errors 1.73×10^{-6} and 1.05×10^{-7}. Root isolation interprets the input floats as rational numbers; the subsequent quadrature and its error estimates remain floating-point. The new grid comprises 145 fixed-gain SDP solves.

D.4. Numerical diagnostics, failures, and timing. We read the source-locked raw artifacts for all 57 heterogeneous and 36 relay ODQ returns, including the nine heterogeneous $m=0$ controls. Table 4 summarizes the diagnostics stored at those returned certificates. These are floating-point residual screens, not interval or rational proofs. The epigraph and dual residuals measure optimization accuracy; they do not replace the independent residual-aware certificate condition. Raw residuals have different scales across systems and dimensions. The additional normalized epigraph diagnostic uses $\max\{1,\|J\|_2\}$ only as a descriptive scale.

The heterogeneous outer exits comprise 30 finite-bundle tolerance exits, 18 oracle-budget exits, and nine vacuous exits with no gain variables; all 36 relay runs reach the finite-bundle tolerance. These are not exact Clarke-stationarity certificates. The stored outer residual is evaluated at a point different from the best returned certificate gain in 33/57 heterogeneous and 9/36 relay runs. Every returned ODQ certificate passes its separate numerical certificate screen, and no reported inner solve changes the requested MOSEK/generalized-Schur backend. The historical files do not retain all oracle evaluations, so they do not support a count of every filtered inner candidate.

The six heterogeneous timeouts of each quartic arm occur on `duffing-chain-3` and `liu-10d`, in all three repeats. The three additional SOS–4(8) failures comprise two returns on `cubic-ring-3` and one on `shear-cubic-a1`; one explicitly fails the prescribed radius-shrink screen and two return no valid certificate. A solver-indeterminate outcome in the relay cohort is not interpreted as mathematical infeasibility.

For each heterogeneous ODQ run, the saved construction time is the sum of lift construction, stable-box screening, the complete inner-oracle calls, gradient sensitivity, and remaining outer logic. The numerical bookkeeping agrees with this partition in all 57 records. Lyapunov solves, matrix assembly, modeling-interface work, SDP solution, and certificate auditing are included within the inner-oracle component. In particular, `t_inner_sdp_total` is not solver-only time. Saved Lyapunov and inner audit counters are nested diagnostics and are not added again. Table 6 reports arithmetic means over the three repeats so the partition remains additive before rounding.

TABLE 5

Outcomes of all predeclared method–case–repeat cells: 57 per heterogeneous method and 36 per relay method. Indet. denotes a solver-indeterminate return; failed denotes another unsuccessful certificate return. Retired relay cells were not executed after an earlier resource-limit event and are counted separately from observed failures.

Cohort	Method	Success	Timeout	Indet.	Failed	Retired
Heterogeneous	Base	57	0	0	0	0
Heterogeneous	Gauge	57	0	0	0	0
Heterogeneous	ODQ	57	0	0	0	0
Heterogeneous	SOS–2	57	0	0	0	0
Heterogeneous	SOS–4(2)	51	6	0	0	0
Heterogeneous	SOS–4(8)	48	6	0	3	0
Relay	Base	36	0	0	0	0
Relay	Gauge	36	0	0	0	0
Relay	ODQ	36	0	0	0	0
Relay	SOS–2	27	6	0	0	3
Relay	SOS–4(2)	11	9	7	0	9
Relay	SOS–4(8)	6	9	8	0	13

TABLE 6

Mean heterogeneous ODQ construction-time partition over three repeats, in seconds. Lift: quadratization; box: stability screen; inner: complete inner oracle; grad.: gradient sensitivity; outer: remaining logic. Geometry, startup, warm-up, and serialization are excluded. Displayed components may differ from the displayed total by rounding.

System	Lift	Box	Inner	Grad.	Outer	Total
`bistable`	0.113	0.0116	0.229	0.0127	0.0439	0.411
`cascade-3d`	0.133	7.29×10^{-4}	2.22	0.0927	0.0846	2.53
`chafee-3`	0.215	7.47×10^{-4}	1.5	0.0695	0.0633	1.85
`cubic`	0.115	3.14×10^{-4}	0.0475	6.69×10^{-3}	0.0268	0.196
`cubic-3d`	0.165	1.18×10^{-3}	4.16	0.151	0.0793	4.56
`cubic-ring-3`	0.245	5.19×10^{-3}	10.2	0.237	0.0666	10.8
`duffing`	0.0955	3.65×10^{-4}	0.0643	7.3×10^{-3}	0.0244	0.192
`duffing-chain-2`	0.145	5.07×10^{-4}	0.93	0.039	0.053	1.17
`duffing-chain-3`	0.178	1.06×10^{-3}	6.52	0.101	0.0733	6.87
`liu-10d`	0.0806	0	0.211	0	1.5×10^{-4}	0.292
`networked-vdp-2`	0.158	6.85×10^{-4}	0.985	0.0354	0.0504	1.23
`planar-quintic`	0.126	4.78×10^{-4}	0.523	0.0285	0.0483	0.727
`quad-2d`	4.93×10^{-3}	0	6×10^{-3}	0	1.07×10^{-4}	0.011
`quad-cross`	3.24×10^{-3}	0	6.81×10^{-3}	0	1.23×10^{-4}	0.0102
`quintic`	0.128	4.13×10^{-4}	0.549	0.0266	0.051	0.755
`septic`	0.153	6.07×10^{-4}	0.966	0.0668	0.0705	1.26
`septic-2d`	0.161	6.16×10^{-4}	1.47	0.0855	0.0818	1.8
`shear-cubic-a1`	0.396	0.0803	2.71	0.138	0.0812	3.4
`vanderpol`	0.112	3.87×10^{-4}	0.142	0.0129	0.0415	0.308

The relay artifacts instead retain launch-to-serialization wall time and separately timed geometry extraction, without an inner-stage partition. We do not infer a missing solver-only time by subtracting available counters, and we retain the two cohorts' different timing conventions.

To inspect matrix-level quantities absent from the historical artifacts, we perform one new fixed-gain inner SDP at each saved ODQ gain across all $19 + 12$ cases. The lifting map, stabilizer factorization, and $Q = I$ remain fixed, and no outer optimization is repeated. Table 7 concerns these newly computed matrices; it does not reconstruct the historical solver matrices or replace their reported certificates and timings.

For $K = H_{\text{returned}} - H_{\text{native}}$, the gauge check evaluates every coefficient K_{jii} and $K_{jik} + K_{jki}$ for $i < k$, rather than testing sampled vectors. The certificate matrix is independently reassembled from the returned P, H. For $L = \left[\begin{smallmatrix} P\otimes I & G \\ G^T & \tau I \end{smallmatrix}\right]$, the scaled

TABLE 7

Fixed-gain audit of newly computed certificates, one per case. The first row gives the minimum positive eigenvalue; all other rows give maximum defects. These floating-point checks are distinct from the historical summaries above.

Diagnostic	Heterogeneous (19)	Relay (12)
$\lambda_{\min}(P)$	0.0306	0.0479
$\|A^TP + PA + I\|_2$	5.54×10^{-15}	3.26×10^{-16}
Coefficientwise gauge identity defect	4.44×10^{-16}	2.22×10^{-16}
Reassembled J discrepancy/ $\max\{1, \|J\|_2\}$	2.15×10^{-16}	2.42×10^{-16}
Model-LMI negative eigenvalue part (scaled)	2.17×10^{-9}	2.07×10^{-9}
Returned-H LMI negative eigenvalue part (scaled)	2.17×10^{-9}	2.07×10^{-9}
Epigraph discrepancy/ $\max\{1, \|J\|_2\}$	9.87×10^{-8}	7.54×10^{-9}
Independently assembled gauge KKT residual	4.45×10^{-12}	2.25×10^{-11}

semidefiniteness defect is $\max\{0, -\lambda_{\min}(L)\}/\max\{1, \|L\|_2\}$; we check both the solver's model G and the G reconstructed from the returned H. Small negative eigenvalues are recorded as numerical defects, not silently rounded to feasibility. These call times may overlap across workers and are diagnostic only.